TS5000 Robot Controller

"VISION SENSOR + CONVEYOR" SYNCHRONIZATION FUNCTION MANUAL

(Compatible Robots: THE Series SCARA Robot)

Rev.3

October 1, 2022

Important

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

This instruction manual applies to the controller TS5000.

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The information contained in this manual is subject to change without notice to effect improvements.

Introduction

Thank you very much for your purchase of SHIBAURA MACHINE's robot system.
This manual describes the Vision Sensor + Conveyor Synchronization function of the TS5000 Robot Controller.
The contents of this manual are subject to change without prior notice to effect improvements. Various instruction manuals are available, including the Operation, Robot Language, Interface, Communications, Transportation and Installation, and Maintenance manuals. Refer to these manuals as necessary.

This manual has the following chapters.

Chapter 1: Overview
・The overview and features of this function is described.

Chapters 2 to 4: Setup
・The connecting method between the TS5000 controller and the conveyor encoder/vision sensor is described.
・The setting procedures for the parameters exclusively used for the Ethernet function (ETHERNET.PAR) and the parameters used for the "Vision sensor + conveyor" synchronization function (CONVEYOR.PAR) are described.
・Various vision sensor setting procedures for communication between the vision sensor and robot controller TS5000 are described.

Chapters 5: Robot Language
・The programming procedures for using the "vision sensor + conveyor" synchronization function are described.

Chapters 6: Operation
・The utility function used for the "vision sensor + conveyor" synchronization function, and the procedures for stopping, resetting and restarting operations are described.

Chapters 7: Troubleshooting
・The self-diagnosis function and troubleshooting of the "vision sensor + conveyor" synchronization function are described.

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Use Range

The products have been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should use these products with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used for these purposes, it will be guaranteed under customer’s responsibility.

Use Method

This describes the operating conditions for the THE series. Usage under different conditions could expose the user to unexpected risks. Be sure to follow the required conditions.

1. Main Specifications of Product Refer to the robot manual for your robot.
2. Restrictions on Use: Tables 1 and 2 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1 Restricted Specification Sheet (1)

1. Purpose and application of using product (restrictions on use)	
1) Intended use	Allow the robot to achieve operation along with the purposes (article conveyance, assembly, inspection, machining, etc.). (1) During teaching and adjustment operation - Operation is likely to be performed by turning on the servo power within the movable range of the robot. - A person who observes teaching and operation checking worker as a co-worker is arranged outside the movable range. (1) Teach the position where the robot operates. (2) Create the operation sequence (program, etc.). (3) Transfer and check signal and data with the external equipment. (4) Check operation at low speed (250 mm/sec or less). (2) During operation - Danger prevention measures that prevent contact with the robot (installation of the fence or enclosure) are taken to keep a person away from the operating robot. - When the operating robot is approached, measures shall be taken to turn off the servo power. (1) Start the robot. (2) Perform operation along with the purpose at the specified speed. (3) Stop the robot. (3) During maintenance and inspection - Contents performed within the movable range of the robot when the primary power is off (1) Check the bolts for looseness and backlash and re-tighten them. (2) Check the cable clamp for looseness. (3) Inspect damage. (4) Clean dust. (5) Inspect and apply rust prevention agent. (6) Check the timing belt for looseness and crack. (7) Check the cable and air tube for wear. (8) Check the operation when each axis is pushed and moved by hand. (9) Replace the consumable parts for the robot and controller. (10) Replace the battery for memory in the controller. - Contents performed within the movable range of the robot when the primary power is on (1) Check the grease state and apply grease (when the servo power is off). (2) Push each axis by hand and check it for backlash (when the servo power is on). (3) Check abnormal vibration and sound during operation (when the servo power is on). (4) Check the teaching point for misalignment (when the servo power is on). (5) Replace the battery for robot position detector (when the servo power is off).

Table 2 Restricted Specification Sheet (2)

2) Predictable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot body, controller, and optional parts and energization with these covers removed - Operation in states other than the specified installation state (placed on floor, suspended from ceiling, mounted to wall, etc.) - Operation on the frame that does not satisfy the specified strength - Operation in other than the specified environment		
3) Unexpected start	- Malfunction caused by excessive noise - Robot start caused by start signal unexpectedly sent from the peripheral equipment to the robot - Unintended start caused by abnormal communication data - Malfunction caused by voltage fluctuation		
2. Replacement of product components (restrictions on time)			
1) Mechanical restriction	Replace consumable parts as required or periodically through daily inspection, regular inspection (every 3 and 6 months and every year) and overhaul (every 5 years).		
2) Electrical restriction			
3 Movable range of product (restriction on space)			
1) Operating range	Each axis	Check the specifications manual for your robot.	
	Additional axis	Compliant with the individual specifications.	
	-	-	
2) Interface	- The product shall be installed at a location suitable for outer shape and operating range. - Space required for maintenance and inspection shall be secured.		
3) Work environment	- Danger prevention measures that prevent contact with the robot (installation of the fence or enclosure) shall be taken. - Space shall be secured to prevent a person from being caught between the robot and the fence or enclosure.		
4. Life cycle of product			
During the life cycle of assembly, shipment test, transportation, installation, teaching, operation, maintenance and disposal, risk assessment is carried out for “installation” stage (transportation and installation), “use” stage (teaching, operation, and maintenance) and disposal.			
5. Person who approaches the machine			
Target	Knowledge, experience, and condition		
1) Transportation and installation worker	Knowledge: Person who has understood the instruction manual		
	Experience: —		
	Condition: Qualified person on forklift or crane (as required)		
2) Teaching/adjustment operation worker (including co-worker)	Knowledge: Person who has understood the instruction manual		
	Experience: Person who has acquired the operation of the robot to be used		
	Condition: Person who has received special training pertaining to the operation such as teaching of the industrial robot and to whom work has been permitted by the employer		
3) Operation worker (operator)	Knowledge: Person who has understood the instruction manual		
	Experience: Person who has understood the work regulation		
	Condition: —		
4) Maintenance and inspection worker	Knowledge: Person who has understood the instruction manual		
	Experience: Person who has acquired the operation of the robot to be used and person who has acquired maintenance work of the target robot		
	Condition: Person who has received special training pertaining to the operation such as inspection of the industrial robot and to whom work has been permitted by the employer		
5) Disposal worker	Knowledge: Person who has understood the instruction manual and the disposal method for each local government		
	Experience: —		
	Condition: —		
6. Other			

Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period

The warranty period of the product is either of the following, whichever is earlier.

(1) 18 months after delivery from our factory
(2) 12 months after installation at your site

2. Warranty scope

(1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
(2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
Any repairing after the warranty period passes will be charged.

3. Disclaimer

The warranty shall not apply to the following cases.

(1) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
(2) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables[*1)]).
(3) Any failure regarding sensuous phenomena such as noise not affecting the functions
(4) When any modification or disassembly without our consent is confirmed
(5) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
(6) Any failure or damage due to act of God, fires, and other force majeure
(7) Internal data such as programs and points created or modified by users
(8) When a product purchased in Japan is taken abroad

1) Consumables are listed in 7 Replacement Parts for Maintenance in this instruction manual.

4. Precautions

(1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
(2) SHIBAURA MACHINE shall not be liable for any injury, equipment damage or failure resulting from failure to observe the warnings and cautions stated in the instruction manual.
(3) Note that warnings, cautions, and other information are those assumed by SHIBAURA MACHINE.

Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.
Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, the ISO/IEC defines the "Guide 51" that groups various standards in a hierarchal structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Safety Precautions

The instruction manual for the controller contains important safety information for ensuring the safe and correct operation of this device and for preventing hazards to the operators and others and property damage.

To ensure that the procedures from installation to operation of the controller are performed safely, be sure to carefully read the separate instruction manual STA1179_THE Series Safety Manual before starting actual operation.

Read and understand the meanings of the following indications and symbols first, and observe these precautions.

[Meanings of Indications]

Indication	Meaning
⚠ DANGER	Indicates the imminent danger that incorrect operation could cause death or serious injury.
⚠ WARNING	Indicates the danger that incorrect operation could cause death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1] to the operator.
NOTICE	Indicates the assumed possibility of only damage to property[2].

1) Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
2) Damage to property indicates extensive damage related to the destruction of assets or materials.

[Meanings of symbols]

Symbol	Meaning
	Indicates a prohibited action. The specific content of the prohibition is shown pictorially or in text form inside the symbol or nearby it.
	Indicates a required action. The specific content of the instructed action is shown pictorially or in text form inside the symbol or nearby it.
	Indicates a danger and precaution. The specific content of the precaution is shown pictorially or in text form inside the symbol or nearby it.

Service Network

Please refer to the materials distributed from the SHIBAURA MACHINE agent.

Overview

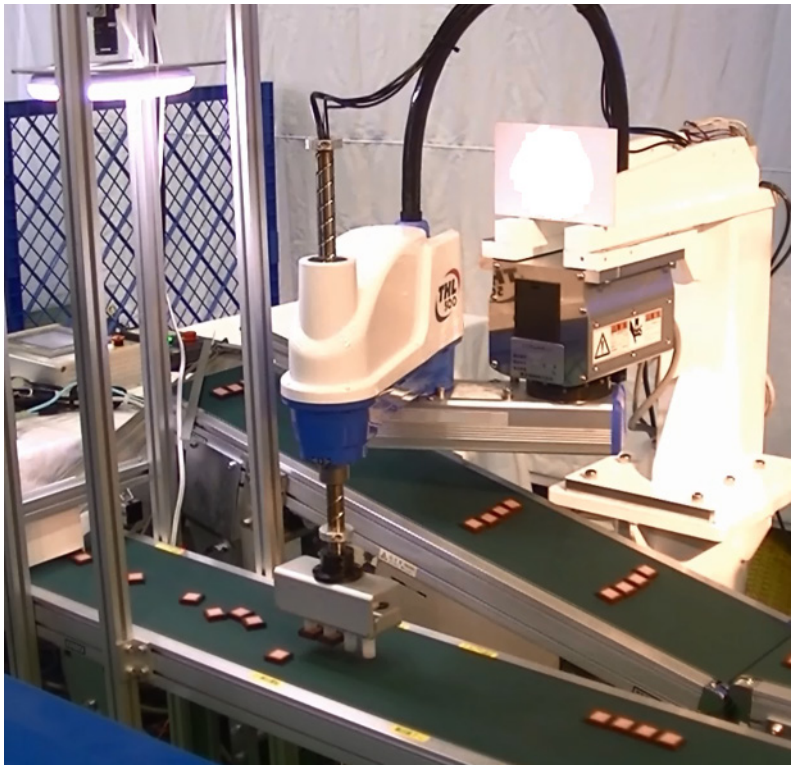
The overview and features of Vision Sensor + Conveyor Synchronization function is described.

1. General Descriptions

1.1 What is the Vision Sensor + Conveyor Synchronization Function?

This is a function that uses a vision sensor to recognize workpieces flowing on a conveyor and is capable of arranging and transferring workpieces while a robot tracks the movement speed of the workpieces (conveyor).

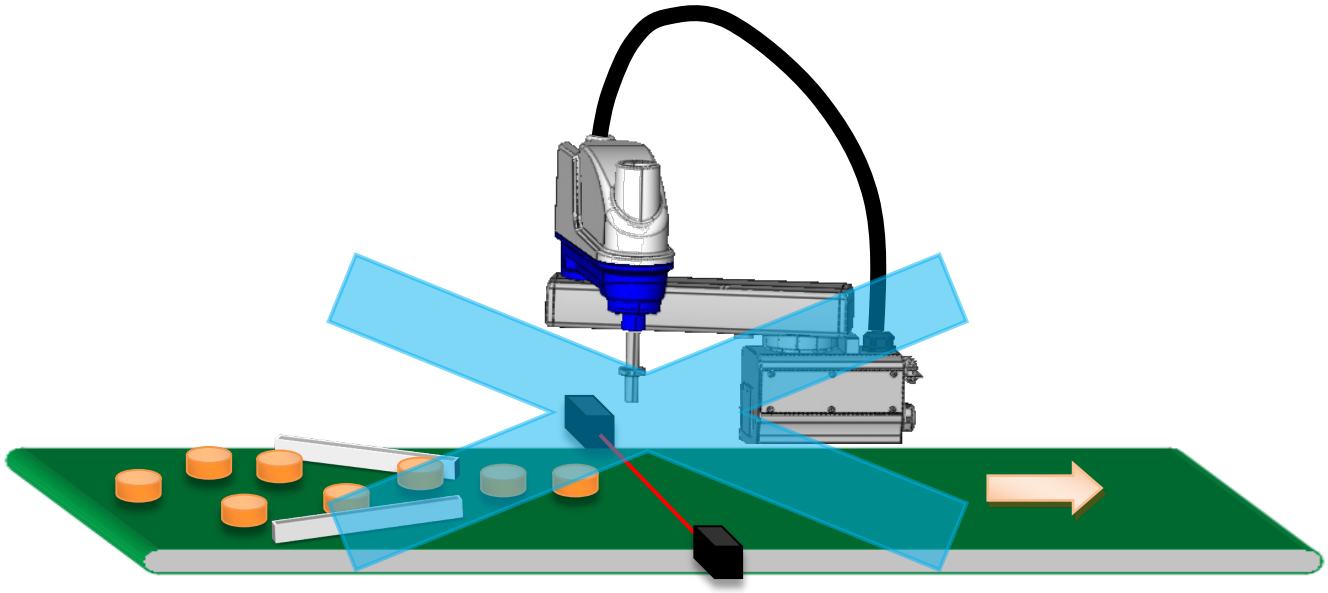
Because the robot operates while tracking the movement speed of the workpieces, operations can be performed without stopping the conveyor.



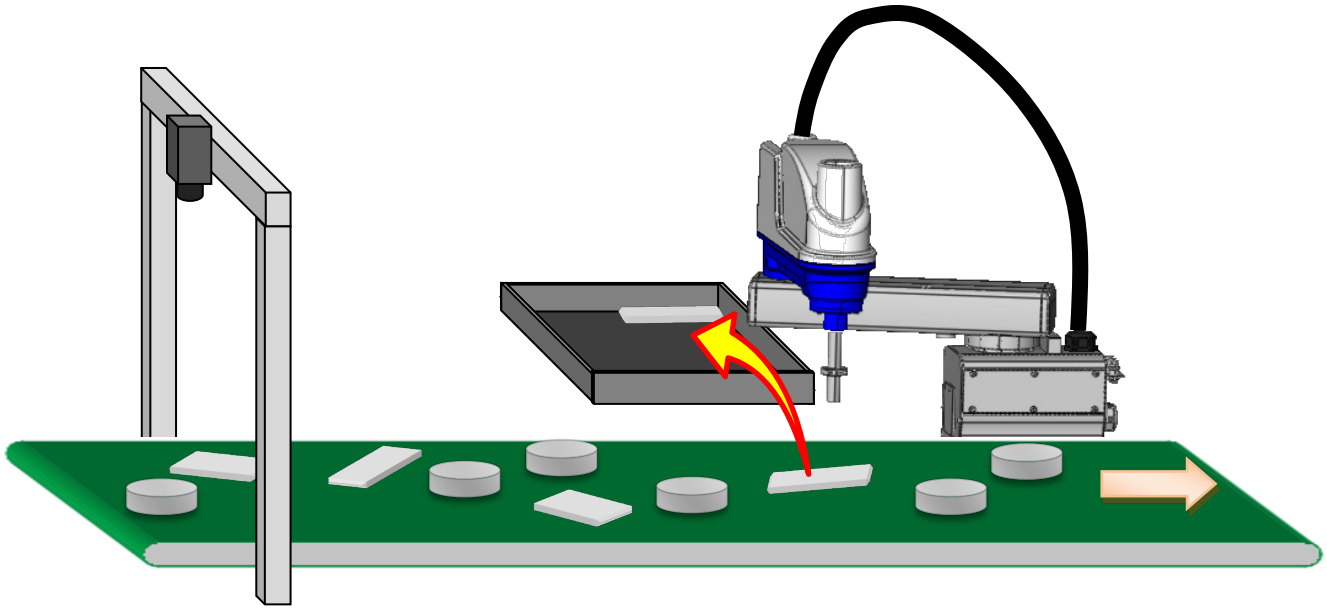
1.2 Advantages of Using the Vision Sensor + Conveyor Synchronization Function

- Commands are provided specifically for the Vision Sensor + Conveyor Synchronization function for enabling clear, simple programming. This eliminates the need for labor-intensive programming such as setting point data imported from the vision sensor, determining whether there is a next workpiece, and so on.

- In contrast to conveyor systems using a laser sensor, this function does not require positioning of the workpieces before moving the laser sensor to a preset position.

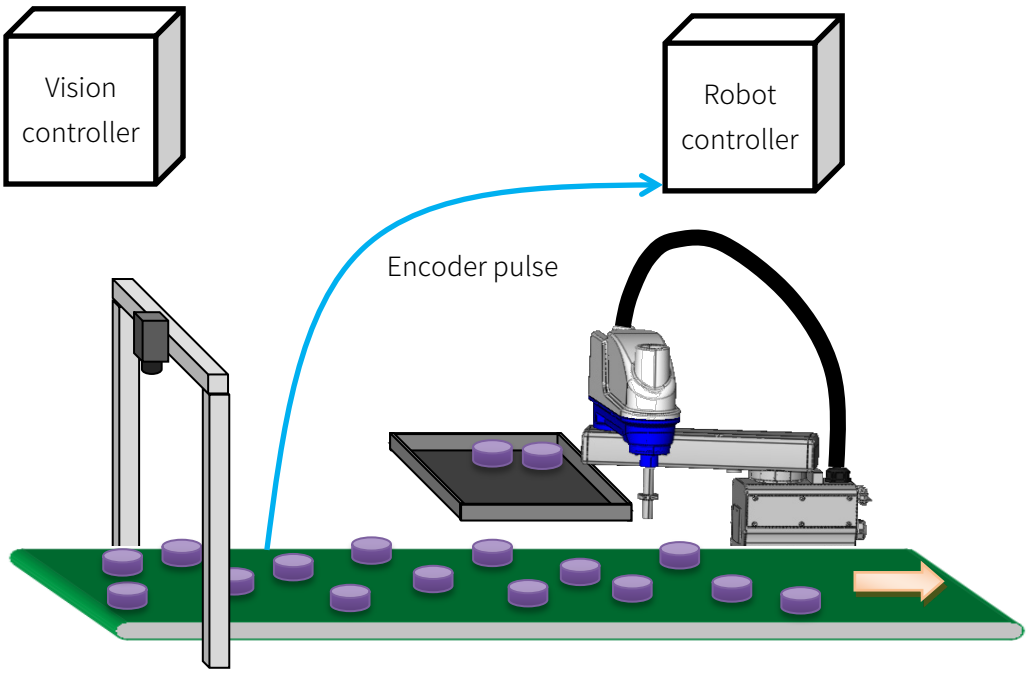


- Because product type data can be received from the vision sensor, specific product types only can be transferred. Also, different robot operations can be used for each product type.

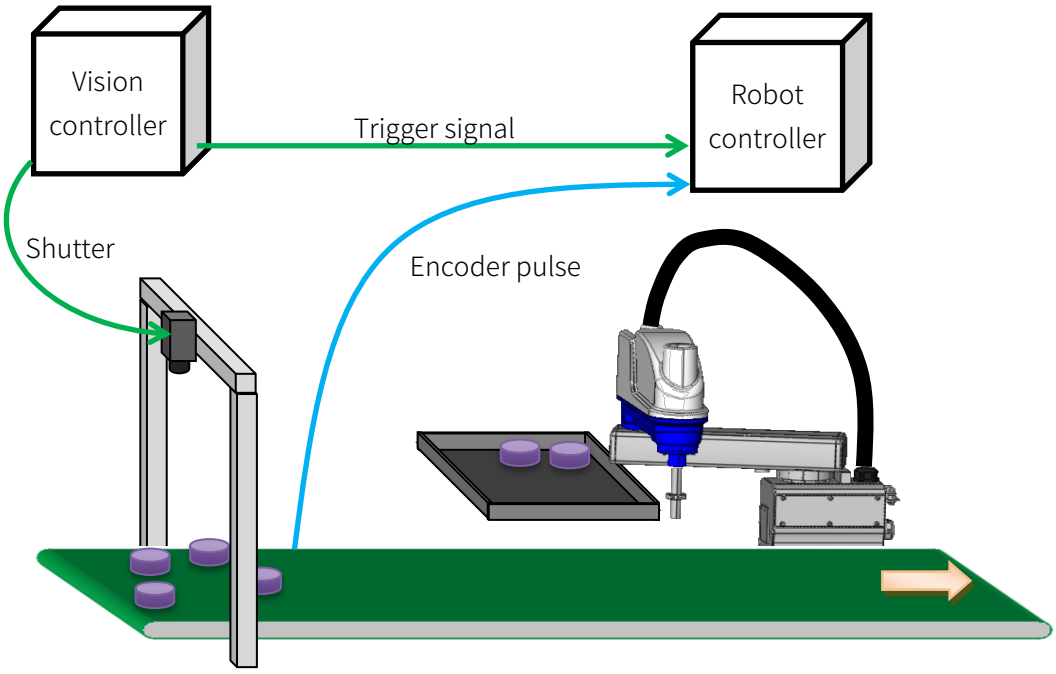


1.3 Flow of Operations in Vision Sensor + Conveyor Synchronization Function

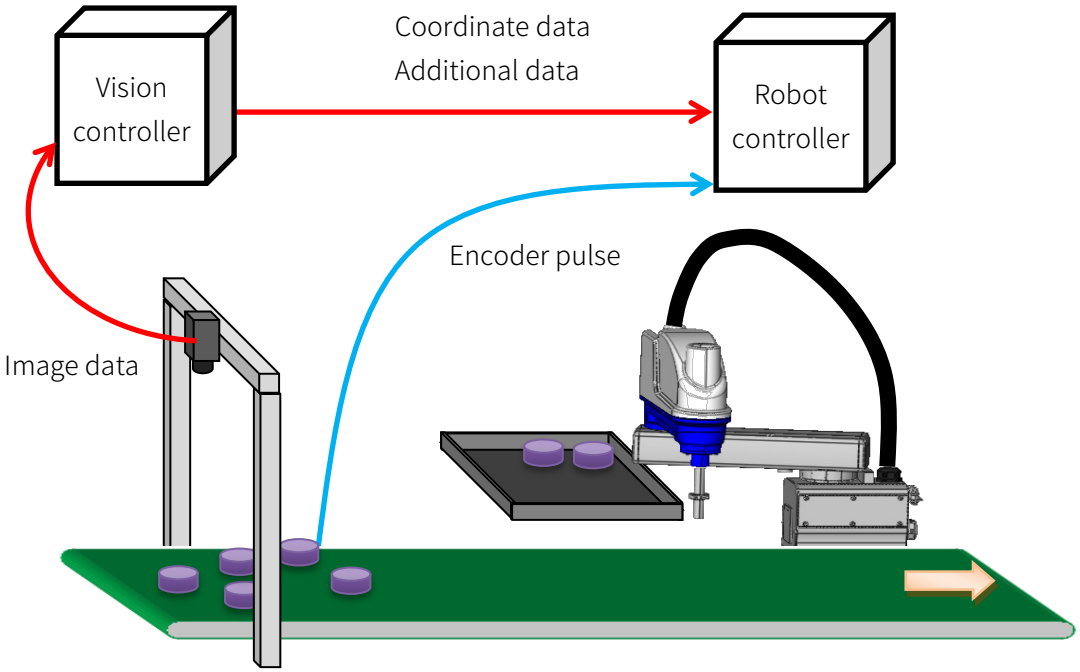
[1] The conveyor runs, and a conveyor encoder pulse is transferred to the robot controller.



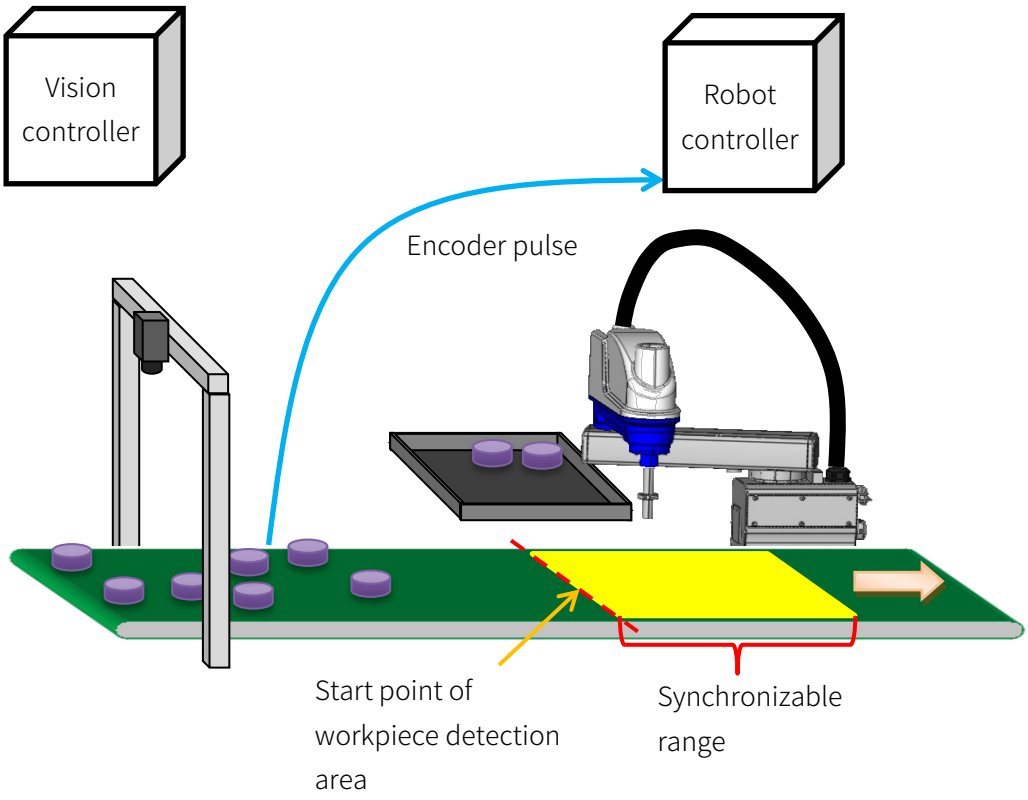
[2] The vision sensor shutter is shut periodically, and at the same time, a trigger signal is output to the robot controller.



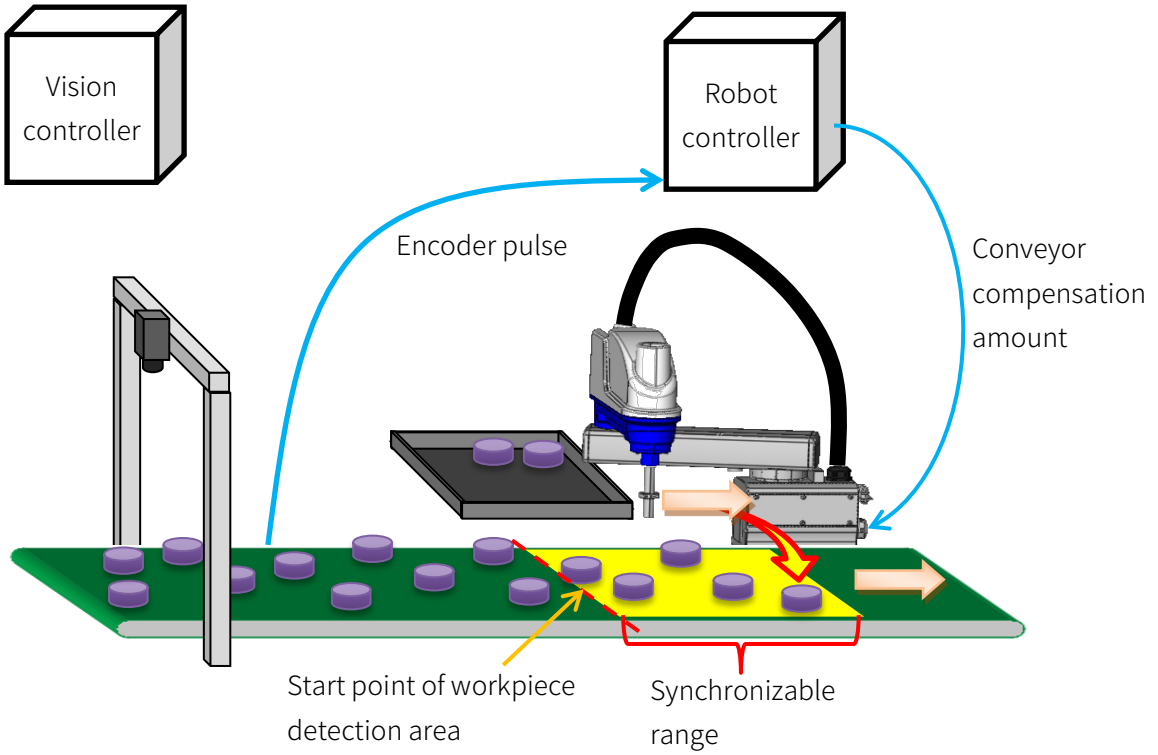
[3] The robot controller imports coordinate data and additional data for the workpiece detected in the capture area.



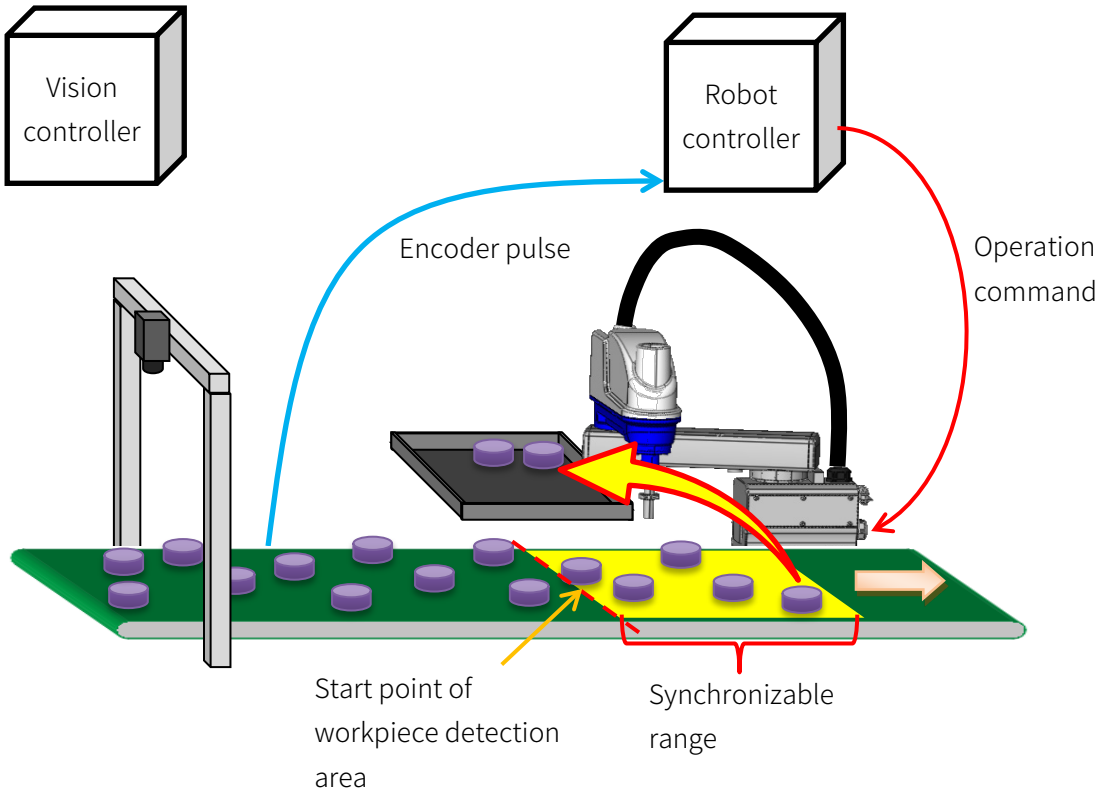
[4] The robot controller monitors the travel distances of workpieces based on the conveyor encoder pulse until the workpieces reach the start point of the workpiece detection area.



[5] Once the workpiece reaches the start point of the workpiece detection area, conveyor synchronization is started, and the operation that was set is executed.

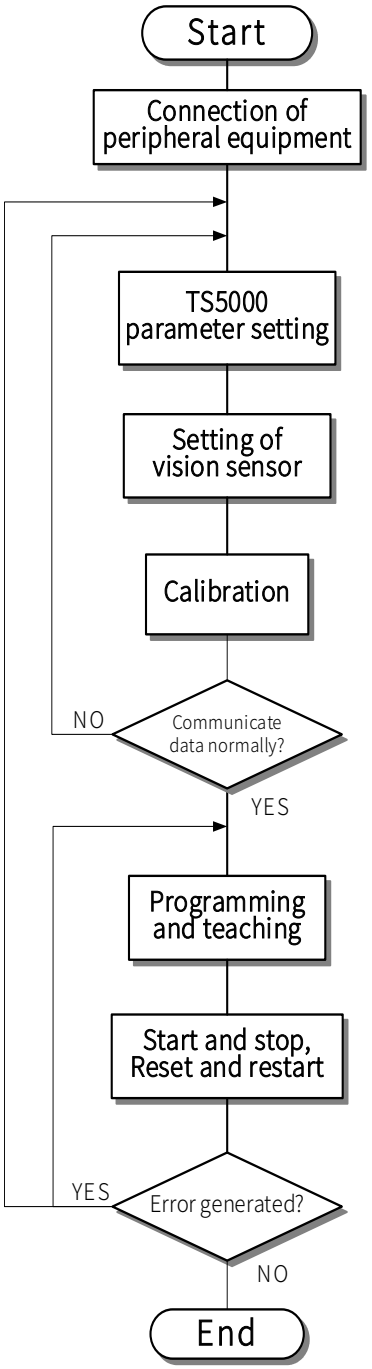


[6] After completing the operation performed during conveyor synchronization, conveyor synchronization is turned off, and the operation is resumed. (Following this, steps [1] to [6] are repeated.)



1.4 Setup Procedures

The Vision Sensor + Conveyor Synchronization System is set up based on the following flowchart.



Connect the TS5000 with peripheral equipment (conveyor encoder, vision sensor, etc.). → Chapter 2

Perform setting of Ethernet parameters (ETHERNET.PAR) and conveyor parameters (CONVEYOR.PAR). → Chapter 4

Set the transmit data format, imaging timing, data transmit protocol, trigger signal output, data output timing, etc. → Chapter 4

Execute calibration (i.e., the work for registering the robot coordinate system in the vision sensor).→ Chapter 4

Make sure of normal data reception, using the test program.

Create a program, using the "vision + conveyor" exclusive language.→ Chapter 5

Operate the robot, using a created program. → Chapter 6

Stop the program. When restarting the program, reset the conveyor as necessary. → Chapter 7

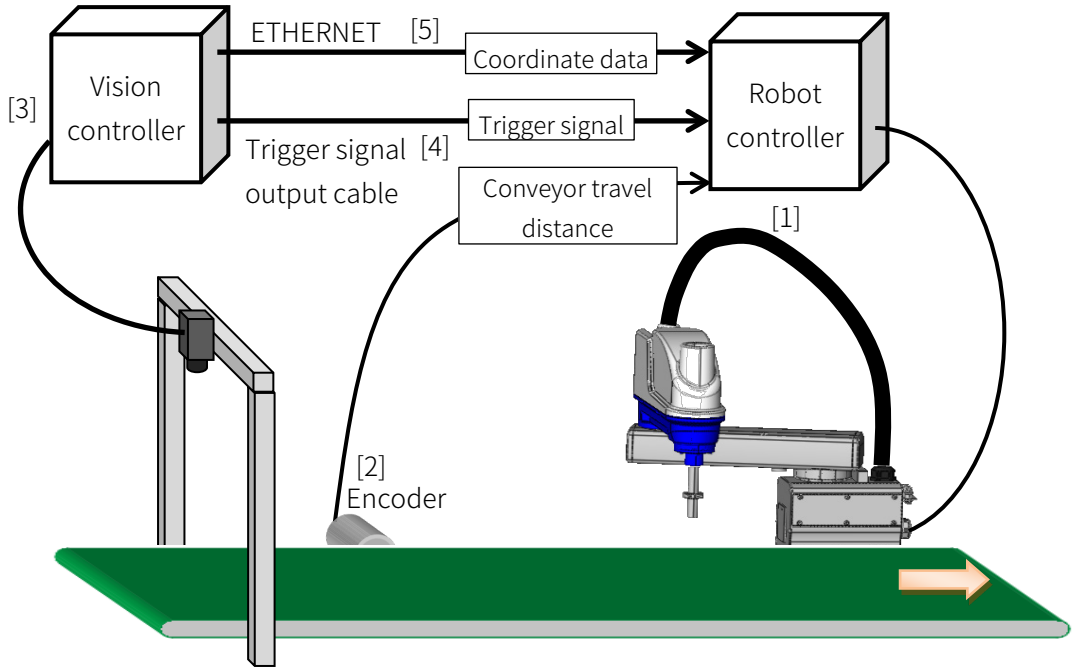
Setup

The connecting method between the TS5000 controller and the conveyor encoder/vision system, miscellaneous parameter and vision sensor settings are described.

2. System Configuration Examples

2.1 System Using One Vision Sensor and One Conveyor

The basic configuration of the Vision Sensor + Conveyor Synchronization System is shown in the figure below.

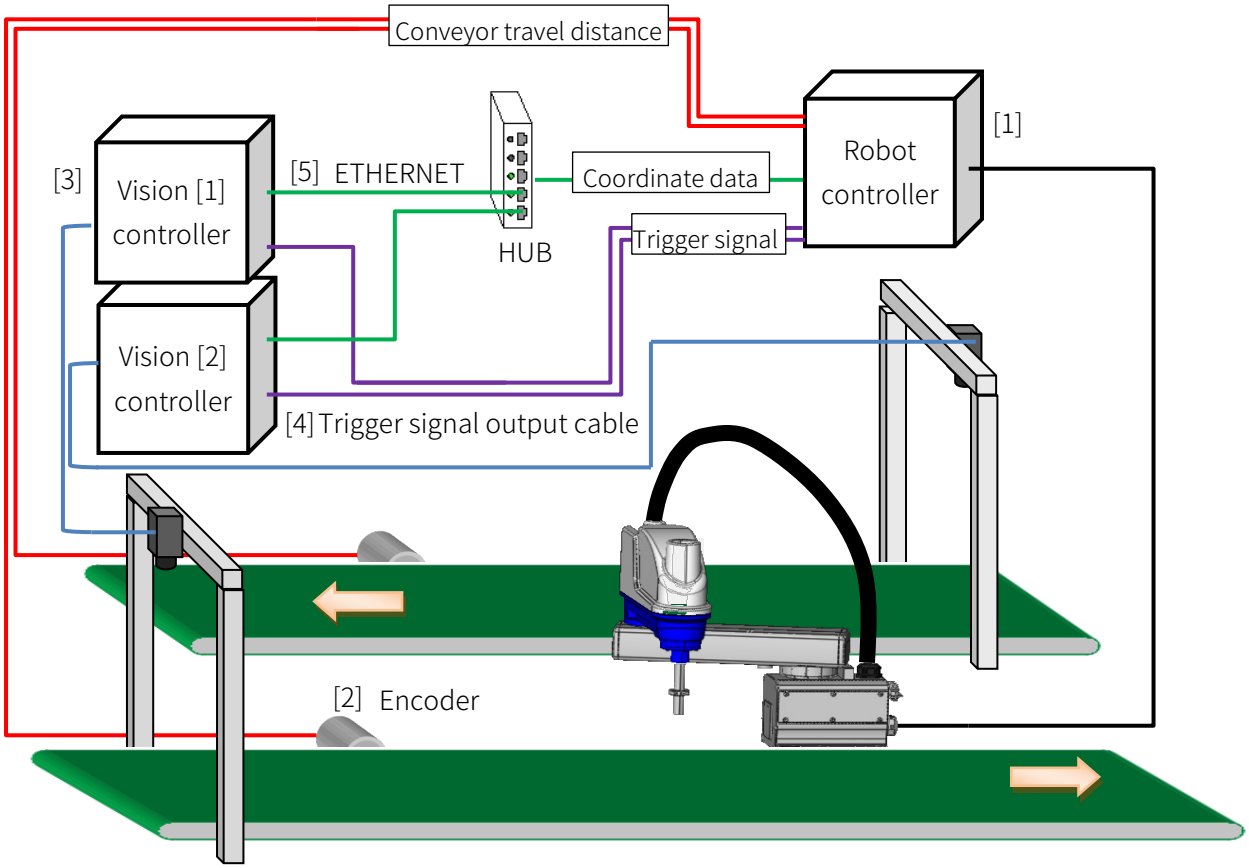


[1]	Robot unit and robot controller
[2]	Conveyor encoder
[3]	Vision sensor and controller
[4]	Trigger signal output cable
[5]	Ethernet cable

2.2 System Using Two Vision Sensors and Two Conveyors

The TS5000 can connect up to two vision sensors and two conveyors.

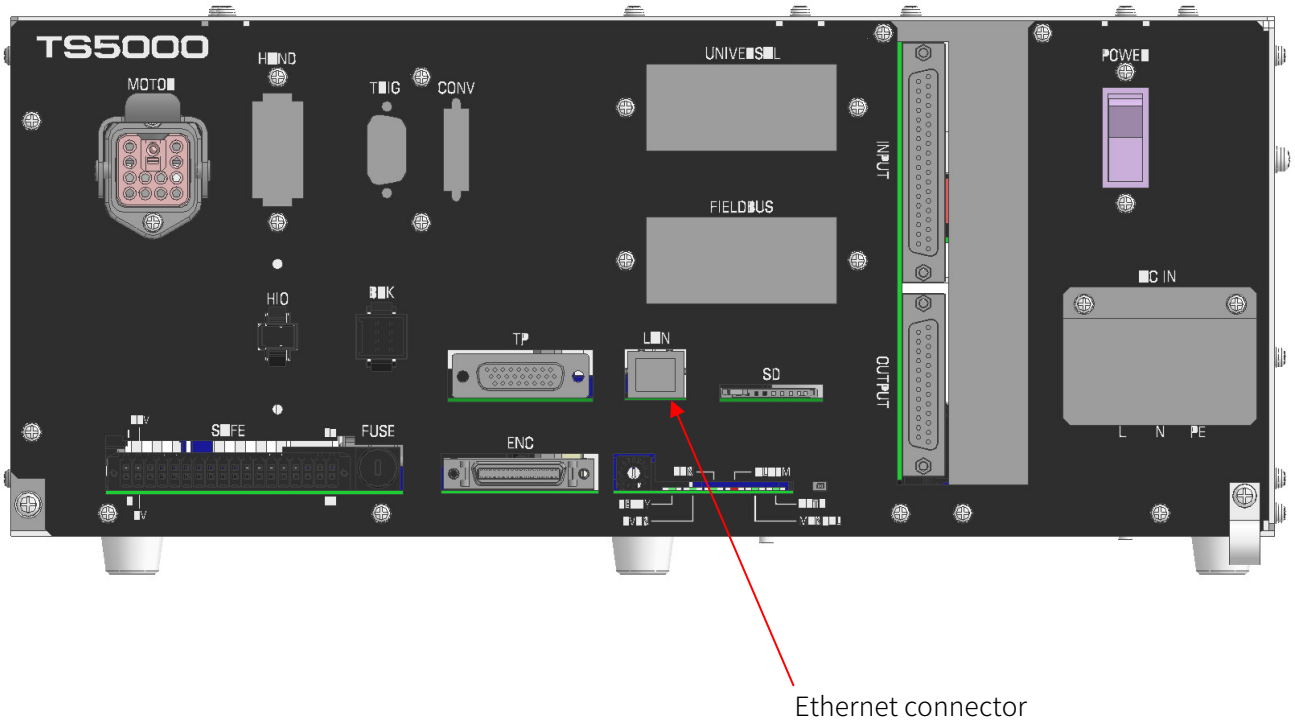
The configuration of the Vision Sensor + Conveyor Synchronization System when using two vision sensors and two conveyors is shown in the figure below.



[1]	Robot unit and robot controller
[2]	Conveyor encoder
[3]	Vision sensor and controller
[4]	Trigger signal output cable
[5]	Ethernet cable

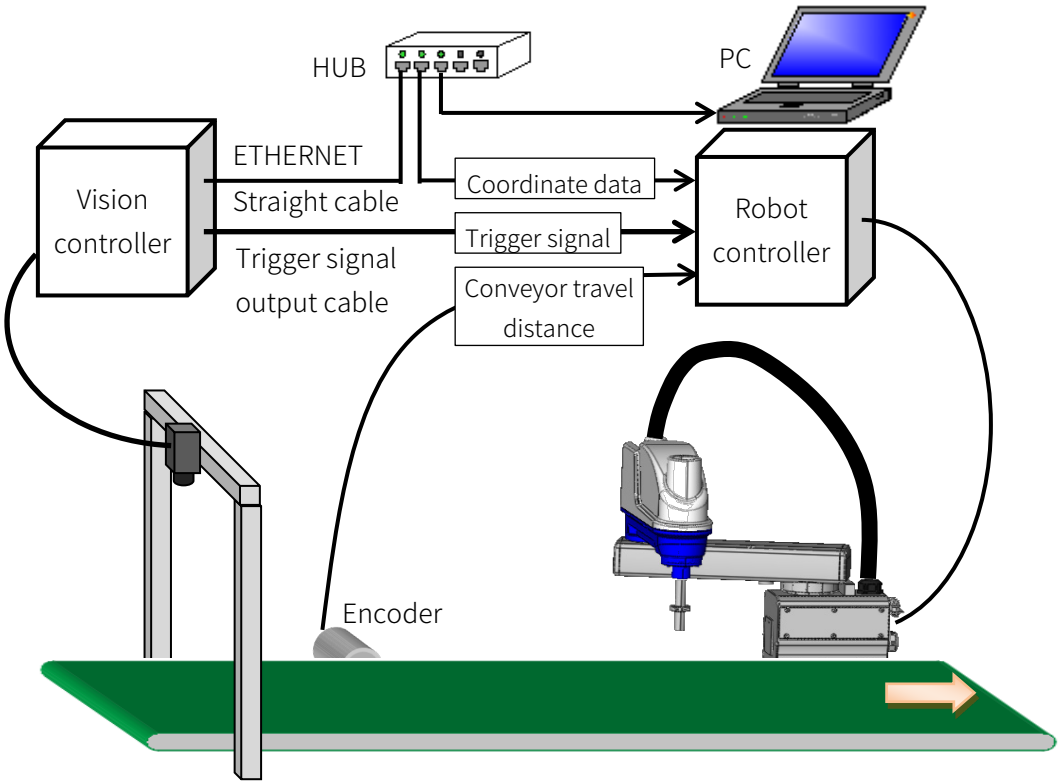
2.3 Connecting Vision Sensor (Ethernet)

The Ethernet function of the TS5000 controller supports the connection pursuant to the "1000BASE-X ".
Communication with a personal computer and vision sensor on the Ethernet is possible in the same local area, using the "TCP/IP" protocol.
The vision sensor is connected to the controller via Ethernet.
The cable is connected to connector "LAN" provided on the front side of the controller.



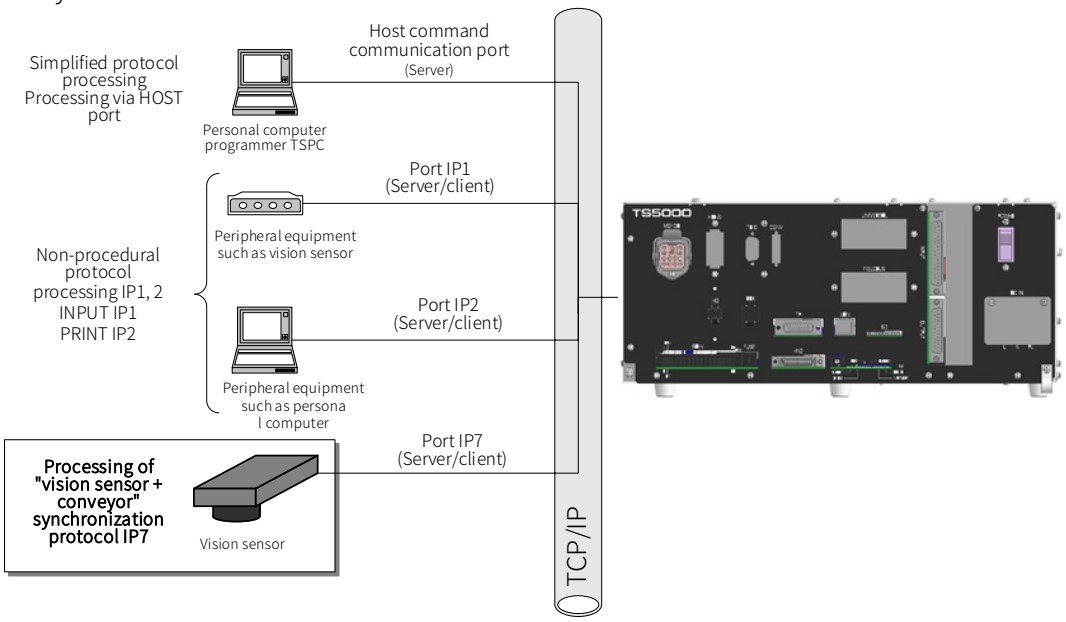
Ethernet connector

When connecting the TS5000 controller with the vision sensor through the Ethernet HUB, use a straight cable.

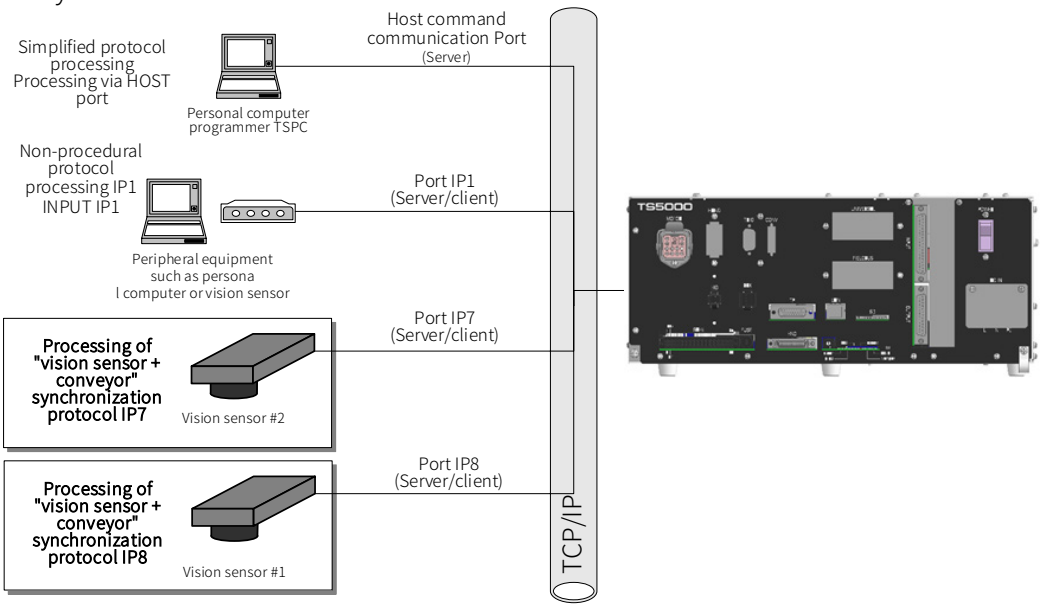


The TS5000 controller has eight Ethernet ports designated as IP1 to IP8. The IP7 and IP8 Ethernet ports can be used for the communication processes in the Vision Sensor + Conveyor Synchronization function.

<1CH System>

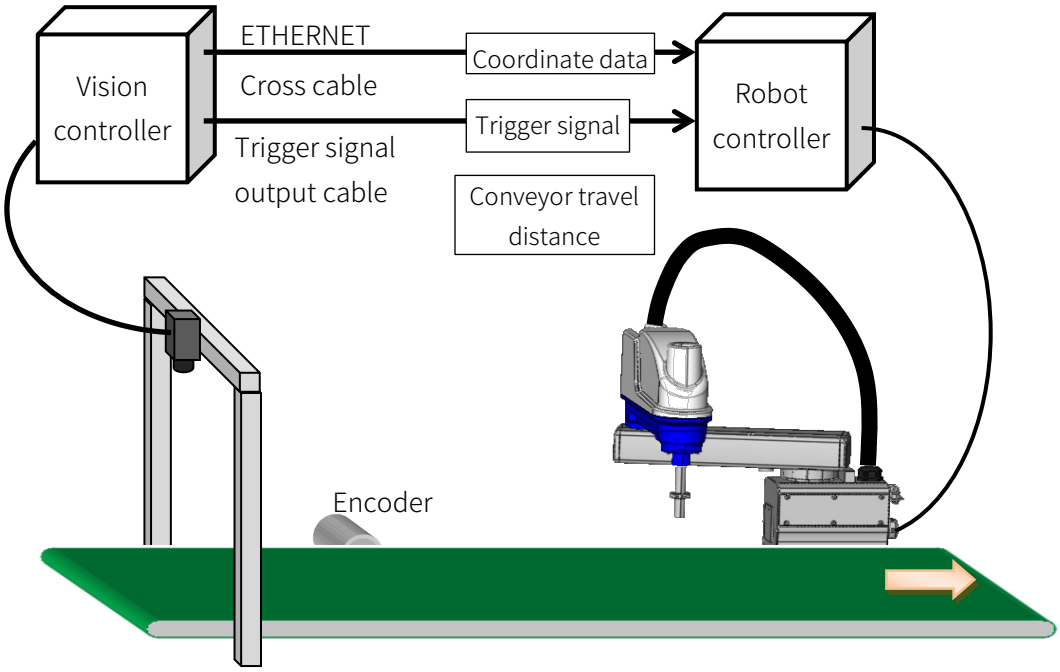


<2CH System>



* For details on setting the Ethernet IP address and port settings, refer to "Section 3 Parameter Setting and Preparations".

When connecting the TS5000 controller directly with the vision sensor, use a cross cable.



2.4 Connecting Trigger Input Signal

The conveyor synchronization function uses trigger input signals for detecting workpieces within the specified area. For this reason, the trigger input signal must be detected periodically. (Note that it is different from the imaging trigger that is output to the vision sensor camera.)
High-speed signal input is used for higher detection accuracy of the conveyor encoder position when the trigger signal is turned on.
The conveyor synchronization start trigger uses two (2) of the four (4) high-speed inputs. Connect the CH1 conveyor trigger to high-speed signal input DIN33 and the CH2 conveyor trigger to high-speed signal input DIN34. (For details, see Fig. 2.1.)
Also, when the high-speed signal input function is used for the conveyor synchronization start trigger switch, the remaining two (2) input switches other than the two (2) input switches become inoperative. (When using four (4) high-speed signal inputs, the position data latch function cannot be used together with the conveyor synchronization function.)

The high-speed signal input specifications of the TS5000 are shown below.

Table 2.1 High-speed signal input specifications

Item	Specifications
No. of inputs	Two (2) inputs (Of total four (4) inputs, only two (2) inputs are used for the conveyor synchronization.)
Input specification	Non-voltage contact input or transistor open collector input
Input signal (See Fig. 2.1.)	CH1: DIN33 CH2: DIN34
Delay time	2.2 µsec (calculated value)

2.4.1 Circuit Structure

The circuit structures are shown below. As a bi-directional photocoupler is used for the input circuit, the N Type or the P Type can be changed over.

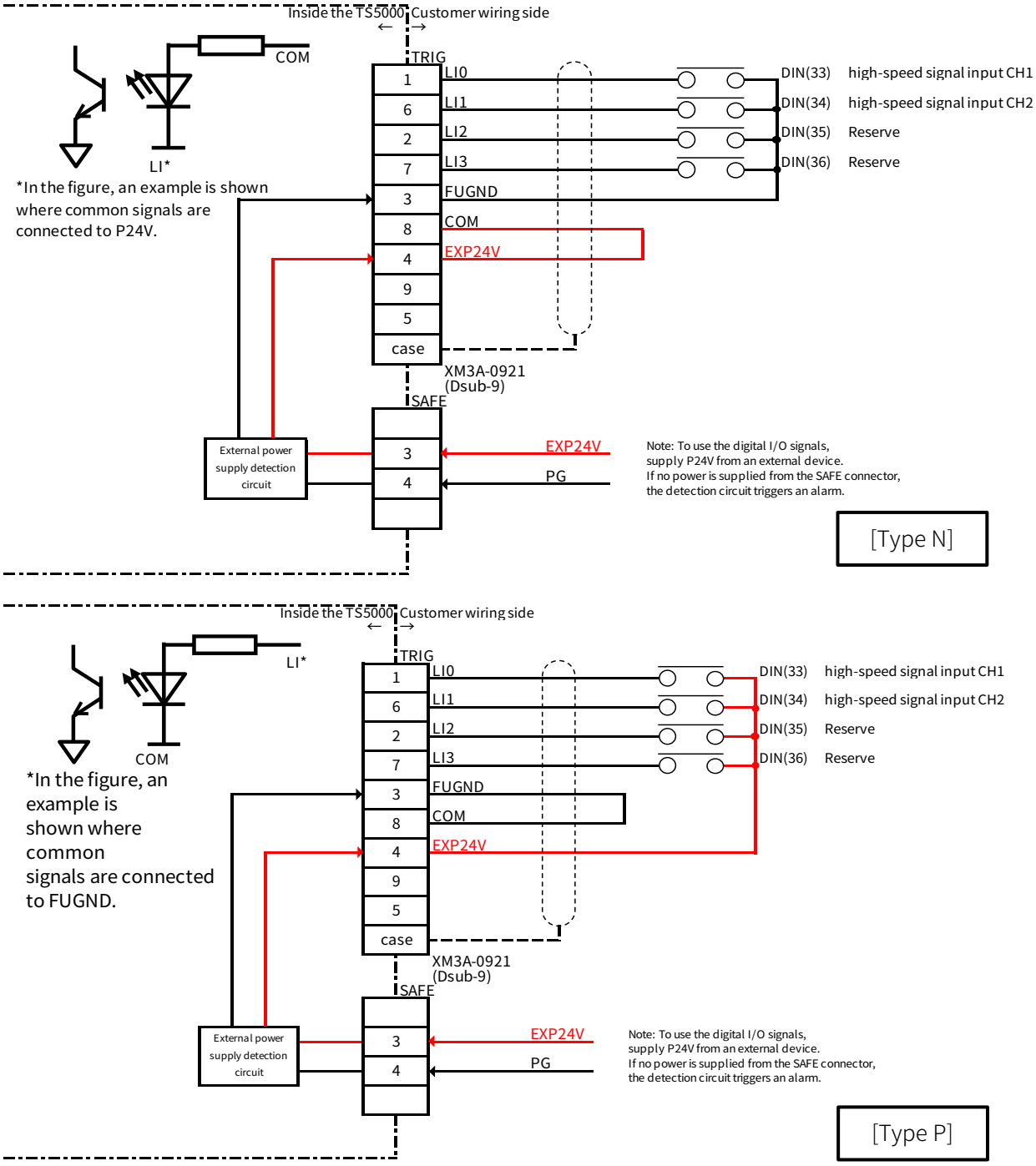


Figure 2.1 Trigger input signal line connections

2.4.2 Contact/Transistor Specifications

The contact/transistor connected with external device should satisfy the following specifications.

Table 2.2 Required non-voltage contact/transistor specifications

Non-voltage contact specifications		Transistor specifications
Contact rating:	24 VDC, 10 mA or over	Voltage between collector and emitter: 30 V or over Current between collector and emitter: 10 mA Circuit current: Approx. 7 mA Leak current between collector and emitter: 100 μA
Circuit current:	Approx. 7 mA	
Minimum current:	24 VDC, 10 mA	
Connected impedance:	100 Ω or less	

2.4.3 Specifications of Conveyor Trigger Input Signal Cable

The specifications of the conveyor trigger input signal cable are shown below.

Cable core: 0.18 mm^2 to 0.32 mm^2 twisted cable
Voltage resistance 60 V, 1 A or less, 75°C
Cable outer diameter: Max. 10 mm
Shield: Shield cluster
Cable length: 30 m or less

2.4.4 Specifications of Conveyor Trigger Input Signal Connector

Connect the conveyor trigger input signal with TRIG[1)].
For the connector signal array, see Para. 2.4.1.

1) Connector on controller side: Model JEZ-9S-3 (J.S.T.MFG. CO., LTD.)

Table 2.3 Connector and manufacturer name

Connector part name	Model and manufacturer name
Connector	XM3A-0921 (OMRON Corporation)
Connector cover	XM2S-0911 (OMRON Corporation)

2.5 Connecting Conveyor Encoder

When the conveyor synchronization function of the TS5000 is used, the signal from the incremental encoder and input signal from the conveyor synchronization start trigger switch are required.

2.5.1 Incremental Encoder Specifications

Use such an encoder having phase A and B signals with 90° phase difference, whose output circuit is of an RS–422 differential output (line driver) type.

Table 2.4 Encoder specifications

Phases A/B maximum response frequency	250 kHz
Cable breakage detection	Provided

2.5.2 Incremental Encoder Connector Specifications

Connect the incremental encoder with CONV[1)].

For the connector signal array, see Para. 2.5.3.

1) Connector on controller side: Model PCR-E20PMRS+ (Honda Tsushin Kogyo Co., Ltd.)

Table 2.5 Connector and manufacturer name

Connector part name	Model and manufacturer name
Connector	PCR-S20FS+ (Honda Tsushin Kogyo Co., Ltd.)
Connector cover	PCR-V20LB (Honda Tsushin Kogyo Co., Ltd.)

2.5.3 Connection between Incremental Encoders

Use and connect a shield twisted pair cable as shown in the figure below. Also, make sure it is connected in pairs, as illustrated below. If it is connected incorrectly, noise resistance deteriorates, causing a malfunction.

When the robot controller external power supply is used, connect according to Fig. 2.2. When the internal power supply is used, see Fig. 2.3. (The encoder input comes in two (2) channels. Use them when necessary.)

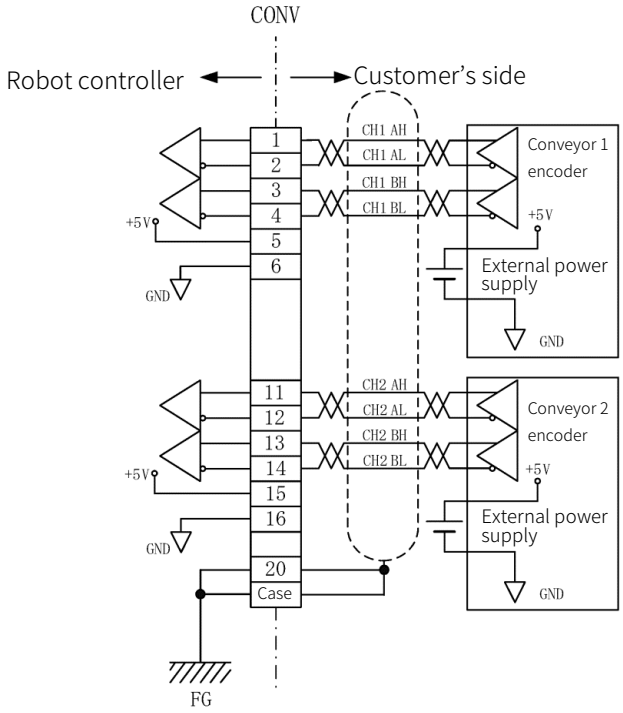


Figure 2.2 Connection between incremental and encoder devices (using external power supply), recommended

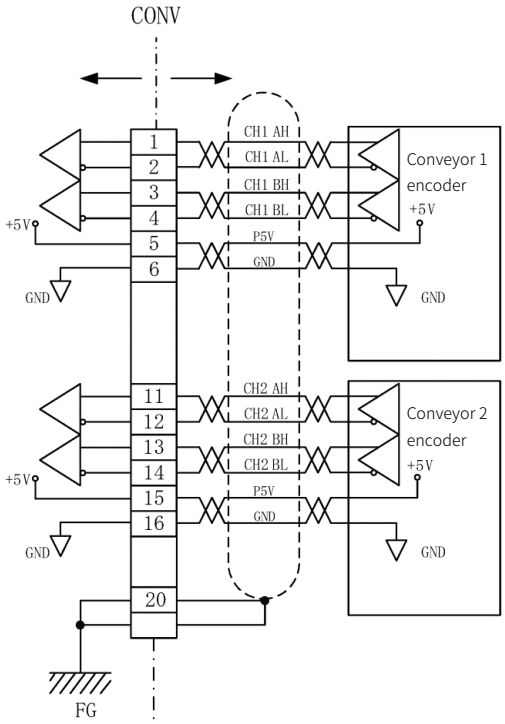


Figure 2.3 Connection between incremental and encoder devices (using internal power supply) , not recommended

2.5.4 Relationship between Encoder Cable Length and Number of Power Supply Twisted Pairs

When +5 V supplied from the robot controller is used for the encoder power supply, connect the cable, considering voltage drop in the cable.

[Example of calculating cable voltage drop]
When the allowable encoder voltage is taken as 4.5 V, relationship between the cable length and number of power supply twisted pairs is as shown below.
(Assume that cable resistance is 110 Ω/km, encoder current consumption 150 mA and controller supply voltage 5 V.)

$$V_e = 5 - \frac{110}{1000} \times l \times 2 \times \frac{150}{1000} \times \frac{1}{p}$$

"Ve" indicates the encoder voltage, "l " the cable length, "2" the counts of reciprocations and "p" the number of twisted pairs.
From the above expression, it can be known that the allowable encoder voltage of 4.5 V can be assured under the following conditions.

No. of power supply twisted pairs	Cable length
1	15 m or less
2	30 m or less
3	45 m or less

When the cable of more than 45 m is required, therefore, the external power supply should be used.

The example of the above calculation is valid under the above conditions only.
Figure out, referring to the example of the above calculation. Especially, note that the current consumption differs depending on the encoder to be used.
If the external power supply is used, the cable length can be longer but no more than 100 m, in terms of the signal.

2.5.5 Fabricating Cable

Fabricate a cable in the following manner.

a) Incremental encoder cable specifications

Core cable: 0.2 mm^2 to 0.52 mm^2, twisted wires
Voltage resistance 60 V, 1 A or less, 75°C
Cable outer diameter: Max. 8.0 mm
Shield: Shield cluster
Cable length: See Para. 2.5.4.

Use such a cable which satisfies the above specifications. Otherwise, the robot may malfunction due to noise entry, in addition to fusing caused by short-circuit or overheating of the core cable.

b) Fabricating a cable

The connector is of a solder cup type which solders wires.
To fabricate a cable, take careful precautions on soldering of the wires to the connector and shield processing, and refer to Fig. 2.4

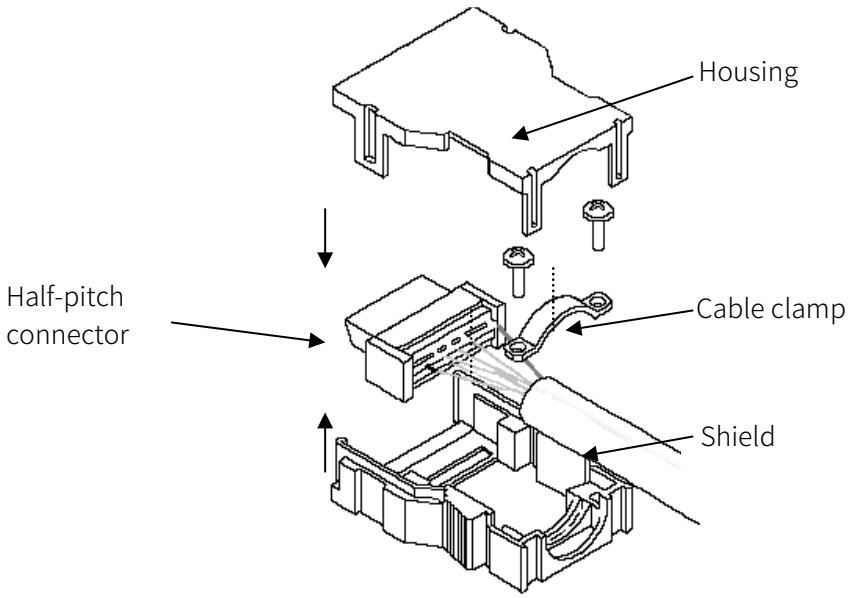


Figure 2.4 Fabricating a cable

MEMO

・The shield of the cable is connected with the grounding terminal of the robot controller through the housing. The shield should be connected to the housing completely in the cable clamping part.

2.5.6 Defining Optimal Resolution of Encoder

The minimum processing period of conveyor synchronization is 3 msec. If the number of pulses of the conveyor encoder that can be sampled in this period is too small, the tracking accuracy to the conveyor will deteriorate.

Select the encoder so that [**Vpulse: Number of sampling pulses per 3 msec**] as calculated from the following equation **is about 10 to 100 pulses**.

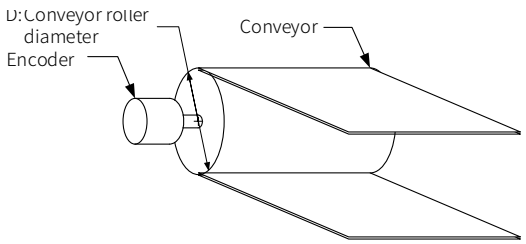
[Vpulse: Number of sampling pulses per 3 msec] can be figured out from the following equation.

$$V_{pulse} = \frac{3}{1000} \cdot R_e \left(\frac{V_{CNV}}{\pi \cdot D} \cdot \frac{1}{G_m} \right) \qquad (1)$$

where,

- Re: Encoder resolution (ppr)
  When the 2000 (ppr) resolution encoder is used, for instance, it is processed as 8000 (ppr) (i.e., 2000 multiplied by 4) in the TS3000 controller.

- Vcnv : Conveyor speed (mm/sec)
  Customer's desired conveyor speed
  Vcnv = 50 (mm/sec)



- D : Conveyor roller diameter (mm)
  Customer's conveyor roller diameter
  D = 100 (mm)

- Gm: Reduction ratio (Conveyor roller speed / Encoder speed)
  This value differs depending on the customer's encoder set conditions.

[Example] When the encoder revolves 25 turns while the conveyor roller revolves one (1) turn:
Gm[1] = 1/25 = 0.04
1) The value of Gm (reduction ratio) should be exactly divisible.

From Expression (1) above, Expression (2) can be obtained.

$$R_e = V_{pulse} \cdot \frac{1000}{3} \cdot \frac{\pi \cdot D \cdot G_m}{V_{CNV}} \qquad (2)$$

When the values are substituted for VCNV, Gm, D and Vpulse in Expression (2), the optimal encoder resolution (Re) is about 840 to 8400 (ppr).

3. Parameter Setting and Preparations

Parameter Setting and Preparations of the Vision Sensor + Conveyor Synchronization function can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

3.1 ETHERNET.PAR

Tabled below are the parameters used in the Ethernet function of the controller. The items marked with a check mark in the table below indicate parameters that must be set in order to use the Vision Sensor + Conveyor Synchronization function.

Check	Data	Description
	[E000] ROBOT CONTROLLER IP ADDRESS	Specifies the robot controller's own IP address.
	[E001] SUBNET MASK	Specifies the subnet mask of the connected network.
	[E002] DEFAULT GATEWAY	Default gateway address
	[E003] ROBOT COMMUNITY NAME	Sets the controller name
✓	[E004] OPEN MODE IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8	Designates the open mode of the Ethernet port.
✓	[E005] OWN PORT NUMBER	Specifies the controller side port number of each port.
✓	[E006] IP ADDRESS OF DESTINATION	Specifies the destination IP address of each port. (To be specified only when the client mode is selected.)
✓	[E007] PORT NUMBER OF DESTINATION	Specifies the destination (i.e., server side) port number of each port. (To be specified only when the client mode is selected.)
	[E008] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8	Sets the alarm level to be generated when the Ethernet connection is lost during program RUN

[E000] ROBOT CONTROLLER IP ADDRESS

Controller IP address

Specify the robot controller's own IP address.

Parameter key	IP-ADDRESS
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data format	Integer type
Data number	4
Data setting range	-
Default	=192，168，0，124

Setting Example (Default)

‘[E000] ROBOT CONTROLLER IP ADDRESS
IP-ADDRESS = 192, 168, 0, 124

MEMO

・Use a comma to separate the numbers, not a dot.

・Be careful not to duplicate IP addresses in the network.

・If the IP address is not set correctly, connection to the controller fails.

・IP addresses "223.255.0.0" to "223.255.255.255" are reserved by the controller. When setting the controller IP address, set an IP address outside the range reserved by the system.

・To apply the setting, power restart is necessary.

[E001] SUBNET MASK

Controller subnet mask setting

Specify the subnet mask of the network to be connected.

Parameter key	SUBNETMASK
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data format	Integer type
Data number	4
Data setting range	-
Default	=255，255，255，0

Setting Example (Default)

```
‘[E001] SUBNET MASK
SUBNETMASK = 255, 255, 255, 0
```

MEMO

・If the subnet mask is not set correctly, connection to the controller fails.

[E002] DEFAULT GATEWAY

Controller default gateway setting

Specify the default gateway of the controller.

Parameter key	GATEWAY
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data format	Integer type
Data number	4
Data setting range	-
Default	=192，168，0，1

Setting Example (Default)

```
‘[E002] DEFAULT GATEWAY
GATEWAY = 192, 168, 0, 1
```

MEMO

・If the default gateway is not set correctly, connection to the controller fails.

・To apply the setting, power restart is necessary.

[E003] ROBOT COMMUNITY NAME

Controller name

Sets the controller name.

Parameter key	ROBO-COMMUNITY-NAME
Parameter data	=”NAME”
Data format	String type
Data number	1
Data setting range	-
Default	=”TS5K”

Setting Example (Default)

```
‘[E003] ROBOT COMMUNITY NAME
ROBO-COMMUNITY-NAME = "TS5K"
```

MEMO

・Set the robot controller name to a unique value on the network.

・To apply the setting, power restart is necessary.

[E004] OPEN MODE IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8

Open mode of each IP

Selects the open mode of IP1 to IP8.

Parameter key	OPEN-MODE	
Parameter data	=(IP1), (IP2), (IP3), (IP4), (IP5), (IP6), (IP7), (IP8)	
Data format	Integer type	
Data number	1	
Data setting range	0 to 3	
Describe	=0	Unused
	=1	Set the controller as a TCP / IP server for communication.
	=2	Set the controller as a TCP / IP client for communication.
	=3	The controller communicates using UDP.
Default	=1,0,0,0,0,0,0,0	

Setting Example (Vision + Conveyor)

```
'[E004] OPEN MODE IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8
OPEN-MODE = 0, 0, 0, 0, 0, 0, 2, 0
```

MEMO

・If the server, client, and UDP settings are not set correctly, the controller cannot be connected.

・When using the vision sensor + conveyor synchronization function, assign it to IP7 or IP8.

[E005] OWN PORT NUMBER

Controller port number

Specify the port number of each port used by the robot controller.

Parameter key	OWN-PORT-NO[num[1)]]
Parameter data	=(Port number)
Data format	Integer type
Data number	1
Data setting range	10000 to 65535

1) num is an integer from 0 to 7.

Setting Example

```
‘[E005] OWN PORT NUMBER
‘{IP1}
OWN-PORT-NO[0] = 10000
‘{IP2}
OWN-PORT-NO[1] = 10001
‘{IP3}
OWN-PORT-NO[2] = 10002
‘{IP4}
OWN-PORT-NO[3] = 10003
‘{IP5}
OWN-PORT-NO[4] = 10004
‘{IP6}
OWN-PORT-NO[5] = 10005
‘{IP7}
OWN-PORT-NO[6] = 10006
‘{IP8}
OWN-PORT-NO[7] = 10007
```

MEMO

・Used when the OPEN MODE set in [E004] is server (= 1).

・Set the port number so that it does not overlap.

・If the port number is not set correctly, connection to the controller fails.

[E006] IP ADDRESS OF DESTINATION

Connection destination IP address

Specify the destination IP address from IP1 to IP8.

Parameter key	DEST-IP-ADR[num[1)]]
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data format	Integer type
Data number	4
Data setting range	-

1) num is an integer from 0 to 7.

Setting Example

```
‘[E006] IP ADDRESS OF DESTINATION
‘{IP1}
DEST-IP-ADR[0] = 192, 168, 0, 150
‘{IP2}
DEST-IP-ADR[1] = 192, 168, 0, 151
‘{IP3}
DEST-IP-ADR[2] = 192, 168, 0, 152
‘{IP4}
DEST-IP-ADR[3] = 192, 168, 0, 153
‘{IP5}
DEST-IP-ADR[4] = 192, 168, 0, 154
‘{IP6}
DEST-IP-ADR[5] = 192, 168, 0, 155
‘{IP7}
DEST-IP-ADR[6] = 192, 168, 0, 156
‘{IP8}
DEST-IP-ADR[7] = 192, 168, 0, 157
```

MEMO

・Used when OPEN MODE set in [E004] is client (= 2).

・If the IP address is not set correctly, connection to the controller fails.

・When using the vision sensor + conveyor synchronization function, assign it to IP7 or IP8.

[E007] PORT NUMBER OF DESTINATION

Connection destination port number

Set the connection port number of IP1 to IP8.

Parameter key	DEST-PORT-NO [num[1)]]
Parameter data	=(Communication Port Number of Connected Equipment)
Data format	Integer type
Data number	4
Data setting range	-

1) num is an integer from 0 to 7.

Setting Example

```
‘[E007] PORT NUMBER OF DESTINATION
‘{IP1}
DEST-PORT-NO[0] = 10000
‘{IP2}
DEST-PORT-NO[1] = 10001
 ‘{IP3}
DEST-PORT-NO[2] = 10002
 ‘{IP4}
DEST-PORT-NO[3] = 10003
 ‘{IP5}
DEST-PORT-NO[4] = 10004
 ‘{IP6}
DEST-PORT-NO[5] = 10005
 ‘{IP7}
DEST-PORT-NO[6] = 10006
 ‘{IP8}
DEST-PORT-NO[7] = 10007
```

MEMO

・Used when OPEN MODE set in [E004] is client (= 2).

・If the port number is not set correctly, connection to the controller fails.

[E008] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8

TCP status alarm

Set the alarm level to be generated when the Ethernet connection is lost during program RUN for IP1 to IP8 of the controller.

Parameter key	TCP-STATUS-ALM
Parameter data	=(IP1), (IP2), (IP3), (IP4), (IP5), (IP6), (IP7), (IP8)
Data format	Integer type
Data number	8
Data setting range	0, 1, 2, 4 or 8
Default	=0, 0, 0, 0, 0, 0, 0, 0

Setting Example (Default)

```
‘[E008] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8
TCP-STATUS-ALM = 8, 4, 2, 0, 0, 0, 0, 0
```

MEMO

If the value is set to 0, no alarm is generated.

3.2 Confirming Connection on Ethernet Utility Display

Before using the Ethernet function, confirm the following matters.

(1) Make sure that the Ethernet cable between the robot controller and external equipment (personal computer, for instance) is connected properly.

(2) Make sure that the Ethernet cable used is correct.

- When directly connecting the external equipment and robot controller, a cross cable should be used.
- When the HUB is used, a straight cable should be used.

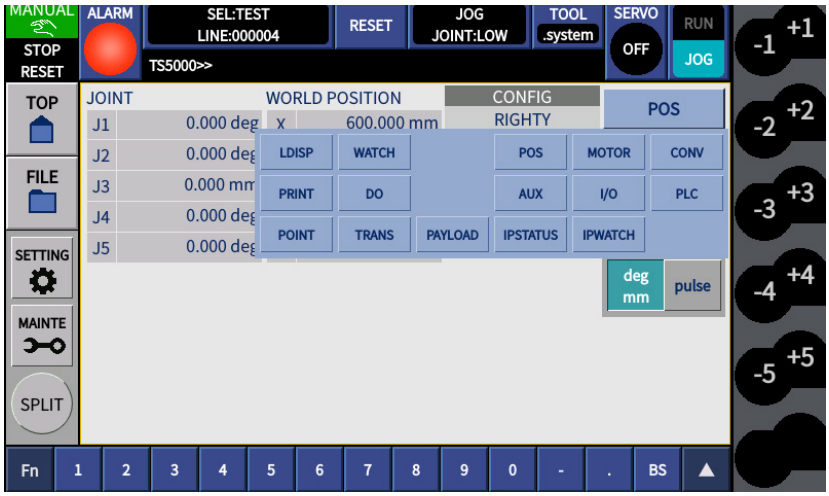
(3) Make sure that the parameters of the controller are set correctly.

(4) Make sure that the controller power was turned off once, after the parameter setting.

<For TP5000>

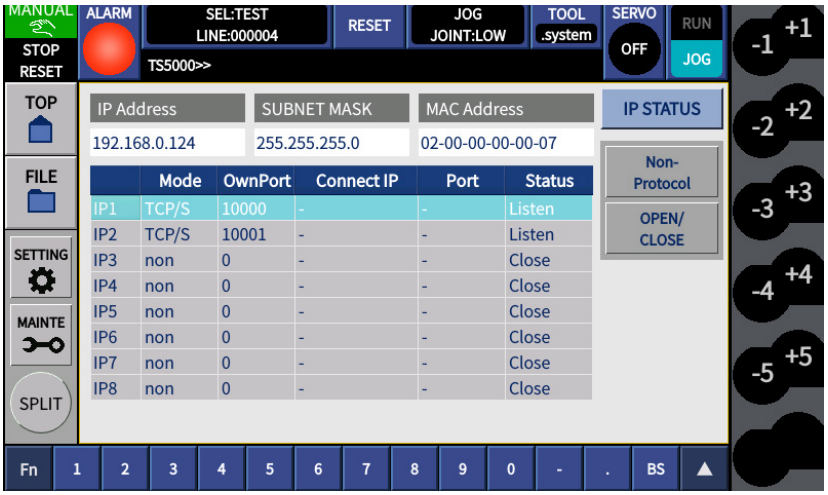
1 Tap the utility menu button.

The utility menu is opened.



2 Tap the "IP STATUS" button in the menu.

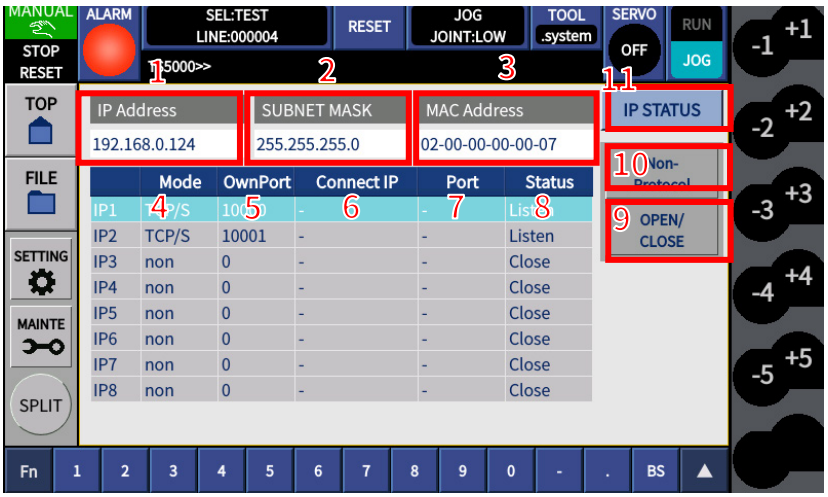
The IP STATUS (non-protocol port) screen is opened.



3 Check that the Ethernet address and subnet mask are set for the controller itself that was set by the parameters in the IP STATUS (non-protocol port) screen.

IP STATUS (Non-protocol Port) Screen

This shows the status of Ethernet communication.



Descriptions of the elements in the screen are shown in the table below.

No. in figure	Description
1	Shows the IP address of the controller.
2	Shows the subnet mask of the controller.
3	Shows the default gateway of the controller.
4	Shows the communication modes of each IP. TCP/S: TCP communication server TCP/C: TCP communication client UDP: UDP communication non: Not used
5	Shows the port numbers of each IP.
6	Shows the connected IP address of each IP.
7	Shows the connection port numbers of each IP.
8	Shows the status of each IP. For the detailed meanings of each status, refer to a book or other reference that explains TCP/IP.
9	Tapping this button switches between open/close for the port where the cursor is located.
10	Tapping this button opens the port switching menu.
11	Tapping this button opens the utility menu.

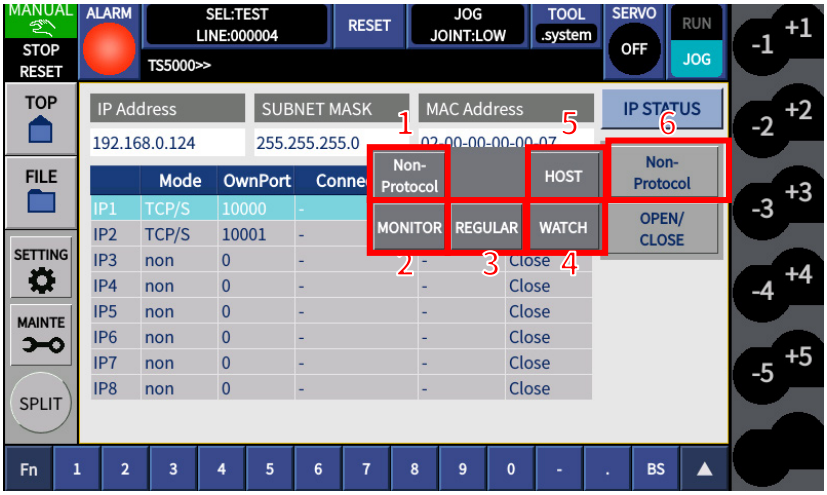
Descriptions of the valid keys in the screen are shown in the table below.

Key name	Action when tapped
↑	Moves the cursor up.
↓	Moves the cursor down.

⚠ CAUTION

MANDATORY	Before closing the port, be sure to check the effect on peripheral devices. Otherwise, a collision could occur between a peripheral device and a robot that performed an unexpected movement.

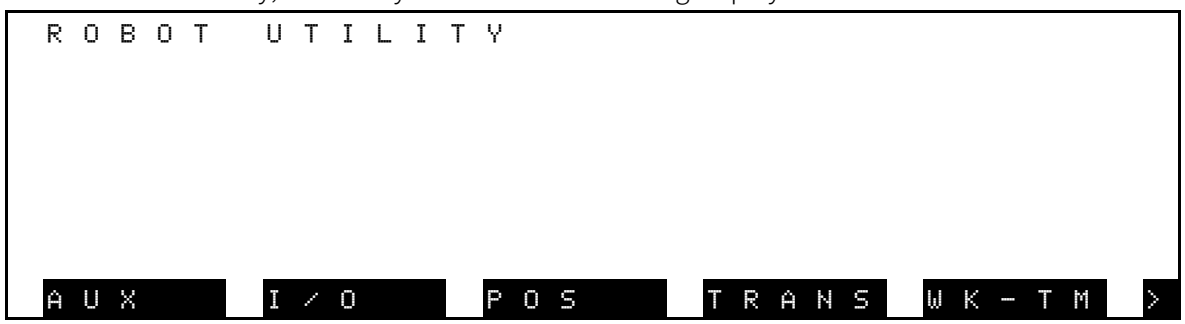
Port Switching Menu



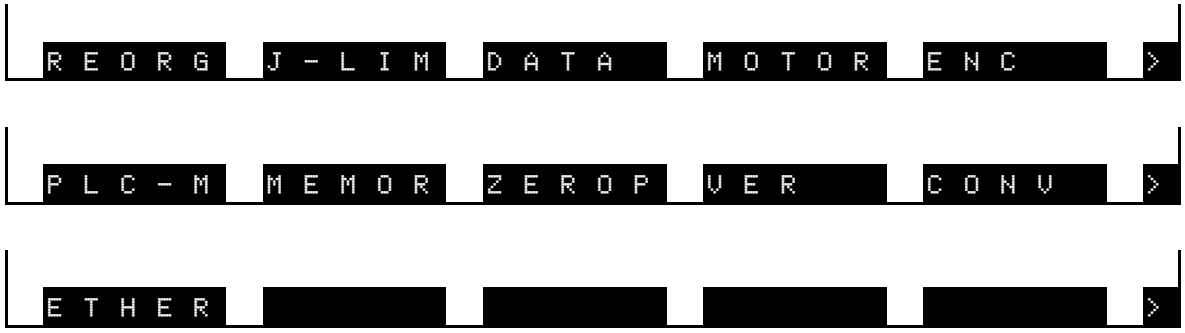
No. in figure	Description
1	Tapping this button switches the displayed port to a non-protocol port.
2	Tapping this button switches the displayed port to a monitor port.
3	Tapping this button switches the displayed port to a periodic monitor port.
4	Tapping this button switches the displayed port to a dump port.
5	Tapping this button switches the displayed port to a control communication port.
6	Tapping this button hides the port switching menu.

<For TP1000>

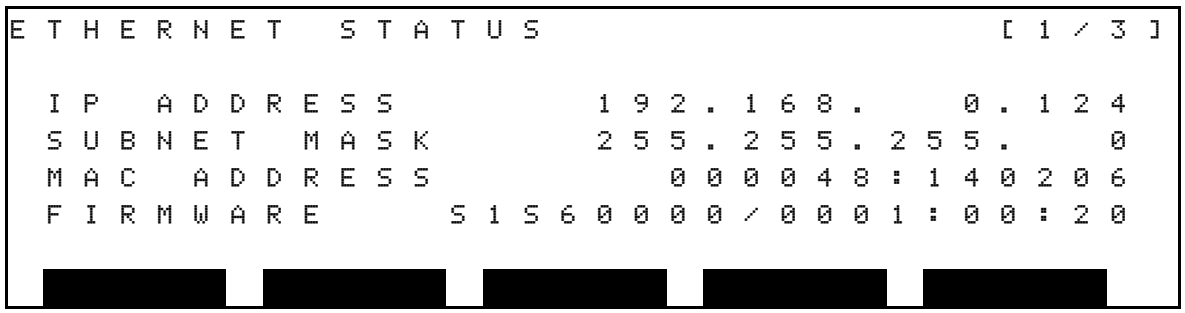
Press the <UTILITY> key, and the system calls the following display.



Press the <NEXT> key three (3) times, and the [ETHER] menu appears.

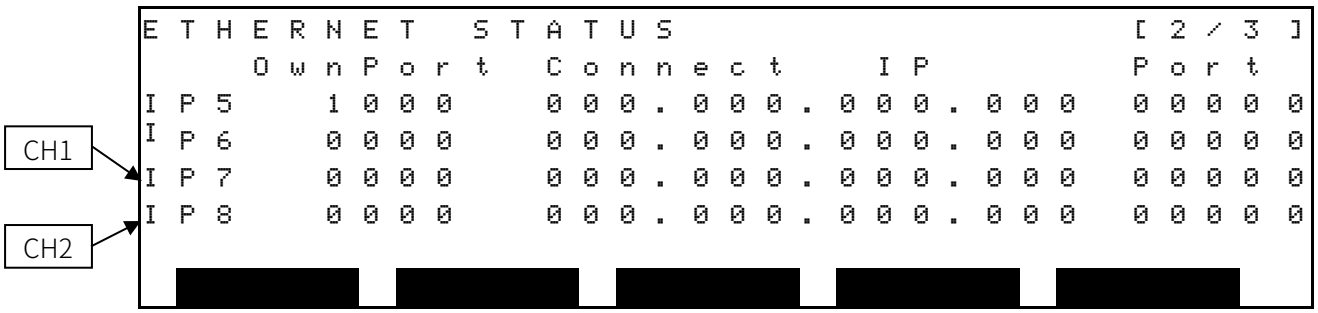


Press the [ETHER] key, and the system calls the following display.



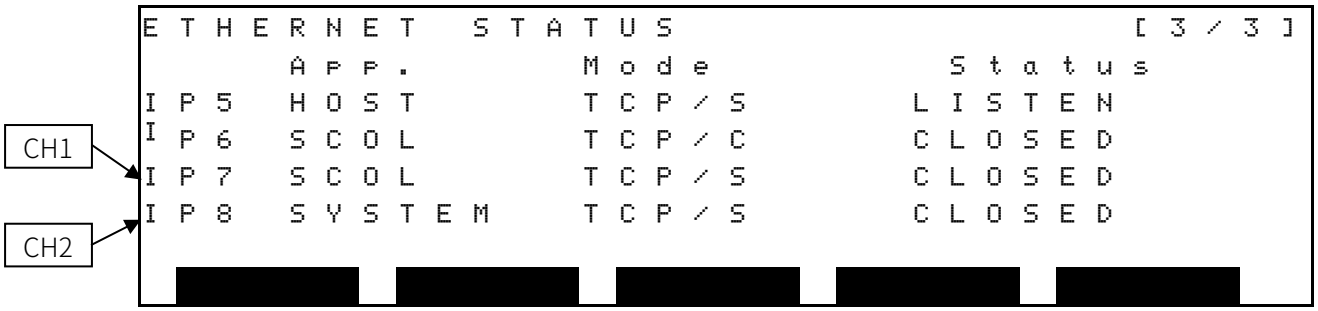
IP ADDRESS: Controller's own Ethernet address as described in Ethernet parameter [E000].
SUBNET MASK: Subnet mask as described in Ethernet parameter [E001].
MAC ADDRESS: MAC address.
FIRMWARE: Firmware version of Ethernet unit.

Press the <NEXT> key, and the following display appears. Make sure that the controller's own Ethernet address and subnet mask as specified by the parameters beforehand are set.



IP1 to IP8: Ethernet channel number.
OwnPort: Port number on the robot side.
Connect IP: IP address of the destination.
Port: Port number of the destination.

On this display, status information on connection when the controller is connected with external equipment is shown.



IP1 to IP8: Ethernet channel number.
App.: Application name using each channel.
MODE: Type of protocol, open status.
Status: Change in Ethernet port status.

・App.(The application name using each port is displayed.)
HOST: Connection pursuant to the simplified protocol.
SCOL: Connection pursuant to the non-procedural protocol.
VCV-1CH: Used for the "vision sensor + conveyor" synchronization function for CH1.[1)]
VCV-2CH: Used for the "vision sensor + conveyor" synchronization function for CH2.[1)]
1) [C000] Dependent on the setting of the type of conveyor synchronization.

・MODE(The open mode of each port is displayed.)
TCP/S: Server connection as per the TCP protocol.
TCP/C: Client connection as per the TCP protocol.

・STATUS（Change in Ethernet status is displayed.）

For details on each status, refer to TCP/IP manuals, etc., commercially available.

1: Closed
2: Listen
3: Syn-Sent
4: Syn-Received
5: Established
6: Fin-Wait1
7: Fin-Wait2
8: Close wait
9: Closing
10: Last ACK
11: Time Wait

3.3 CONVEYOR PAR

[C000] CONVEYOR PARAMETER

The conveyor parameters (CONVEYOR.PAR: [C000] Conveyor Parameter) are exclusively used for the conveyor synchronization function. The items marked with a check mark in the table below indicate parameters that must be set in order to use the Vision Sensor + Conveyor Synchronization function.

Check	No.	Data	Description	Initial value (CH1/CH2)
✓	1	Encoder rotating direction	Encoder rotating direction	0 0
	2	Encoder sampling period [msec]	Encoder sampling period	3 3
	3	Conveyor stop judgment time [msec]	Time for judging conveyor stop.	0 0
✓	4	Type of conveyor synchronization	Type of conveyor synchronization (0 : Linear 1 : Circular (CW) 2 : Circular (CCW) 3 : Vision + Conveyor)	0 0
	5-7	Dummy data	Dummy data	0 0
✓	8	"Travel value per pulse of encoder[um/pulse]	Travel distance per one (1) pulse of encoder	0.0 0.0
	9	Rotation center position X of arc conveyor [mm]	X coordinate of arc conveyor rotation center position	0.0 0.0
	10	Rotation center position Y of arc conveyor [mm]	Y coordinate of arc conveyor rotation center position	0.0 0.0
	11	Limitation width of synchronization(inner)	Conveyor synchronization limit area (inner circumference) (mm)	0.0 0.0
	12	Limitation width of synchronization(outer)	Conveyor synchronization limit area (outer circumference) (mm)	0.0 0.0
	13-15	Dummy data	Dummy data	0.0 0.0

Encoder rotating direction

Encoder rotating direction

The value of this parameter is decided according to the encoder specifications used and the rotating direction of the encoder set to the conveyor.

Data format	Integer type	
Data format	2	
Data setting range	0 or 1	
Description	=0	When the encoder rotation direction is positive [1)]
	=1	When the encoder rotation direction is negative [1)]
Default	=0 , 0	

Vision + Conveyor function setting example

```
'{Encoder rotating direction}
CONV-DIRECTION = 0, 0
```

1) Check the rotation direction using TP5000 or TP1000.

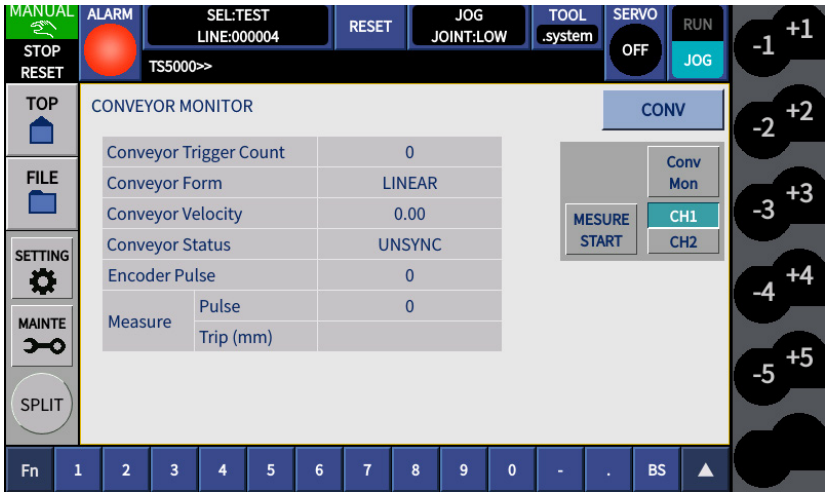
<For TP5000>

1 Tap the utility menu button in the main window.

The utility menu switching window is opened.

2 Tap the "CONV" button in the menu.

The conveyor monitor screen is opened.



Run the conveyor, check the change in the "Encoder pulse value", and set this parameter.

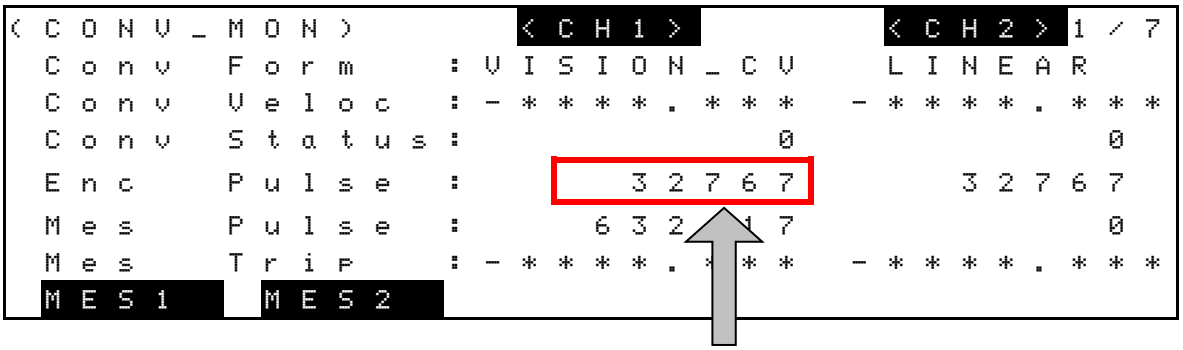
MEMO

・Encoder rotation direction setting value: When the "Encoder pulse value" increases→ Specify zero (0).

・Encoder rotation direction setting value: When the "Encoder pulse value" decreases→ Specify one (1).

<For TP1000>

Press the <UTILITY> key, then select the [CONV] menu. The following menu display appears.



Run the conveyor and set the parameter checking whether the value of [Enc Pulse] increases or decreases.

MEMO

・When the value of [Enc Pulse] increases while "0" is specified for the encoder rotating direction → Specify zero (0).

・When the value of [Enc Pulse] decreases while "0" is specified for the encoder rotating direction → Specify one (1).

Encoder sampling period

This parameter specifies the period[1)] during which the counter value from the encoder attached to the conveyor is obtained.

1) As the arithmetic operation period of the conveyor synchronization function is 3 ms, the actual parameter value is a multiple of this arithmetic operation period (3 ms) whose fractions are ignored.

Data format	Integer type
Data number	2
Data setting range	3 to 30000
Data unit	ms
Default	=3 , 3

Vision + Conveyor function setting example

```
'{Encoder sampling period [ms]}
CONV-SMP-PERIOD = 3, 3
```

MEMO

The conveyor synchronization function calculates the synchronization travel distance at minimum periods of 3 msec (default). It is also possible to specify a larger value, however. Determine the value, taking note of the following.

■When the value is decreased (minimum value 3 msec):

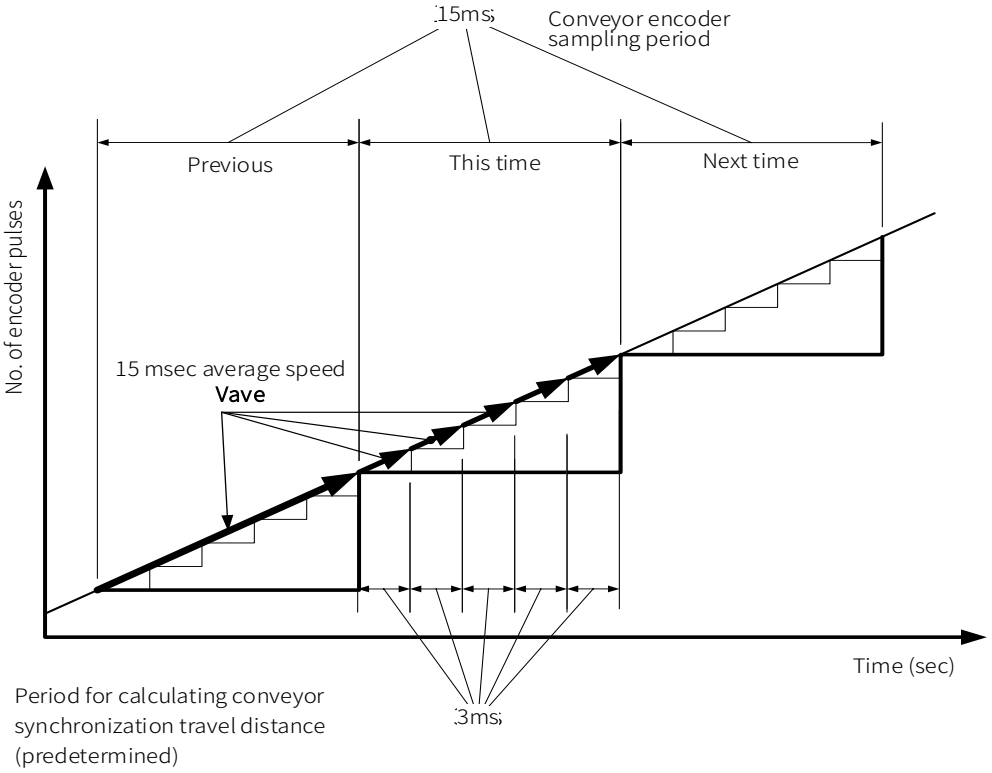
- *The trackability for the conveyor speed improves.*
- *If this value is too small, however, the synchronization speed changes even with conveyor pulsation, which may result in vibration of the robot.*

■When the value is increased:

- *The synchronization speed is less affected by conveyor pulsation.*
- *The conveyor rotating angle and arithmetic operation error can be absorbed.*
- *The trackability for the conveyor speed deteriorates.*
- *Unless the pulsation is caused, specify the default value (3 msec).*

Example:

When the conveyor sampling period is set to 15 msec, the synchronization travel distance is figured out based on the previous conveyor sampling period average speed (Vave).



Conveyor stop judgment time

This parameter specifies the conveyor stop judgment time. Unless the encoder counter value changes during the time set by this parameter value, the system interprets that the conveyor is stopped, and outputs the error of "001–048: Conveyor Stop". As this is the error of level 1, programmed operation continues.

Data format	Integer type
Data number	2
Data setting range	0 to 30000
Data unit	ms
Default	=0 , 0

Vision + Conveyor function setting example

```
'{Conveyor stop judgment time [ms]}
CONV-TIMEOUT = 0, 0
```

MEMO

・Normally, the conveyor stop time is specified by the 2nd argument of the SYNC command of the SCOL2 language. Unless the 2nd argument is specified, the conveyor stop is judged with this parameter value.

・When the 2nd argument of the SYNC command is not specified and this parameter value is zero (0), the system will not judge the conveyor stop.

Type of conveyor synchronization

This parameter specifies the conveyor stop judgment time. Unless the encoder counter value changes during the time set by this parameter value, the system interprets that the conveyor is stopped, and outputs the error of "001–048: Conveyor Stop". As this is the error of level 1, programmed operation continues.

Data format	Integer type	
Data number	2	
Data setting range	0 to 3	
Description	=0	Linear conveyor (Can be set for CH1 and CH2)
	=1	(Reserve)
	=2	(Reserve)
	=3	"Vision+ conveyor" synchronization function (Can be set for CH1 and CH2.)
Default	=0 , 0	

Vision + Conveyor function setting example

```
'{Type of conveyor synchronization}
CONV-FORM = 3, 0
```

MEMO

・If this parameter is not set, "001-0147: COM Buffer Over" will occur.

Dummy data

Reserved area

Area reserved for the system.

Data format	Integer type
Data number	2
Default	=0 , 0

Default setting

```
'{Dummy data}
CONV-ERR-COMP = 0, 0
'{Dummy data}
CONV-ERR-COMP-VELGAIN = 0, 0
'{Dummy data}
CONV-ERR-COMP-ACCGAIN = 0, 0
```

MEMO

Do not change this setting.

Travel value per pulse of encoder [um/pulse] [mdeg/pulse]

Travel value per pulse of encoder

This parameter specifies the conveyor travel distance[1] (μm) per one (1) pulse of the encoder set to the conveyor.

Data format	Real number type
Data number	2
Data unit	μm
Default	=0 , 0

Vision + Conveyor function setting example

```
'{Travel value per pulse of encoder [um/pulse] [mdeg/pulse]}
CONV-ENC-GEAR = 12.000000, 0.000000
```

1) the conveyor travel distance (μm) per one (1) pulse of the encoder
If the total number of encoder pulses is ΔP (pulse) when the conveyor moves by ΔL (mm), the conveyor travel distance per one (1) pulse of the encoder α (μm) can be calculated from the following equation.

$$\alpha = (\Delta L / \Delta P) * 1000 \quad [1]$$

The total number of encoder pulses (ΔP) can be obtained in the following manner using TP5000 or TP1000.

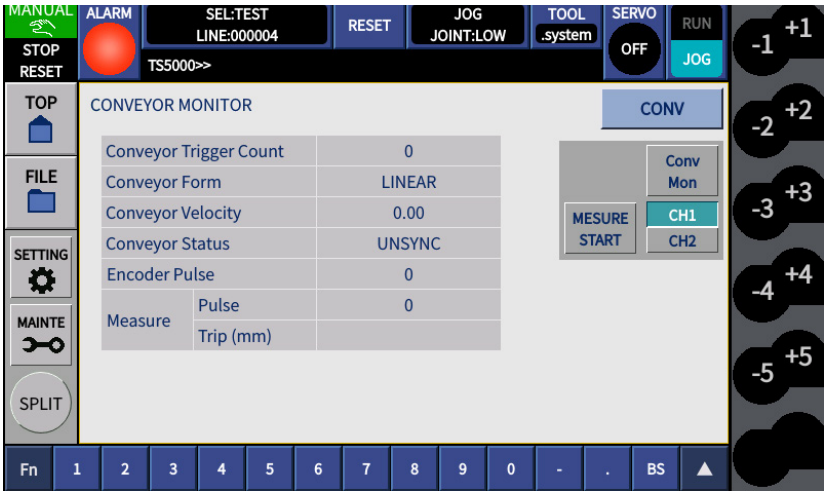
<For TP5000>

1 Tap the utility menu button in the main window.

The utility menu switching window is opened.

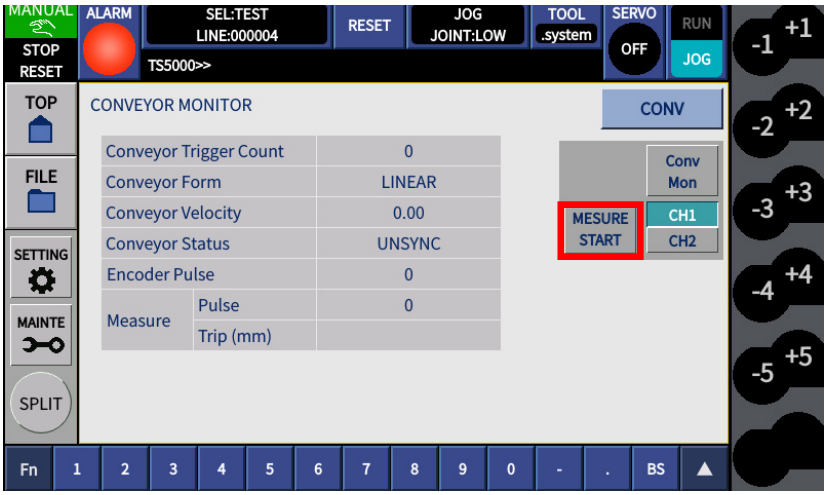
2 Tap the "CONV" button in the menu.

The conveyor monitor screen is opened.



3 Tap the channel to be measured.

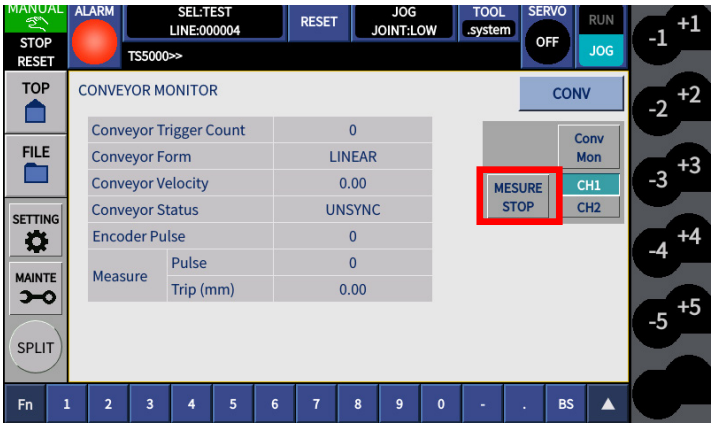
4 Tap the "Start Measurement" button.



5 Move the conveyor by an amount of ΔL only.

6 Tap "End Measurement".

A numerical value is displayed for "Pulses" in "Measurement". This value is ΔP.



7 Substitute into formula [1] to find the travel α (µm) per pulse.

<For TP1000>

1 Press the <MES1> (F1) or <MES2> (F2) function key corresponding to the channel (CH1 or CH2) through which you wish to measure, using the cursor.

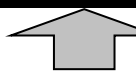
```
( CONV_MON )          < CH1 >          < CH2 > 1 / 7
  Conv  Form    : VISION_CV      LINEAR
  Conv  Veloc   : -****.***   -****.***
  Conv  Status  :          0            0
  Enc   Pulse   :      32767        32767
  Mes   Pulse   :     632417            0
  Mes   Trip    : -****.***   -****.***
  MES1    MES2
```

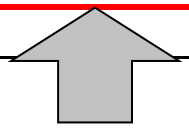


To count the number of pulses of CH1, press [MES1].

2 To count the number of pulses of CH1 or CH2, select [MES1] or [MES2], respectively. After moving the conveyor by ΔL, press the <SET> key.

```
( CONV_MON )          < CH1 >          < CH2 > 1 / 7
  Conv  Form    : VISION_CV      LINEAR
  Conv  Veloc   : -****.***   -****.***
  Conv  Status  :          0            0
  Enc   Pulse   :      32767        32767
  Mes   Pulse   :     632417            0
  Mes   Trip    : -****.***   -****.***
  MES1    MES2
```





This value is ΔP.

3 When you press the <SET> key, the present total number of pulses (Mes Pulse) is shown on the display. Then figure out the travel distance per one (1) pulse (α μm) by substituting this value for "ΔP" in the equation [1]. (To measure the value again, press the [MES1] or [MES2] key.)

Rotation center position X of arc conveyor [mm][1)]
Rotation center position Y of arc conveyor [mm] [1)]

Rotation center position X and Y coordinates of arc conveyor

Set these parameters when using arc conveyor[1)].

1) TS5000 is not supported.

Default setting

```
'{Rotation center position X of arc conveyor [mm]}
CONV-CIRC-XCENT = 0.000000, 0.000000
'{Rotation center position Y of arc conveyor [mm]}
CONV-CIRC-YCENT = 0.000000, 0.000000
```

MEMO

・Do not change this setting.

Limitation width of synchronization(inner)
Limitation width of synchronization(outer)

Conveyor synchronization limit area

This parameter indicates the distance from the inner circumference and outer circumference operation limit circles of the robot.

Data format	Real number type
Data number	2
Data unit	mm
Default	=0 , 0

Vision + Conveyor function setting example

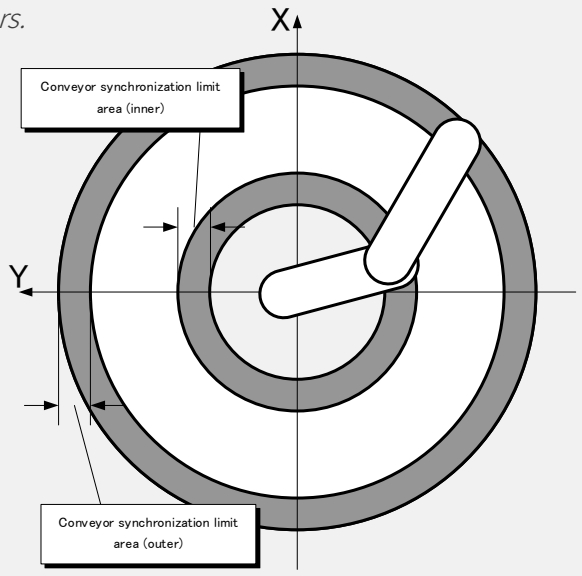
```
'{Limitation width of synchronization(inner)}
CONV-SLIM-WIDTH[0] = 50.000000, 0.000000
'{Limitation width of synchronization(outer)}
CONV-SLIM-WIDTH[1] = 600.000000, 0.000000
```

MEMO

・When the robot enters this "Conveyor synchronization limit area" during conveyor synchronization, the error "004-051: Conveyor sync limit area" occurs, and the robot decelerates to a stop. Conveyor synchronization is also stopped when this occurs.

・Limitation width of synchronization(inner)
→ Distance from robot inner circumference operation limit circle.

・Limitation width of synchronization(outer)
→ Distance from robot outer circumference operation limit circle.



Dummy data

Reserved area

Area reserved for the system.

Data format	Real number type
Data number	2
Default	=0.000000, 0.000000

Default setting

```
'{Dummy data}
CONV-DUMY-F[0] = 0.000000, 0.000000
'{Dummy data}
CONV-DUMY-F[1] = 0.000000, 0.000000
'{Dummy data}
CONV-DUMY-F[2] = 0.000000, 0.000000
```

MEMO

・Do not change this setting.

[C001] VISION + CONVEYOR TRACKING PARAMETER (CH1, CH2)

These parameters are exclusively used for the "vision sensor + conveyor" synchronization function. This function is operative only when [3: "Vision sensor + conveyor" synchronization function] is specified for the type of conveyor synchronization of conveyor parameter [C001] described above. The items marked with a check mark in the table below indicate parameters that must be set in order to use the Vision Sensor + Conveyor Synchronization function.

Table 3. 7 CONVEYOR PARAMETER 2 (CH1,CH2)

Check	No.	Data	Description	Initial value (CH1/CH2)
	1	Delimiter Character	Delimiter of communication data 0 : Comma	0 0
	2	Work sort function enable SW	Work data sorting function enable SW	0 0
	3	Virtual vision conveyor tracking enable SW[1)]	Dummy data (integer)	0 0
	4	Data reception timeout [ms]	Capture data receive timeout [ms]	0.0 0.0
✓	5	Calibration offset [mm]	Calibration offset [mm]	0.0 0.0
✓	6	Detection area start point [mm]	Start point of workpiece detection area [mm]	0.0 0.0
✓	7	Detection area end point [mm]	End point of workpiece detection area [mm]	0.0 0.0
✓	8	Conveyor Sync limit point [mm]	Limit point of conveyor synchronization [mm]	0.0 0.0
✓	9	Overlap allowance [mm]	Allowance for workpiece overlap [mm]	0.0 0.0
✓	10	Capture width [mm]	Capture width [mm]	0.0 0.0
✓	11	Calibration reference position (X)	Calibration reference point X coordinate [mm]	0.0 0.0
✓	12	Calibration reference position (Y)	Calibration reference point Y coordinate [mm]	0.0 0.0
	13	Minimum Trigger Interval [msec]	Minimum trigger detection time [ms]	0.0 0.0

1) Not supported in TS5000.

Delimiter Character

Delimiter character for communication data

Specifies the delimiter of communication data.

Data format	Real number type	
Data number	2	
Description	=0	, (Comma)
	=1	(space)
Default	=0, 0	

Default setting

'{Delimiter Character (0:' , ' 1:' ')}
VCV-DLMT-CHAR = 0, 0

MEMO

・As only comma is used as the delimiter of communication data, this parameter is not used.

Work sort function enable SW

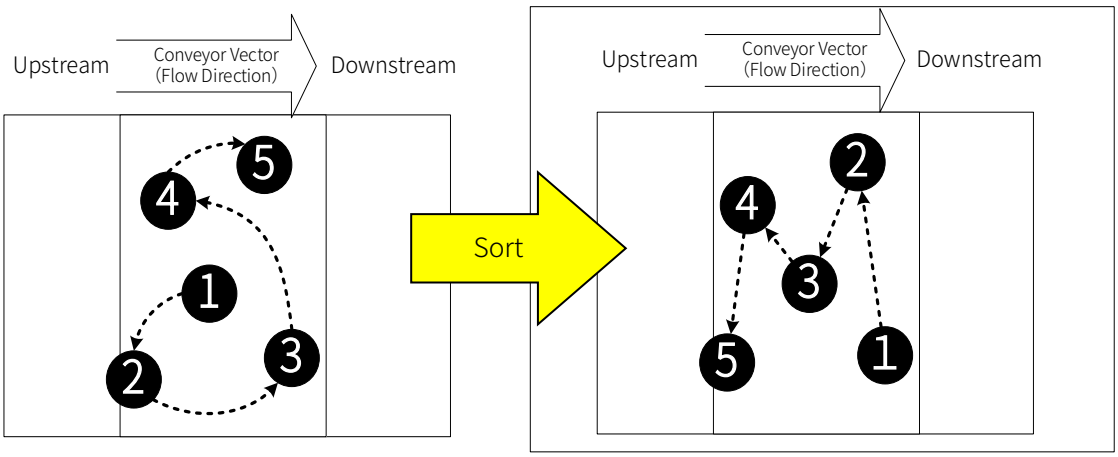
Data sorting function

"Workpiece data sorting function" is a function to rearrange the order of the received workpiece data from downstream to upstream of the conveyor. This is enabled by setting to "1". This enables workpieces to be processed in order from the downstream, and so the optimal picking operation with minimum waste can be performed.

Data format	Integer type	
Data number	2	
Description	=0	Disable
	=1	Enable
Default	=0, 0	

Setting example

```
'{Work sort function enable SW (0:Disable, 1:Enable)}
VCV-WSORT-SW = 1, 0
```



Virtual vision conveyor tracking enable SW1)

Virtual vision conveyor tracking function[1)]

1) TS5000 is not supported.

Data format	Integer type	
Data number	2	
Description	=0	Disable
	=1	Enable
Default	=0, 0	

Default setting

'{Virtual vision conveyor tracking enable SW (0: Disable, 1: Enable)}
VCV-VRVCVENABLE = 0, 0

MEMO

・Do not change this setting.

Data reception timeout

Capture data reception timeout

If data reception is not completed within the pre-set time, the error "1-180: VCV Comm Process Timeout" occurs.

Data format	Real number type
Data number	2
Data unit	ms
Default	= 0.000000, 0.000000

Default setting

'{Data reception timeout [ms]}
VCV-RCV-TIMEOUT = 0.000000, 0.000000

MEMO

・This function is disabled when the conveyor parameter [C001] {Data reception timeout[ms]} of this function is set to "0.000000".

Calibration offset

Calibration offset

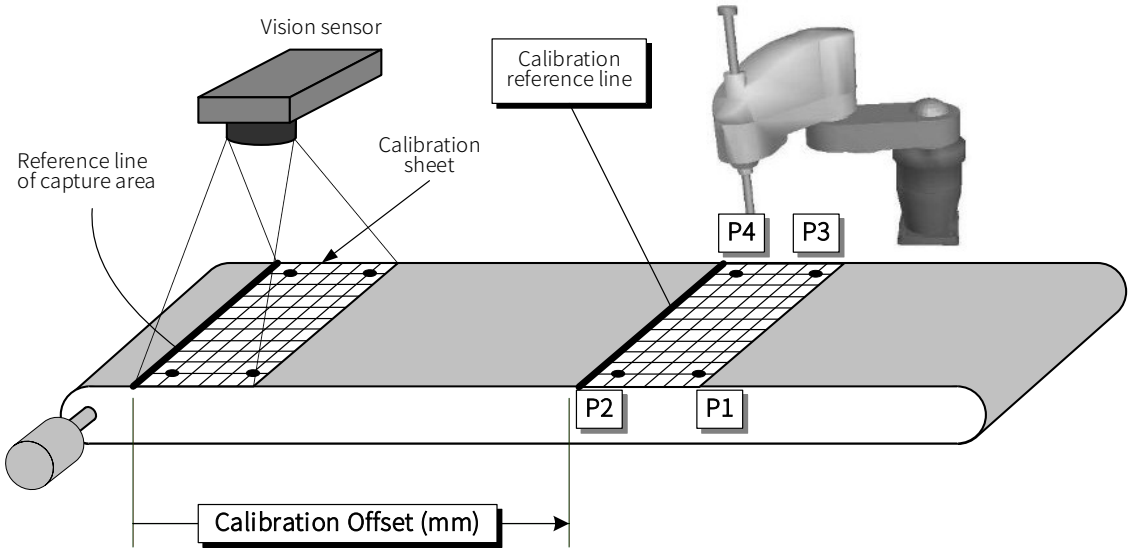
Calibration Offset is the parameter for specifying the length from the capture area reference line to calibration reference line in mm.

Data format	Real number type
Data number	2
Data unit	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Calibration offset [mm]}
VCV-CALIB-OFST = 425.000000, 0.000000
```

1) "Calibration" is the work for transforming the vision sensor coordinate system to the robot's counterpart. Once the calibration is executed, using the vision sensor, all coordinates sent from the vision sensor are based on the world coordinate system of the robot.



MEMO

・It can also be set by directly editing parameter CONVEYOR.PAR. Using the UTILITY function, however, makes parameter setting easy. For details, see Para. 6.1.

<Outline of calibration work>

1 Capture the calibration sheet by the vision sensor.

2 Register four (4) calibration points (P1 to P4) on the calibration sheet to the vision sensor.

3 Move the conveyor to a position where the robot can reach the calibration sheet.
⇒ This position is called the "calibration position".

4 Move the end effector of robot to each calibration point (P1 to P4) in order to measure the point (P1 to P4) in the world coordinate system.

5 Register these four (4) calibration points (P1 to P4) taught to the vision sensor.

MEMO

・For details on the calibration procedures, see Section 4.3.

Detection area start point
Detection area end point
Conveyor Sync limit point

Start point of workpiece detection area

Specify the distance from the capture reference point to the start point of workpiece detection area in mm.

End point of workpiece detection area

Specify the distance from the capture reference point to the end point of workpiece detection area in mm.

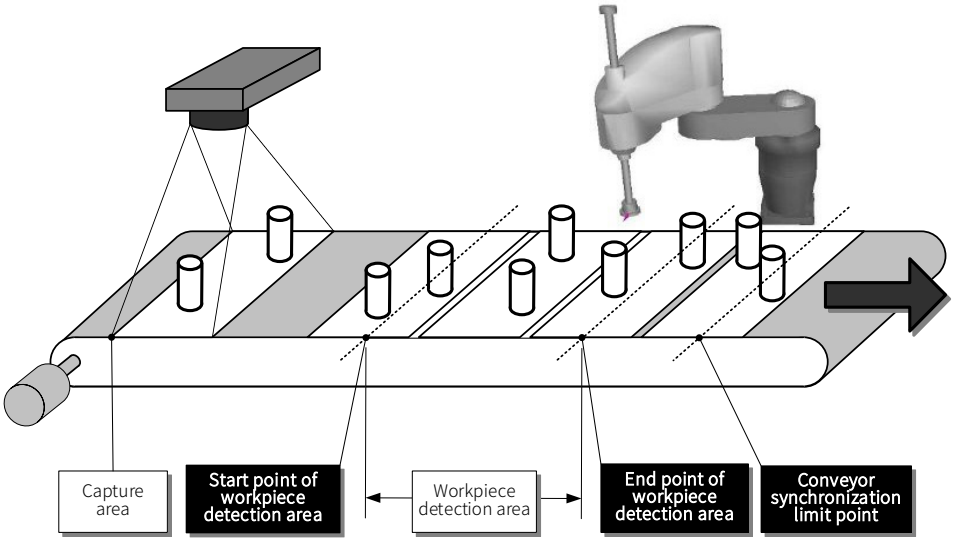
Conveyor synchronization limit point

Specify the distance from the capture reference point to the conveyor synchronization limit point in mm.

Data format	Real number type
Data number	2
Data unit	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Detection area start position [mm]}
VCV-DETECT-STR = 425.000000, 0.000000
'{Detection area end position [mm]}
VCV-DETECT-END = 650.000000, 0.000000
'{Conveyor Sync limit position [mm]}
VCV-SYNC-LIMIT = 850.000000, 0.000000
```



♦ About the workpiece detection area:

When workpieces running on the conveyor have entered the workpiece detection area, the robot starts tracking the conveyor.

At this moment the robot starts to move to the position of a workpiece which is located most downstream in the workpiece detection area.

Unless a workpiece exists in the workpiece detection area, the robot will not start synchronizing with the conveyor. The robot waits for the next workpiece entering the workpiece area.

♦ About conveyor synchronization limit point:

When the robot apex position has passed the conveyor synchronization limit point after the robot started synchronizing with the conveyor, it finishes the synchronization.

NOTICE	
CAUTION/ WARNING	・If the robot's speed is low, the robot gets out of the conveyor synchronization limit and collides with the workpieces. Be careful when the speed of the robot is low.

Overlap allowance

This function ignores the workpiece detected this time if the distance "Δw" between the workpiece detected previously and the workpiece detected this time is the setting value (L) or less.

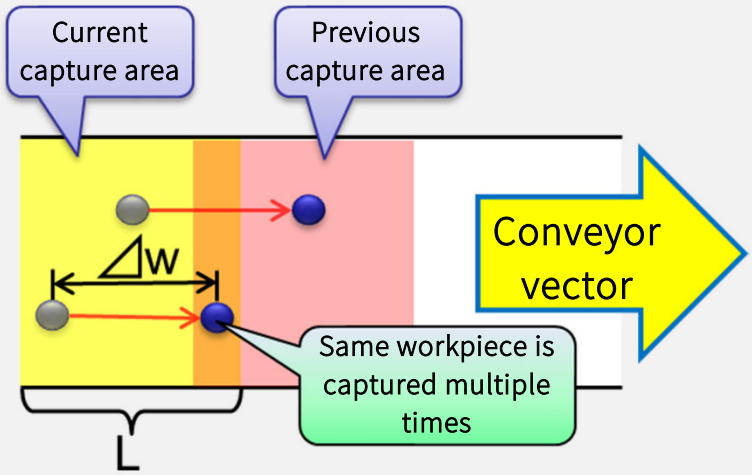
Data format	Real number type
Data number	2
Data unit	mm
Default	= 0.000000, 0.000000

Setting example

'{Overlap allowance [mm]}
VCV-OVERLAP-ALW = 30.000000, 0.000000

MEMO

・As shown in this figure, this is used when the same workpiece is inadvertently captured multiple times.



Capture width

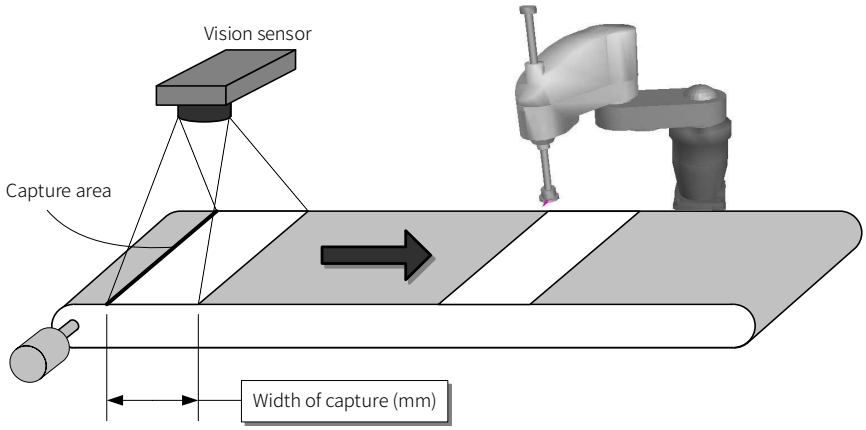
Capture width

Specify the width of an image captured by the vision sensor in millimeters.

Data format	Real number type
Data number	2
Data unit	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Capture width [mm]}
VCV-CAP-WIDTH = 170.000000, 0.000000
```



Calibration reference position (X)
Calibration reference position (Y)

Calibration reference position (X coordinate)
Calibration reference position (Y coordinate)

The calibration reference position (X, Y) is any position in the world coordinate system, which is located on the calibration reference line. By registering this reference position, it is possible to make the robot controller recognize the relationship between the capture area reference line of vision sensor and the calibration position.

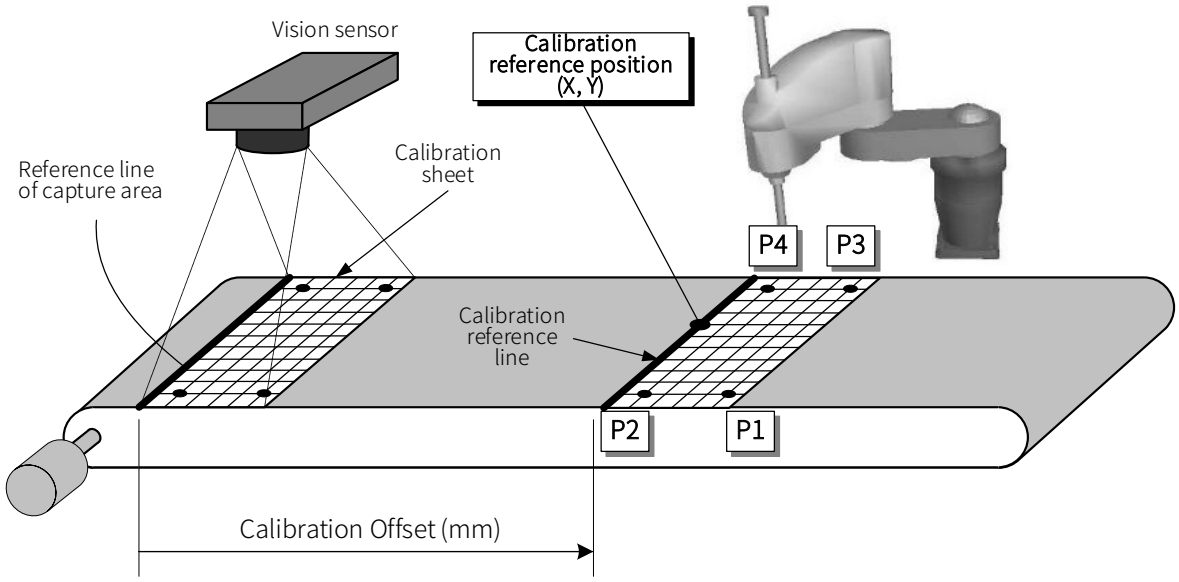
Data format	Real number type
Data number	2
Data unit	mm
Default	= 0.000000, 0.000000

Setting example

'{Calibration reference point (X)}
VCV-CAP-ORG-X = 450.000000, 0.000000
'{Calibration reference point (Y)}
VCV-CAP-ORG-Y = -290.000000, 0.000000

MEMO

・For the calibration reference position (X, Y), specify the position (X, Y) in mm in the world coordinate system of the robot. (For the setting procedures, see Para. 4.3.)



Minimum Trigger Interval

Minimum trigger interval

Data format	Real number type
Data number	2
Data unit	ms
Default	= 0.000000, 0.000000

Setting example

```
'{Minimum Trigger Interval [ms]}
VCV-MINTRIGINTERVAL = 0.000000, 0.000000
```

[C002] CONVEYOR TRACKING DOUT FUNCTION

This parameter is the "Conveyor synchronization pulse output function" setting parameter. It is enabled when one of the conveyor synchronization types is set in "Type of conveyor synchronization" for the conveyor parameter [C000] described above.
This is used when the trigger signal is wired as shown in the figure below.

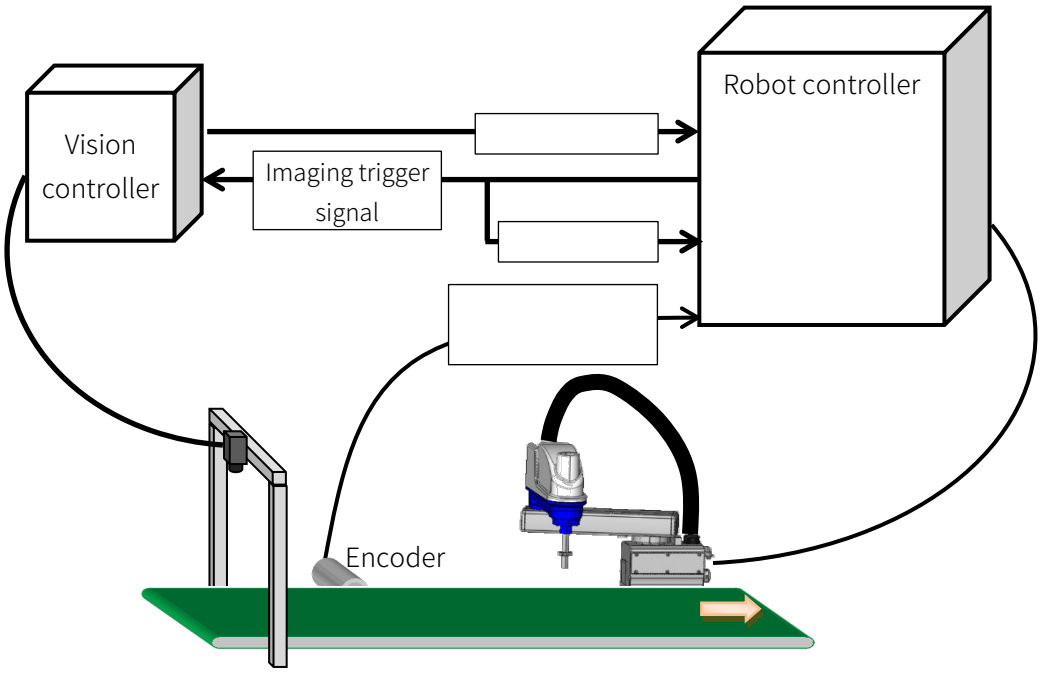


Table 3.8 CONVEYOR TRACKING DOUT FUNCTION

No.	Data	Description	Initial value (CH1/CH2)
1	Output signal number	Output signal number	0 0
2	Width of signal on [mm]	Signal ON width	0.0 0.0
3	Width of one cycle [mm]	Width of one cycle	0.0 0.0

Output signal number

This sets the signal that outputs pulses as imaging trigger signals. Set to any general-purpose output signal number.

Data format	Integer type
Data number	2
Data setting range	0～25
Default	=0, 0

Setting example

```
'{Output signal number}
CONV-CYCLE-IONO = 1, 0
```

MEMO

・When using DOUT9 to 25, use an extended IO board.

・The function is disabled when this parameter is set to 0.

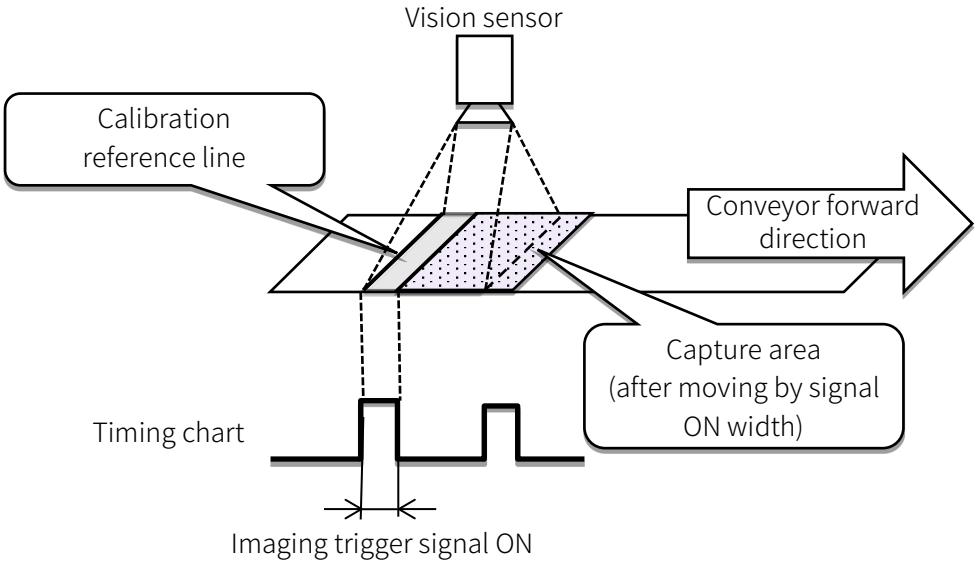
Width of signal on

This sets the conveyor travel distance where the imaging trigger signal is held at ON.

Data format	Real number type
Data number	2
Default	=0.000000, 0.000000

Setting example

```
'{Width of signal on [mm]}
CONV-CYCLE-ON-WIDTH = 50.000000, 0.000000
```



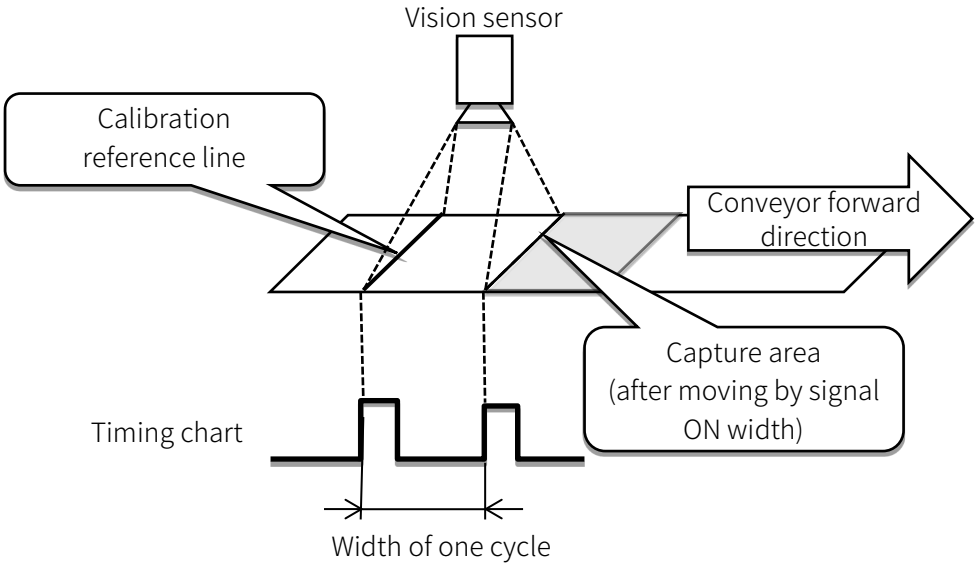
Width of one cycle

This sets the conveyor travel distance during one signal output cycle.

Data format	Real number type
Data number	2
Default	=0.000000, 0.000000

Setting example

```
'{Width of one cycle [mm]}
CONV-CYCLE-WIDTH = 125.000000, 0.000000
```



[C003] CONVEYOR FUNCTION SWITCH

Do not change this setting. TS5000 is not supported.

[C004] CONVEYOR FUNCTION SWITCH2

Do not change this setting. TS5000 is not supported.

[C005] DATA SET GROUP FOR VIRTUAL VISION CONVEYOR TRACKING

Do not change this setting. TS5000 is not supported.

4. Setting Vision Sensor

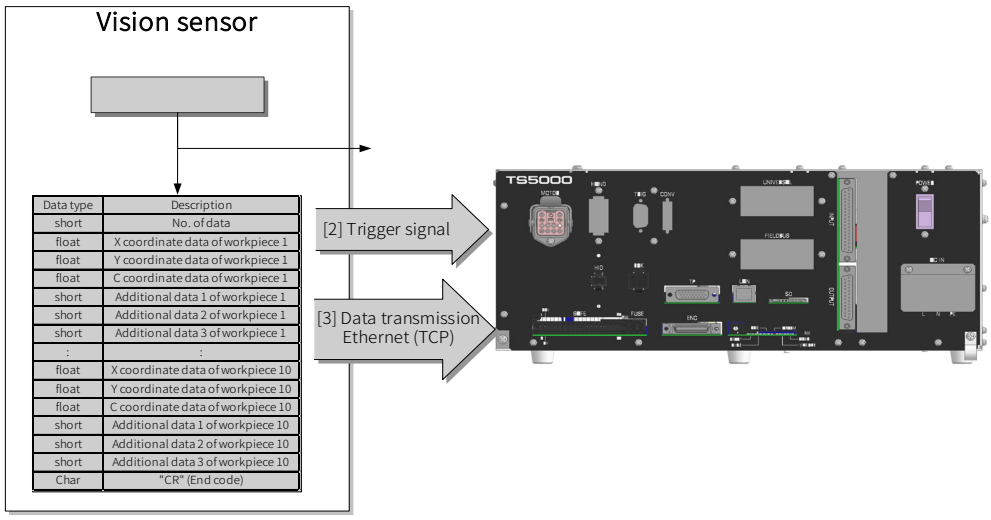
Vision Sensor setup of the Vision Sensor + Conveyor Synchronization function can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

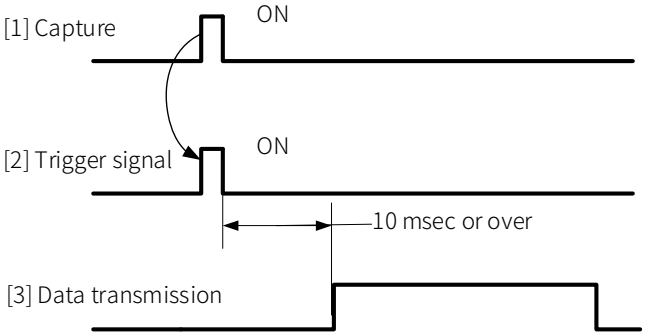
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

4.1 Trigger Signal and Data Transmission Process

Set the vision sensor so that the trigger signal output and data transmission to the robot controller are done, according to the following process.





The vision sensor captures a workpiece.

The trigger signals are output to the robot controller.

(It is ideal that the trigger signal is output with the timing that the vision sensor has obtained coordinate data.)

Coordinate data are sent to the controller via Ethernet.

(**Send the data no shorter than** 10 msec after the trigger signal is sent.)

If the Ethernet communication port (IP7 or IP8) is closed during vision sensor + conveyor synchronization, the error "004-052: VCV Comm port closed " occurs.

4.2 Transmit Data Format

The data format is predetermined as shown below.

Data format: ASCII code

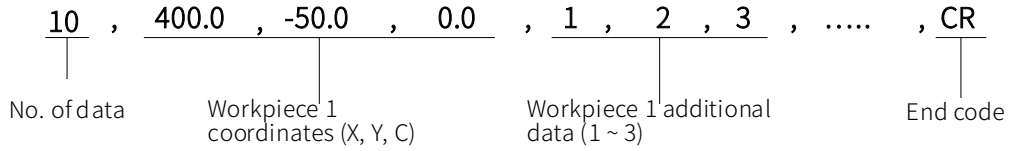
Delimiter: Separation by comma (,)

End code: CR (End code) * Only the end code "CR" is enabled.

Data type	Name	Description	
Integer	Num	No. of data	
Real number	x	X coordinate data of workpiece 1	Workpiece coordinate data (X, Y, C)
Real number	y	Y coordinate data of workpiece 1	
Real number	c	C coordinate data of workpiece 1	
Integer	Opt1	Additional data 1 of workpiece 1	Workpiece additional data (workpiece type, etc.)
Integer	Opt2	Additional data 2 of workpiece 1	
Integer	Opt3	Additional data 3 of workpiece 1	
:	:	:	
Real number	x	X coordinate data of workpiece 10	
Real number	y	Y coordinate data of workpiece 10	
Real number	c	C coordinate data of workpiece 10	
Integer	Opt1	Additional data 1 of workpiece 10	
Integer	Opt2	Additional data 2 of workpiece 10	
Integer	Opt3	Additional data 3 of workpiece 10	
Char	"CR"	"CR" (End code)	

Max. 10 (per each capture)

<Example format>

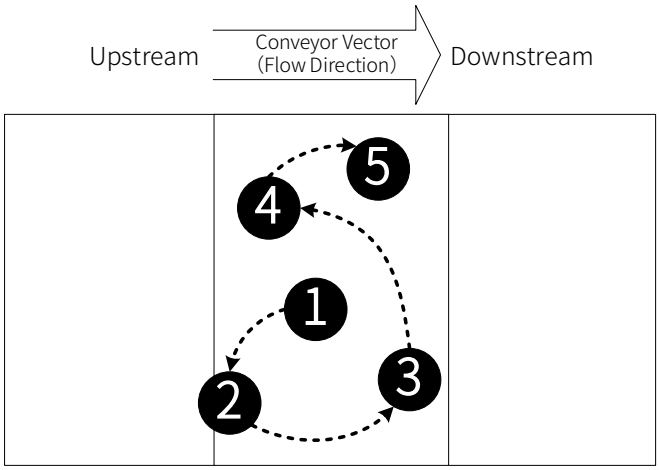


<Number of transmittable workpieces>

10 workpieces of workpiece 1 to workpiece 10 can be sent per one transmission. The error "001-178: VCV Comm data format error" occurs when over 10 data is sent, and the sent data is canceled automatically. Robot program does not stop even if this error occurs.

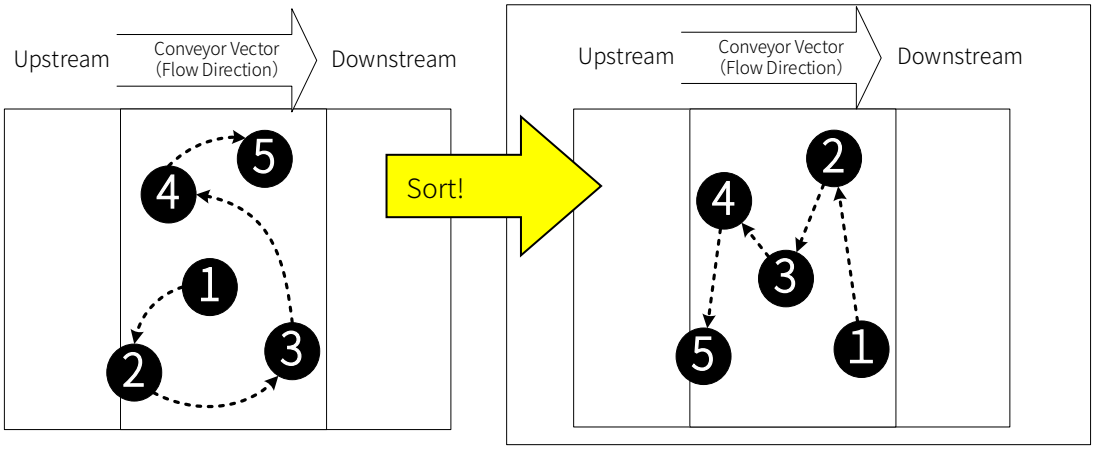
<Work Data sorting function>

As shown in the figure below, the robot controller normally performs indexing in sequential order from the first workpiece.



Enable the workpiece data sorting function to process the workpiece from the downstream of the conveyer sequentially.

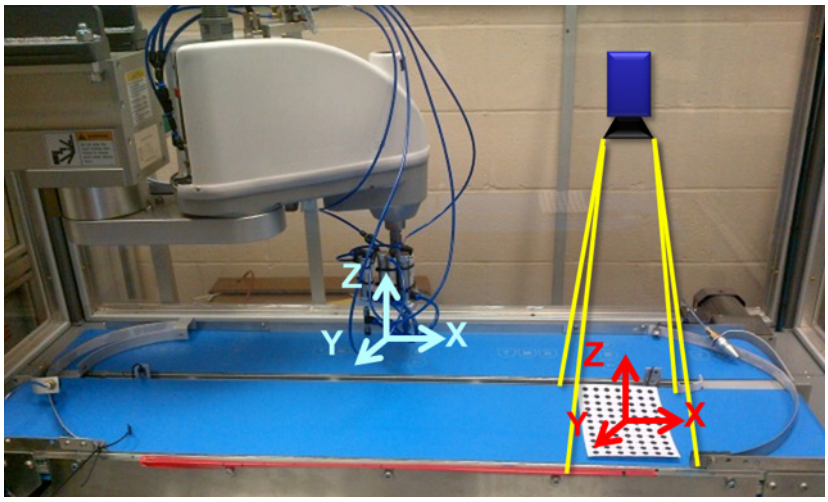
"Workpiece data sorting function" is a function to rearrange the order of the received workpiece data from downstream to upstream of the conveyor. This is enabled by setting to "1". This enables workpieces to be processed in order from the downstream, and so the optimal picking operation with minimum waste can be performed.



Please enable this function referring to the paragraph of "3.3. [C001] VISION + CONVEYOR TRACKING PARAMETER (Work sort function enable SW)".

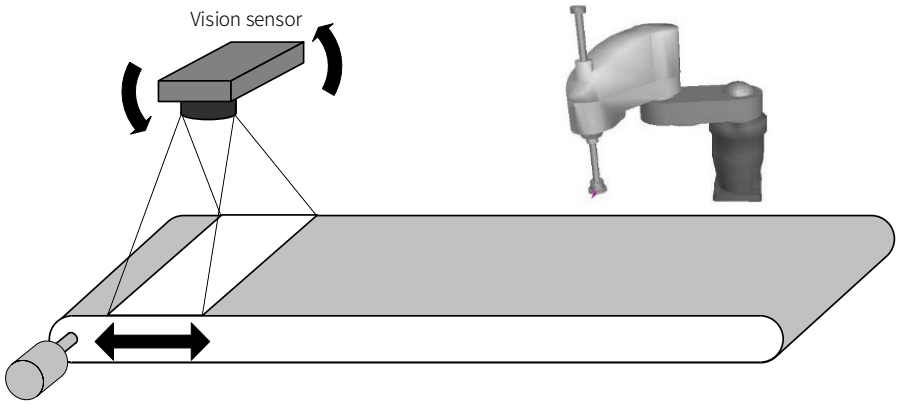
4.3 Calibration Procedures

"Calibration" is the work for registering the robot's coordinate system in the vision sensor. When the calibration is performed using the vision sensor, all coordinates sent from the vision sensor are based on the world coordinate system of the robot.



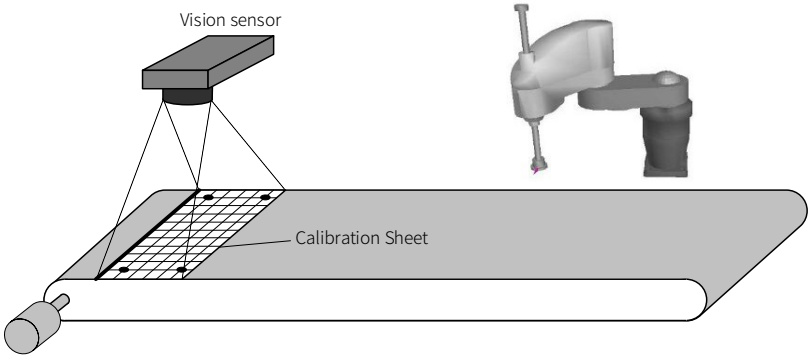
4.3.1 Determining Vision Sensor Capture Position

Determine the capture position of the vision sensor.



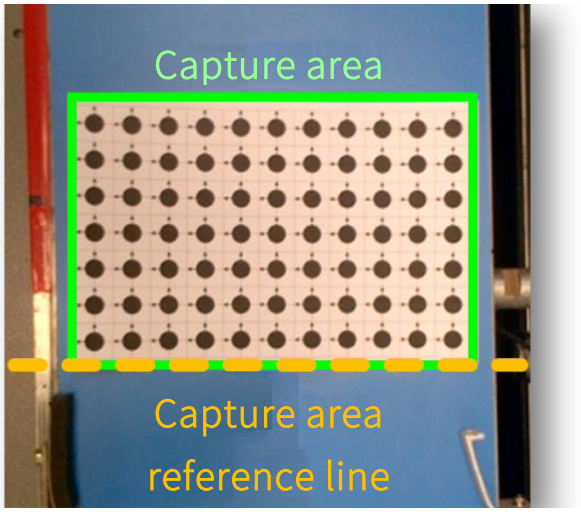
4.3.2 Preparing Calibration Sheet

Cut off the calibration sheet according to the size and width of the conveyor and place it on the capture position. For the calibration sheet, copy and use the sheet in Para. 4.5.



4.3.3 Setting Capture Area

Set the capture area to a desired size on the vision sensor setting display. At this time, fine-adjust the capture area so that it can align with the reference lines of the calibration sheet.

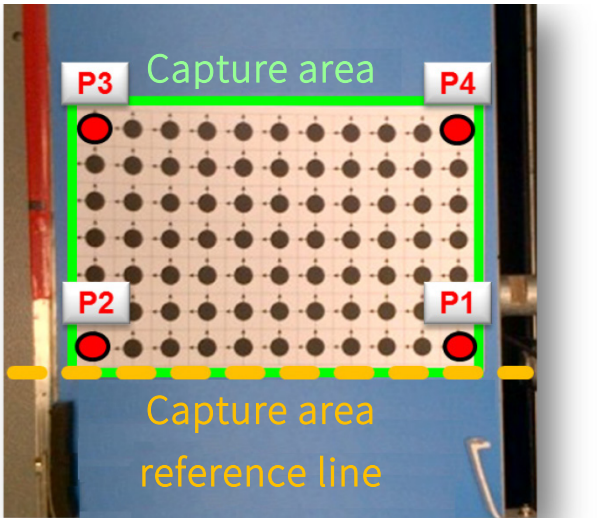


MEMO

・The capture area of the vision sensor should be set to an appropriate width based on the workpiece size, conveyor speed, vision sensor specifications and performance. (If many workpieces have been captured at once, using the vision sensor with low processing ability, the analysis time of the vision sensor becomes long and data transmission to the robot controller may be delayed.)

4.3.4 Register of Vision Sensor Calibration Points

Select and determine four (4) reference points (P1 to P4) from a group of points on the calibration sheet, then register them in the vision sensor.



MEMO

・The information in 4.3.1 to 4.3.4 varies depending on the vision sensor manufacturer. Set based on the calibration procedure for the vision sensor that you are using.

4.3.5 Register of Calibration Offset (mm)

Register the calibration offset (mm) of conveyor parameter "[C01] Vision + Conveyor Tracking Parameter" in the following manner.

> *MEMO*
>
> *・Before registering the calibration offset, be sure to stop the operation. (If it is executed during programmed operation, target motion positions of the robot will change.)*

<For TP5000>

1 Tap the utility menu button in the main window.

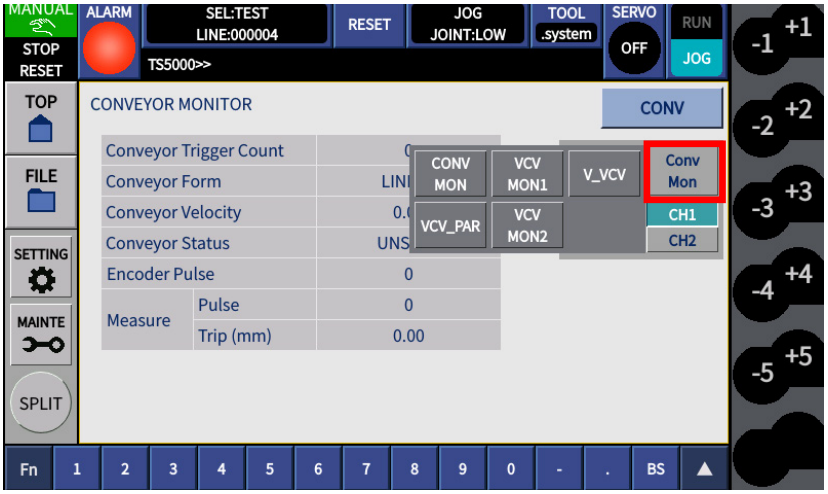
The utility menu switching window is opened.

2 Tap the "CONV" button in the menu.

The conveyor monitor screen is opened.

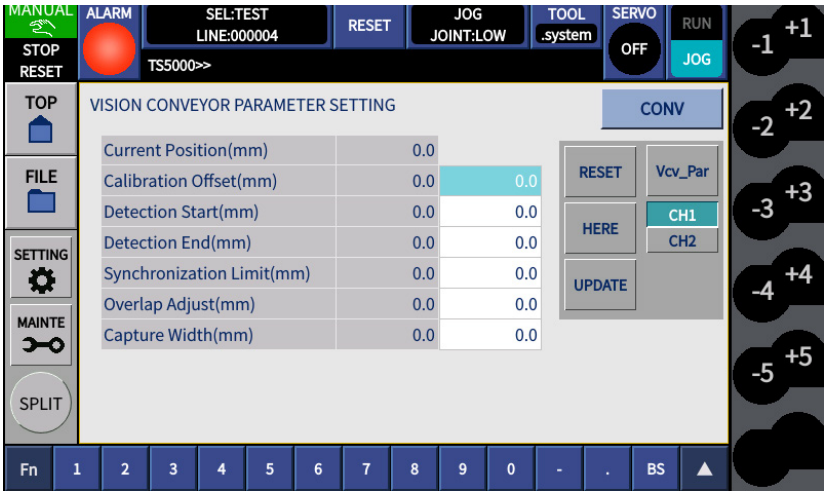
3 Tap the CONV screen selector button.

The CONV screen selection menu is displayed.



4 Tap the "VCV_PAR" button.

The vision conveyor parameter setting screen is opened.



5 Tap the text box for the line for the calibration offset.

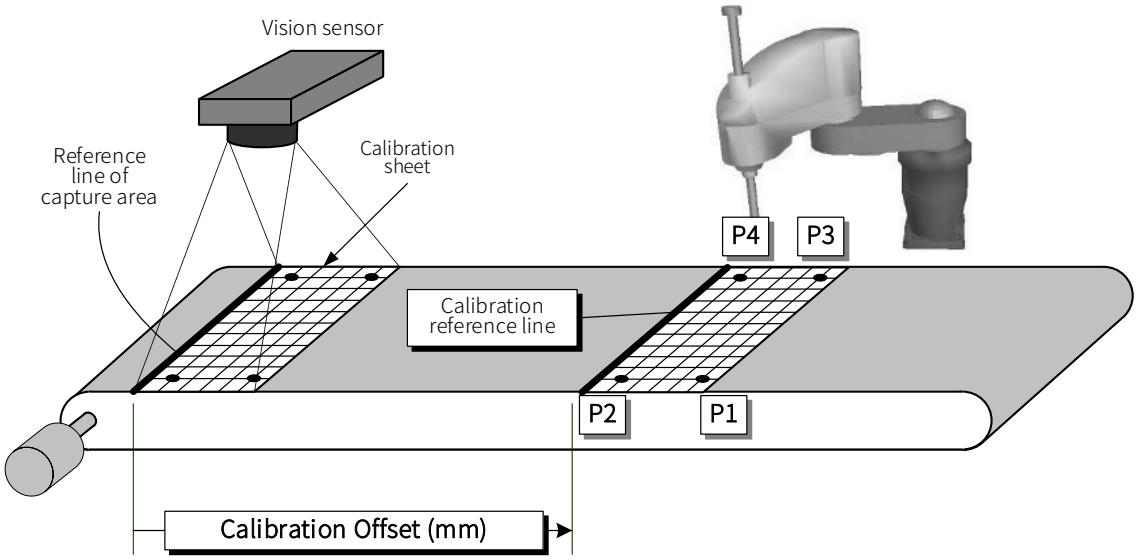
The cursor moves to the text box.

6 Tap the "Reset" button.

The current position is reset, and measurement of the conveyor travel is started.

7 Move the conveyor to the position where the robot reaches the calibration sheet.

The value for the current position changes.

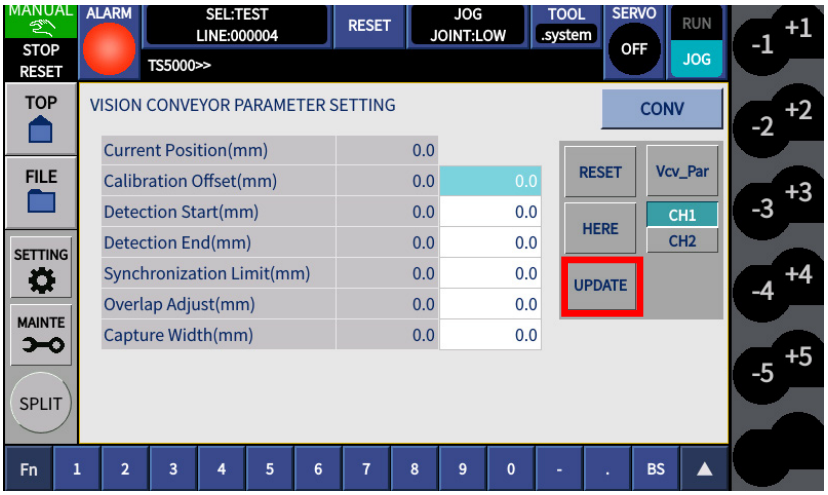


8 After stopping the conveyor, tap the "Read current value" button.

The value in the text box is updated to the value of the current position.

9 Tap the "Update" button.

The value for the calibration offset (mm) is updated to the value in the text box.



MEMO

- *To enable the setting value, reset the program, and then perform RUN for the program again.*
- *To change the value, tap the text box for the calibration offset so that it changes to an editable state.*

After directly entering the numerical value, tap the "Update" button again.

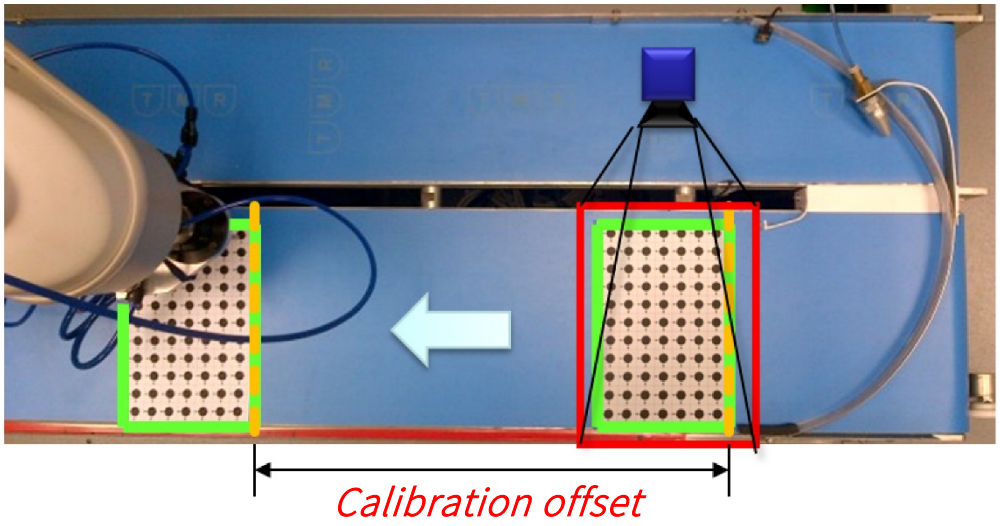
<For TP1000>

Press the <UTILITY> key followed by the [CONV] (F4) key to start up the conveyor utility.

Open the conveyor synchronization utility (VISION+CV_PAR1) display on page 2, and the following display is called.

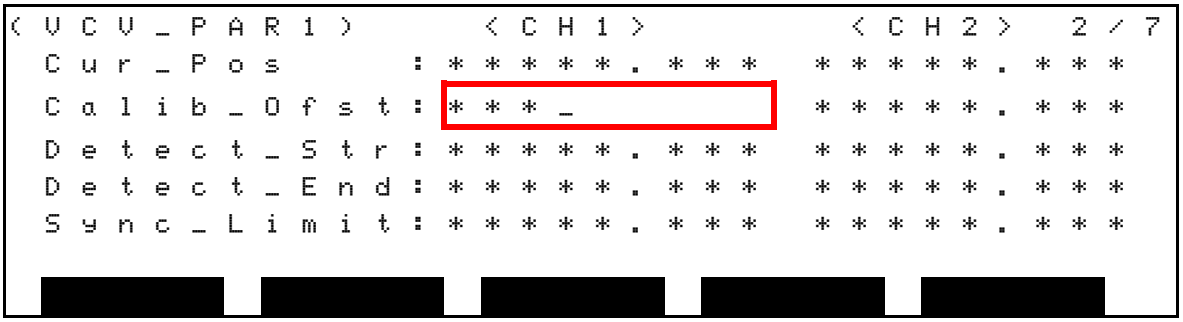
```
( VCV_PAR1 )          < CH1 >                < CH2 >    2 / 7
  Cur_Pos       : ******.***   ******.***
  Calib_Ofst    : ******.***   ******.***
  Detect_Str    : ******.***   ******.***
  Detect_End    : ******.***   ******.***
  Sync_Limit    : ******.***   ******.***
```

Press the <F1：RESET> key and move the conveyor to a position where the robot can reach the calibration sheet.



After the conveyor has moved to the position where calibration is to be done, stop the conveyor and press the [HERE] (F2) key. Now, the calibration offset (mm) has been registered. To make the set value valid, reset the program, then run the program again.

To change the value directly, press the <EXE> key. Then the numerical input mode takes effect. Enter a desired numerical value directly through the teach pendant, press the <EXE> key again.



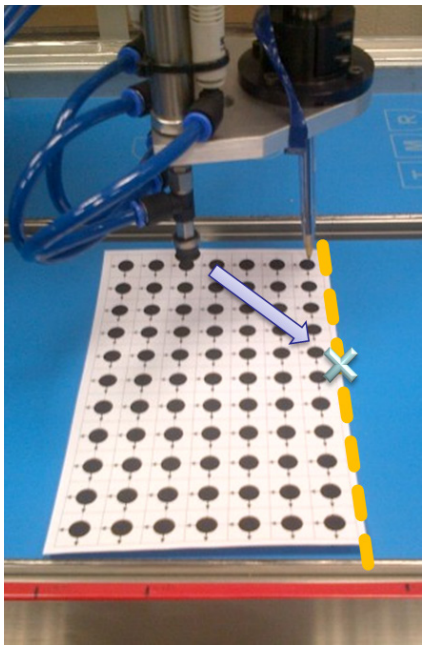
Align the end effector of the robot with P1 to P4, measure the X and Y coordinates of P1 to P4 based on the world coordinate system position, then register these values in the vision sensor.

4.3.6 Register of Calibration Reference Position

Register the calibration reference position (X, Y) of conveyor parameter "**[C001] Vision + Conveyor Tracking Parameter**".

```
[C001] Vision + Conveyor Tracking Parameter
        :
'{Calibration reference point (X)}
VCV-CAP-ORG-X = 0.000000, 0.000000   ←Calibration reference position X (mm)
'{Calibration reference point (Y)}
VCV-CAP-ORG-Y = -0.000000, 0.000000  ←Calibration reference position Y (mm)
```

The calibration reference position (X, Y) is **any position in the world coordinate system, which is located on the calibration reference line.** By registering this "reference position", it is possible to make the robot controller recognize the relationship between the capture area reference line of vision sensor and the calibration position.



As in the measurement of coordinate values of P1 to P4 made in Para. 4.3.5 above, move the robot tip to any position on the calibration reference line and set the coordinate value (X, Y) of this position in the parameter, to set the calibration reference position.

Set master mode to MANUAL, and then edit **[C001] Vision + Conveyor Tracking Parameter** in CONVEYOR.PAR.

4.3.7 Setting Capture Interval (sec) of Vision Sensor

Specify the time intervals (sec) at which the vision sensor captures data.

<For TP5000>

1 Tap the utility menu button in the main window.

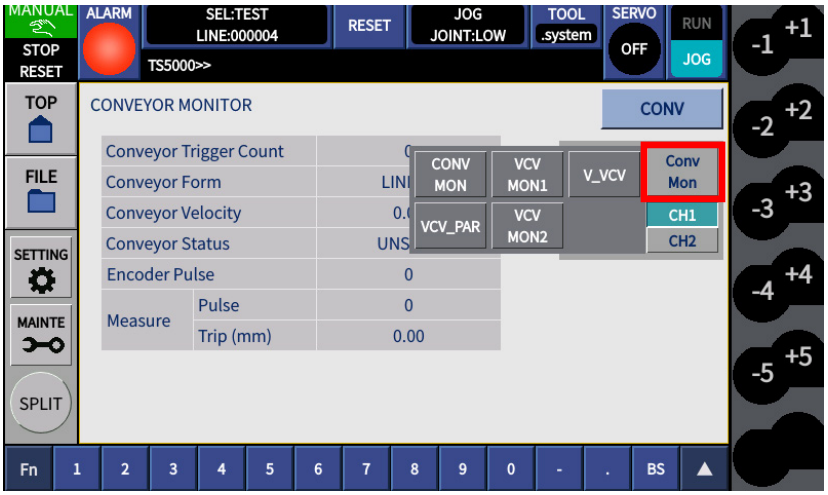
The utility menu switching window is opened.

2 Tap the "CONV" button in the menu.

The conveyor monitor screen is opened.

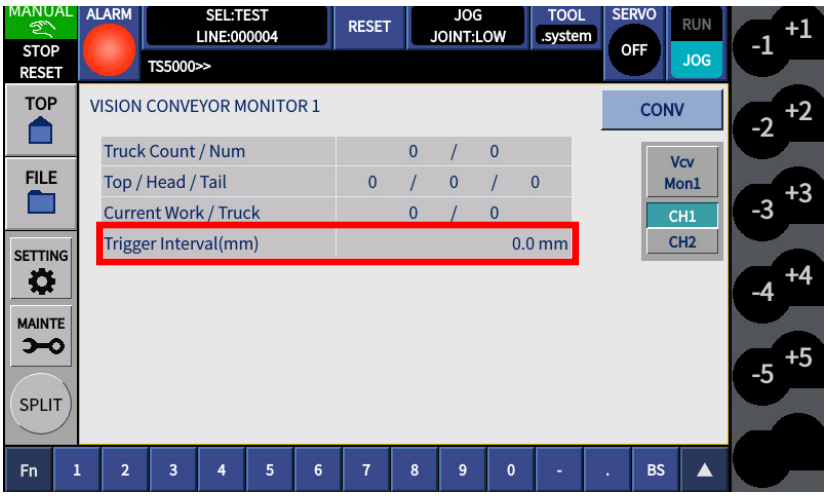
3 Tap the CONV screen selector button.

The CONV screen selection menu is displayed.



4 Tap the "VCV MON1" button.

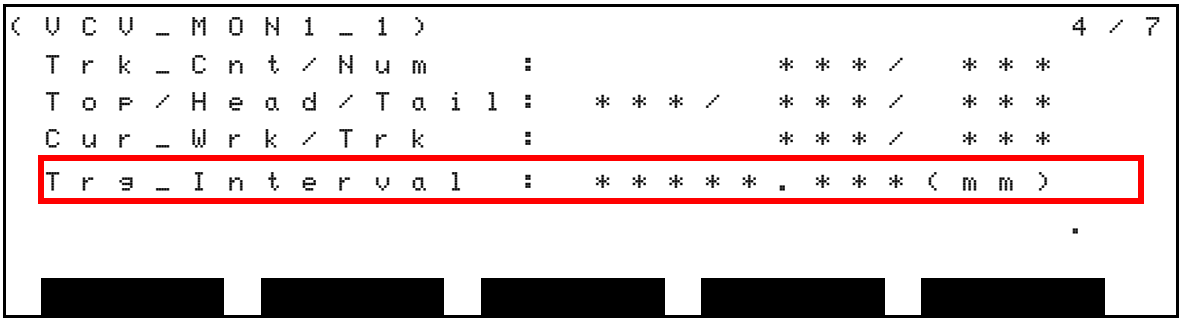
The vision conveyor monitor 1 screen is opened.



<For TP1000>

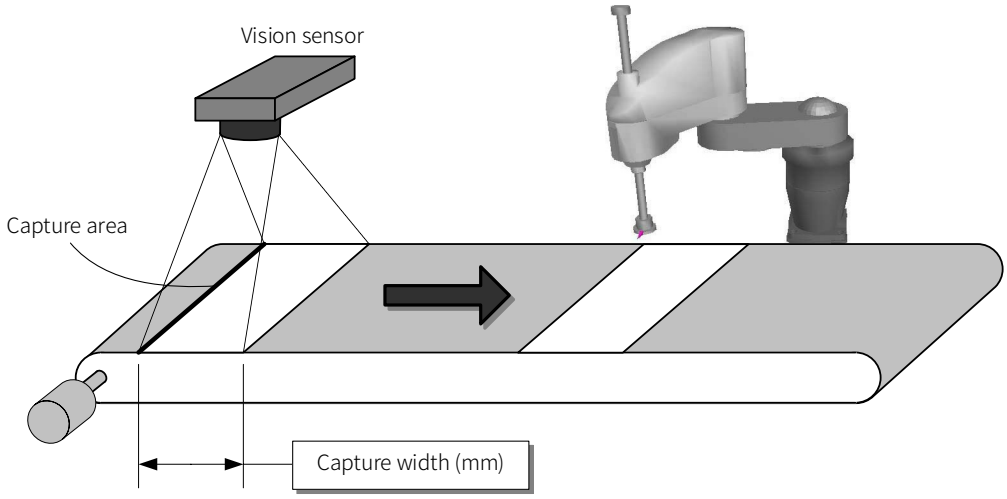
Press the <UTILITY> key followed by the [CONV] (F4) key to start up the conveyor utility.

Open the **conveyor synchronization utility** display on page 4 and 5, and the following display is called.



■Trg_Interval : Trigger interval (mm)

The conveyor travel distance until the current trigger signal is accepted after the previous trigger signal is displayed in mm. Adjust the capture interval (**sec**) of the vision sensor so that this trigger interval (mm) can be identical with the value of "Capture width (mm)" of "[C001] Vision + Conveyor Tracking Parameter".



MEMO

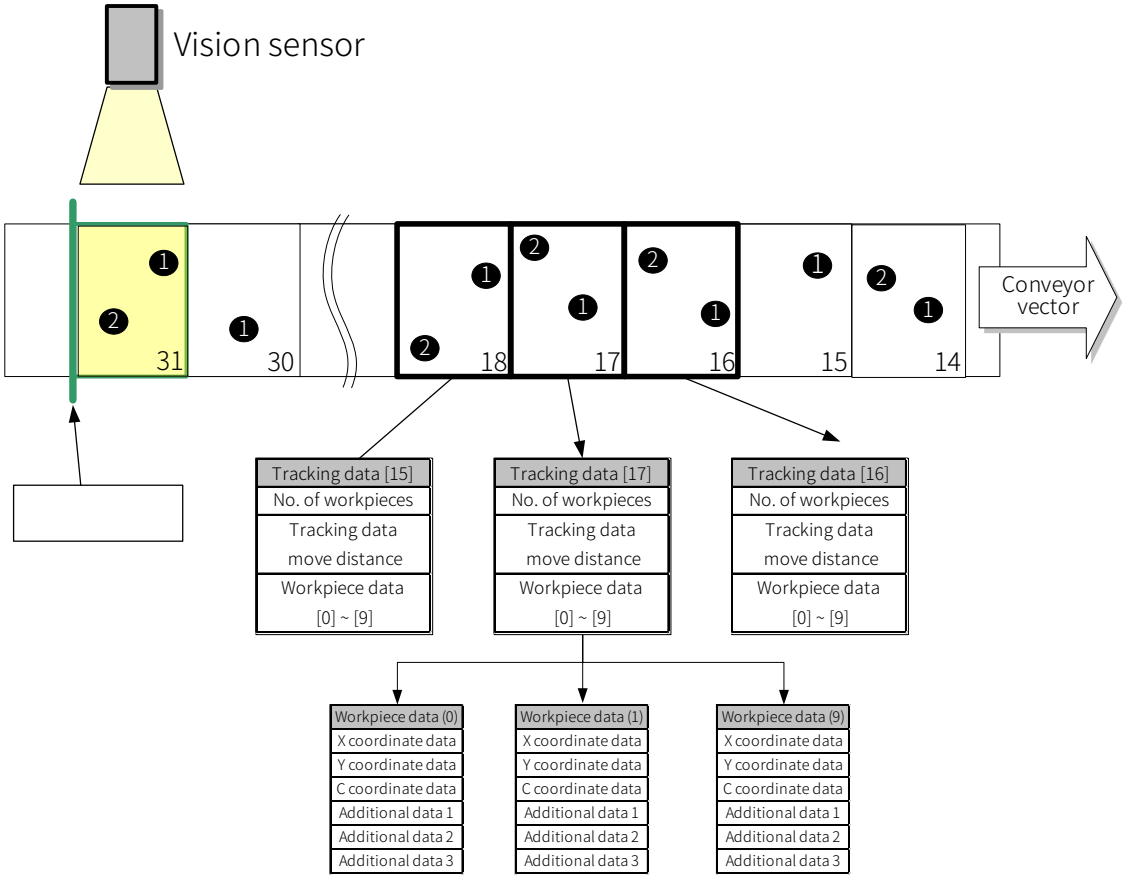
Unless the capture interval (sec) is set properly, the following phenomenon will occur.

・The capture interval is too long. → Missing of workpiece.

・The capture interval is too short. → Duplicated detection of workpiece.

4.4 Tracking Data

When the robot controller has detected the trigger signal sent from the vision sensor, it creates the tacking data internally and automatically. Up to 100 tracking data can be saved. (The number of tracking data that can be saved varies with the capture width and conveyor length. Design the system so that the number of tracking data will not exceed the maximum value.)



Each of these tracking data is assigned with an index number, as shown above. The robot controller tracks the workpiece position on the conveyor by using such tracking data.

<Contents of tracking data>

1 No. of workpieces

Number of workpieces collected during one (1) count of capture.

2 Tracking data move distance

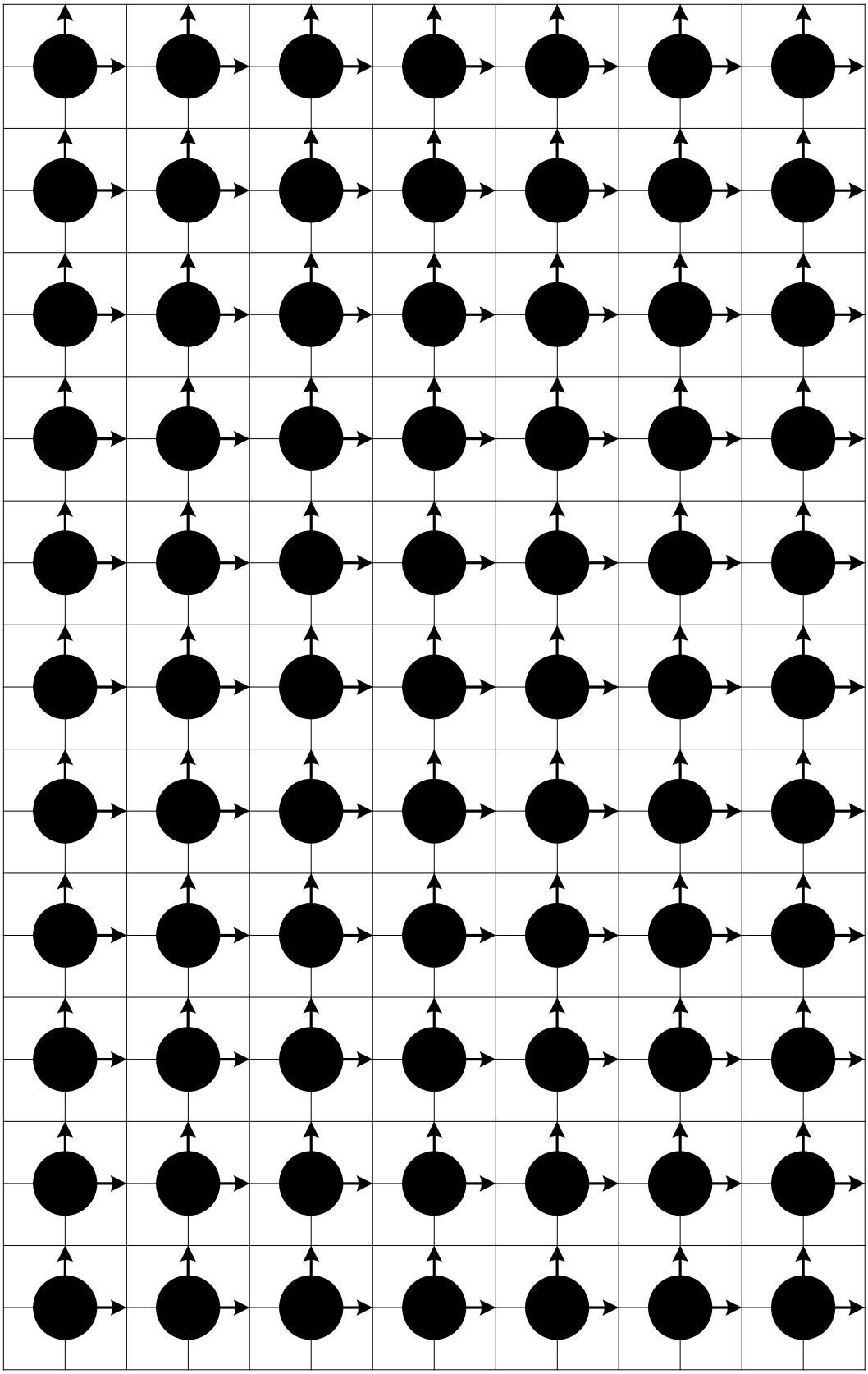
The move distance of tracking data as viewed from the capture area reference line is stored.

3 Workpiece data 0 to 9

Workpiece coordinates and other information sent from the vision sensor are stored here. (They are created for each collected workpiece.)

4.5 Calibration Sheet

Copy this sheet, cut it off to a desired size and use it for the calibration purpose.



Robot Language

Robot language and the programming procedures for the "vision sensor + conveyor" synchronization function are described.

5. Programming and Teaching

Programming and Teaching can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

5.1 Exclusive Language for Conveyor Synchronization

The commands used for the conveyor synchronization function are tabled below.

Command	Function
CONV	Defines the conveyor synchronization parameters.
SYNC	Starts the conveyor synchronization.
UNSYNC	Finishes the conveyor synchronization.
XIN	Monitors the encoder cable breakage.
CNVSTAT	System variable denoting conveyor status.

CONV

Function

Defines the conveyor synchronization parameters.

Classification

Option function Command

Format

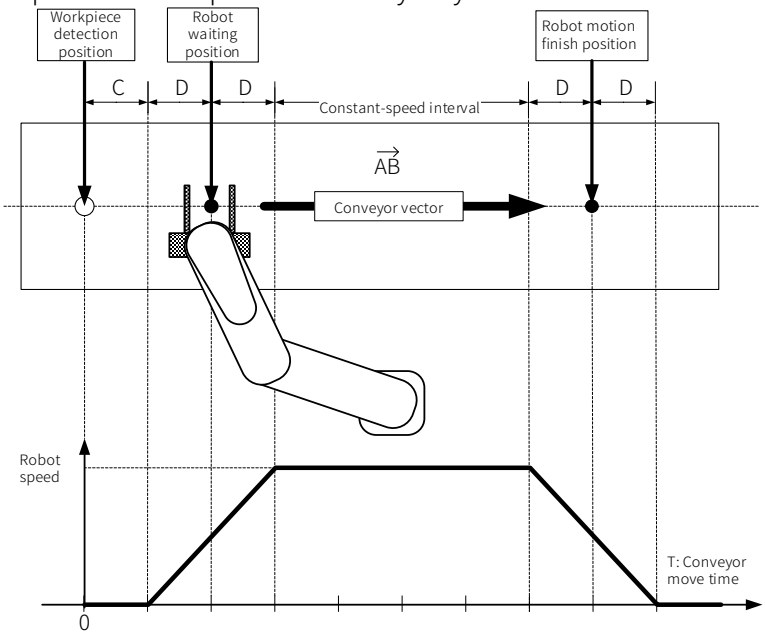
CONV (<Number>, <Position 1>, <Position 2>, <Real number 1>, <Real number 2>, <Real number 3>)

Example

CONV(N,A,B,C,D,E)

Explanation & caution

Defines the parameters required for conveyor synchronization.



<Number; N> Conveyor channel number

This controller allows synchronization of conveyors on a total of two (2) lines. Specify the channel number for desired conveyor synchronization.

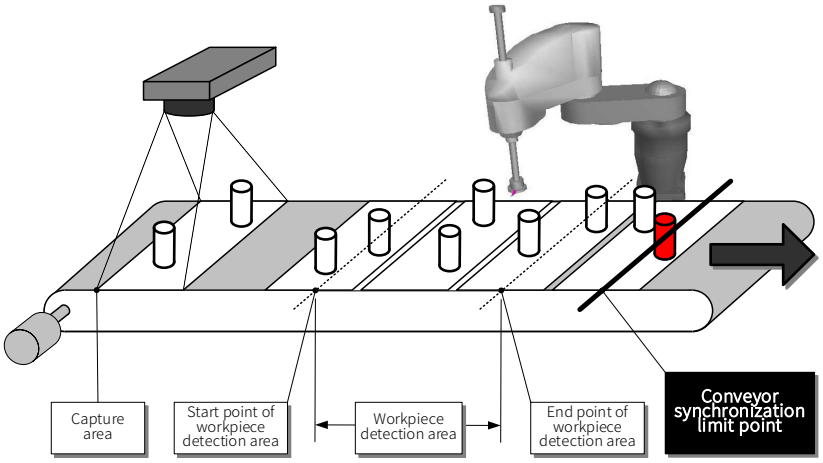
<Position 1; A> Conveyor vector start position (position type data)

Specify the start position of conveyor vector. (Only vector type position data can be specified.)

<Position 2; B> Conveyor vector end position (position type data)

Specify the end position of conveyor vector. (Only vector type position data can be specified.)

The parameters given above (i.e., <Position 1; A> and <Position 2; B>) are handled in a different manner between the conveyor synchronization function and "vision sensor + conveyor" synchronization function, as shown below.

Conveyor synchronization	When the conveyor has moved by $\lvert\overrightarrow{AB}\rvert$ after synchronization, the conveyor synchronization finishes.
"Vision sensor + conveyor" synchronization	When the robot apex position has passed the conveyor synchronization limit point in the figure below, the conveyor synchronization finishes. 

* The conveyor synchronization also finishes when program language "UNSYNC" has been executed, or program reset has been performed during program stop.

<Real number 1; C> Conveyor synchronization delay distance (mm)

This parameter should be specified to compensate for the physical synchronization delay. If this value is too large, however, an error will occur. To avoid this, this value should be used for fine adjustment and the system should be configured so as to minimize this distance, where possible.

<Real number 2; D> Conveyor synchronization acceleration/deceleration distance (mm)

Specify the acceleration/deceleration distance (unit: mm) required for conveyor synchronization.

<Real number 3; E> Travel distance per pulse of encoder (μm)

This parameter is the same as conveyor parameter "[C000] Conveyor Parameter (Travel value per pulse of encoder)". For how to figure out the value, see "3.3. [C000] CONVEYOR PARAMETER (Travel value per pulse of encoder".

This parameter can be omitted.

• When it is omitted (or E = 0), the conveyor parameter value is used instead.

• When the value of the conveyor parameter is zero (0), the value of CONV (,,,,E) is used.

<Caution>
When the value of the conveyor parameter is 0.0 with the E parameter omitted (or E = 0), the error saying "2–033: CONVEYOR Parameter Error" occurs at the time of program execution and the robot stops.

MEMO

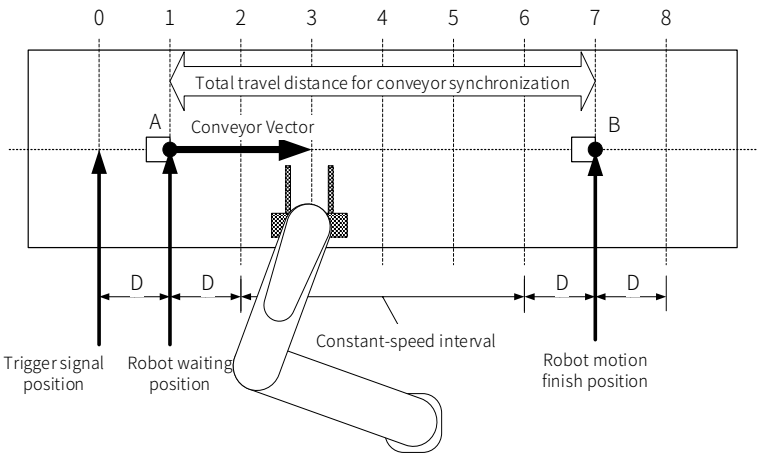
・*Once defined, the conveyor synchronization parameter data are stored in the internal memory until the controller power is turned off. To change the conveyor synchronization parameter value, be sure to describe the new CONV command and reset.*

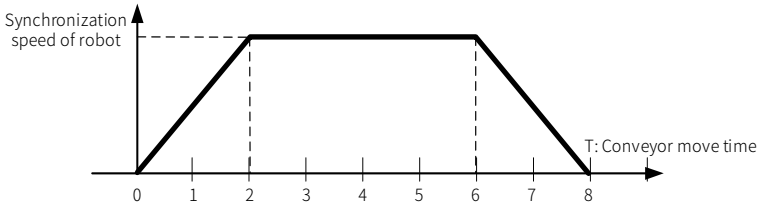
NOTICE

CAUTION/ WARNING

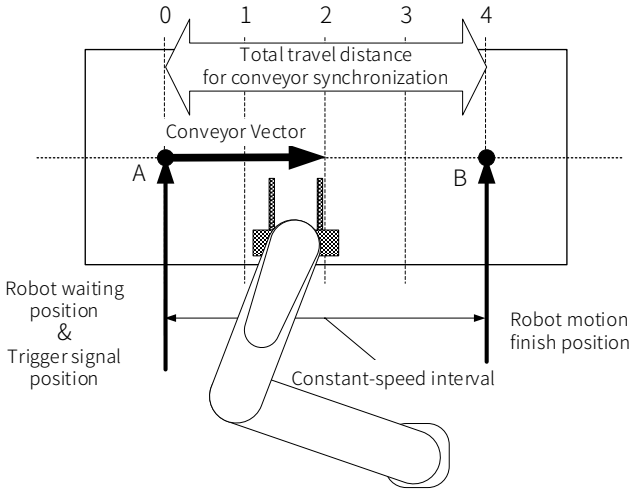
- Zero (0) can be specified for each of parameters C and D. When D (acceleration/deceleration distance) = 0, however, the synchronization speed of the robot is sharply increased to the conveyor speed, as shown below, which results in an excessively large load exerted on the robot.

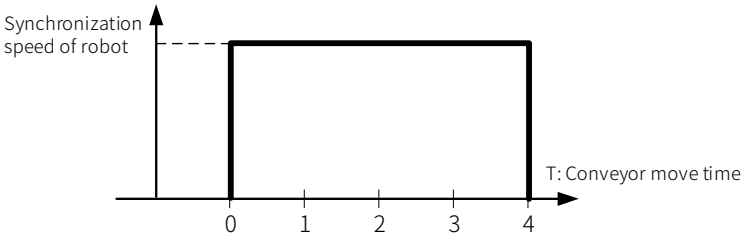
<C=0>





<C=D=0>





SYNC

Function

Starts the conveyor synchronization.

Classification

Option function Command

Format

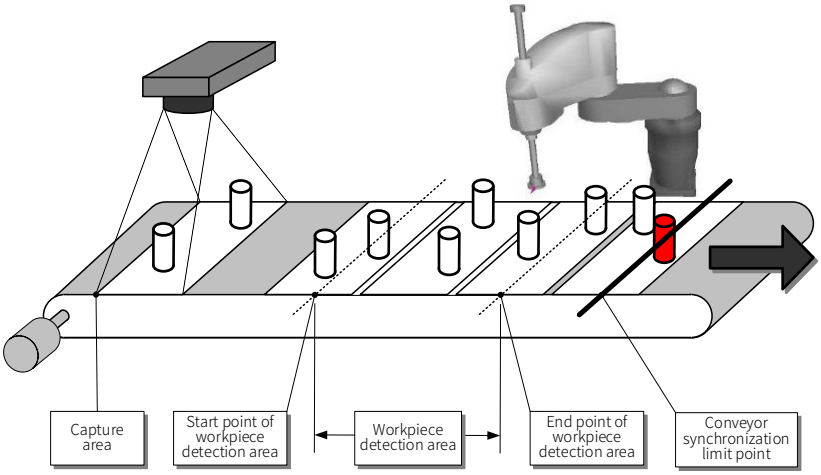
SYNC（<Number>， ［<Stop judgment time>］）

Example

SYNC(N,T)

Explanation & caution

This command is handled in a different manner between the "conveyor synchronization function" and "vision sensor + conveyor" synchronization function, as shown below.

Conveyor synchronization	At the same time that the trigger signal has turned on, conveyor synchronization starts.
"Vision sensor + conveyor" synchronization	 •When a workpiece is present in the workpiece detection area, conveyor synchronization starts. •When VCSEEK1 to 3 ("vision sensor + conveyor" synchronization detecting conditions (data on type of workpiece and workpiece status) is used, it is possible to start synchronization by detecting only workpieces of specified type. (For details, see the descriptions on VCSEL1 to 3 given later.)

<Number; N> Conveyor channel number

Specify the channel number of the conveyor to be synchronized. Either 1 or 2 can be designated. Constant or variable scalar type data can be used.

<Stop judgment time; T> Conveyor stop judgment time

Specify the conveyor stop judgment time. Unless this value is specified, the stop judgment is executed according to the value of conveyor parameter "[C000] Conveyor Stop Judgment Time".

Unless the conveyor synchronization parameters are defined by the CONV command, an error occurs upon execution of this command.

When starting the "vision sensor + conveyor" synchronization, be sure to describe the SYNC command at the beginning and the UNSYNC command, which is described later, at the end of the synchronization. Otherwise, an error occurs upon execution of this command.

Nesting is not possible for the SYNC command or UNSYNC command.

If the stop operation is executed during waiting for workpiece detection, the conveyor synchronization will not start and the program stops. To restart the program, be sure to perform the reset operation, irrespective of the execution state (reset, retry or continue), then execute the program. For details, see Section 6.

MEMO

Following 4 motion commands are available during the conveyer synchronization

MOVES： Straight line interpolation

MOVEC： Circular interpolation

MOVE ： PTP(point to point) motion command

MOVEJ： Arch motion command

- *MOVE or MOVEJ motion commands make more high-speed handling possible even during the conveyer synchronization. Please be careful of the following points.*
- *If a value other than FREE is specified for CONFIG or if a value different from the CONFIG of the previous operation is specified, "2-069：CONFIG Error" occurs.*
- *The MOVE and MOVEJ operation commands generate a path so that the axis that requires the longest travel time is set to the maximum speed (when SPEED=100). Also, during conveyor synchronization, the conveyor speed is added to this operation speed. As a result, if each joint operation direction matches the conveyor travel direction, "8-041 to 8-046: Axis1 to Axis6 Exceed Speed (Joint Axis Speed Limit Exceeded)" or "8-461 to 8-466: Axis1 to Axis6 Motor speed over (Motor speed exceeded)" occurs. If one of these errors occurs, use the SPEED command to reduce the robot operation speed.*

UNSYNC

Function

Finishes the conveyor synchronization.

Classification

Option function Command

Format

UNSYNC（<Number>）

Example

UNSYNC（N）

Explanation & caution

The conveyor synchronization finishes.

<Number; N> Conveyor channel number
Specify the logical number of the conveyor for which synchronization is to be finished.

- When a value other than zero (0) is specified for [the acceleration / deceleration distance for conveyor synchronization <Relative distance 2; D>], the robot decelerates and stops while the conveyor moves by "D" from the timing of this command execution.

- When the robot hand tip has reached the conveyor synchronization limit point during conveyor synchronization, the conveyor synchronization finishes automatically, irrespective of description of the UNSYNC command. (This does not mean that the UNSYNC command is unnecessary. Rather, the robot movement synchronized with the conveyor speed is stopped to assure safety.)

- Unless this command is used together with the SYNC command described on the preceding pages, an error occurs.

XIN

Function

Monitors the encoder cable breakage.

Classification

Option function Command

Format

XIN (<Signal name>)

Example

ON XIN(1) DO GOTO L100

Explanation & caution

Breakage of the encoder cable is monitored. If a fault is detected, the fault is reported.

<Signal name; N> : Specify the logical number of the conveyor for which encoder cable breakage is monitored.

- This command is used to instantly detect a cable breakage. Use this command together with the ON command so that cable breakage can be detected at any time.
- When the ON–DO command is used together with this command, the number of counts that can be simultaneously monitored should not exceed ten (10). If the value exceeds ten (10), an error occurs and the robot program stops accordingly.
- To cancel the monitoring of encoder cable breakage, use the IGNORE command. This command can be used together with the IF command and WAIT command.
- If the encoder cable has broken, the error saying "001–047 Conveyor Line Breakage" occurs. The robot will not stop, however, and continuously move, synchronized with the conveyor (encoder) speed effective just before the detection of cable breakage.
- The error of "001–047 Conveyor Line Breakage" is not reset as long as the power is ON. When this error has occurred, turn the power OFF, identify and remedy the broken cable, then turn the power on again.
- Even if an encoder cable breakage has been detected during conveyor synchronization, the robot continues tracking the conveyor at the same speed because it memorizes the speed just after the cable breakage.

NOTICE



REQUIRED

●If the above error (001–047) has occurred, however, be sure to recess the robot to the safe place. If the operation is continued without taking any necessary steps, the robot and workpiece, or conveyor may collide. Recess the robot to the safe place by referring to the retraction program listed below.

Sample

```
PROGRAM XINSAMPLE
   CONV( 1, A, B, 40.0, 10.0 )

L000:
   ON XIN(1) DO GOTO L100          Monitoring of encoder cable breakage before
                                   the start of synchronization.
                                   (If the monitoring is cancelled just after the XIN
                                   command, breakage detection check cannot
                                   be executed.)
   IGNORE XIN(1)                   Cancel of monitoring before the start of
                                   synchronization.
   SYNC(1)
   ON XIN(2) DO GOTO L200          Monitoring of encoder cable breakage during
                                   synchronization.
   IGNORE XIN(2)                   Cancel of monitoring during synchronization.
   UNSYNC(1)
   GOTO L000
L100:
   PRINT “C/V ERROR(1)”,CR
   STOP
L200:
   MOVES P1                        After the robot is recessed to the safe position
                                   with the linear interpolation control command,
                                   stop the program execution.
   UNSYNC(N)
   PRINT “C/V ERROR(2)”,CR
   STOP
END
```

CNVSTAT

Function

Accesses the information on conveyor synchronization status.

Classification

System variable (Option function)

Format

CNVSTAT

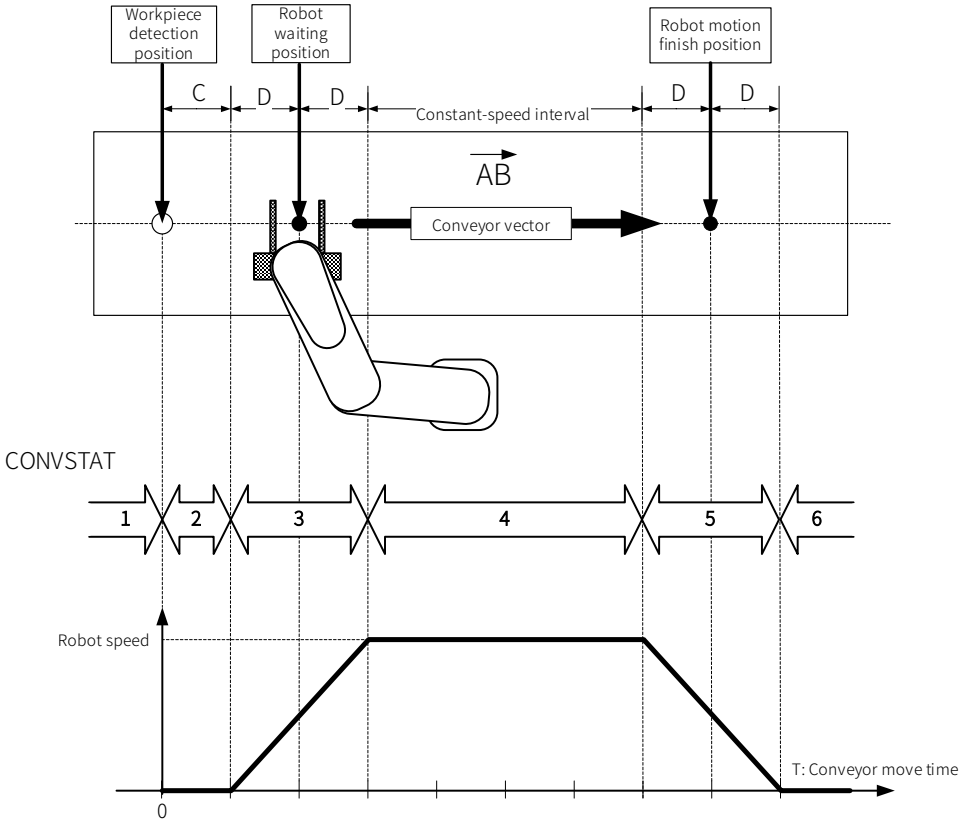
Example

```
WAIT CNVSTAT > = 4
ON CNVSTAT,.4 BREAK DO GOTO L1
```

Explanation & caution

The information on conveyor synchronization status is accessed.

- This command can be used during conveyor synchronization (i.e, after the SYNC command is executed).
- A value cannot be substituted for CNVSTAT.
- Only when the CNVSTAT conditional statement is used as the ON condition, the BREAK and PAUSE commands can be used.



Conveyor status table

0	Conveyor asynchronization	Signifies that the conveyor is not synchronized. (As this command can be used only after execution of the SYNC command, "1" is not set in the actual program.)
1	Waiting for trigger signal	The conveyor is waiting until the trigger signal is turned on.
2	Conveyor moving in delay interval	The conveyor is moving in the delay interval.
3	Conveyor moving in acceleration interval	The conveyor is moving in the acceleration interval. During this time, the robot catches up with the conveyor speed.
4	Conveyor moving in constant-speed interval	The conveyor is moving in the constant-speed interval. The robot is moving by tracking the conveyor. Handling of a workpiece on the conveyor should be performed while CNVSTAT = 4.
5	Conveyor moving in deceleration interval	The conveyor is moving in the deceleration interval. During this time, the robot catches up with the conveyor speed.
6	Finish of conveyor synchronization	The conveyor synchronization has finished, and the conveyor is waiting for the UNSYNC command.

Sample

```
PROGRAM MAIN
    CONV(1,A,B,100.0,200.0)
    MOVE P1
    SYNC(1)
    WAIT CNVSTAT>=4
    MOVES P2
    MOVES P3
    WAIT CNVSTAT>=6
    UNSYNC(1)
END
```

5.2 "Vision Sensor + Conveyor" Exclusive Language

In the Vision Sensor + Conveyor Synchronization function, **a special exclusive language** is used to enable easy creation of complex vision sensor + conveyor application programs.

Language	Example of statement	Description
VCFUNC	VCFUNC=ON	Data receiving start switch ON: This starts the receiving of data from the vision sensor. OFF: Receiving of data from the vision sensor finishes.
VCMARK	IF DIN(-203,203) THEN VCMARK(1) IF VCMARK(1)==1 THEN GOTO LP	SYNC, VCNEXT, and VCSEEK enable performing of the manual marking process on indexed workpieces.
VCNEXT	IF VCNEXT(1)==OFF THEN GOTO HMPS	During conveyor asynchronization, this searches for a workpiece and returns the result. (1: Workpiece found, 0: Workpiece not found)
VCPOS	MOVES VCPOS MOVES VCPOS + POINT(,,100)	Target motion position during "vision sensor + conveyor" synchronization. All target positions during "vision sensor + conveyor" synchronization are calculated internally and automatically, which are set then in this system variable.
VCPSWLD1-2	P0=VCPSWLD1 MOVE VCPSWLD2	This outputs point data in the world coordinate system for workpieces that were detected by SYNC, VCNEXT, or VCSEEK.
VCSEEK	IF VCSEEK == ON THEN GOTO LP	This returns whether a next workpiece was found that meets the indexing conditions (VCSEL1 to VCSEL3) within the workpiece detection area. (ON: Next workpiece found, OFF: Next workpiece not found) If the next workpiece is found, the target position of the next workpiece is set to "VCPOS".
VCSEL1-3	VCSEL1 = 1 VCSEL2 = 0	Used to specify the next workpiece detecting conditions by means of VCSEEK or SYNC command.
VCVTYPE1-2	VCVTYPE1 = 1	When using virtual vision conveyor synchronization, this selects the table where the workpiece data is set.
VCWSTAT1-3	IF VCWSTAT1 == 1 THEN GOTO LP	Used to refer to additional information of currently detected workpiece.

VCFUNC

Function

Vision + Conveyor synchronization function command

Specifies the start of data reception.

Classification

Option function Command

Format

VCFUNC = <Value>

Example

VCFUNC = ON

Explanation & caution

ON: Receiving of data from the vision sensor starts.

OFF: Receiving of data from the vision sensor ends.

- When VCFUNC = ON, receiving of data sent from the vision sensor starts.
- After input of trigger signal has been detected, analysis of received data starts. For details, see Para. 4.1.
- When VCFUNC = OFF, receiving of data sent from the vision sensor ends.

MEMO

- *Even when VFUNC = OFF, conveyor synchronization will not finish.*
- *As long as the workpiece received at "VCFUNC=ON" remains on the conveyer, the conveyer synchronization is continued.*
- *To cancel the data of a workpiece now on the conveyor, stop the program, then reset it.*

Sample

```
PROGRAM VCPROG1
      :
VCFUNC = ON                    Receiving of data starts.
CONV( N, A, B, C, D )
LP:
      SYNC(1)                  If a workpiece exists in the workpiece detection
      :                        area, synchronization starts.
      USYNC(1)
      :                        Synchronization ends.
GOTO LP
VCFUNC = OFF
END                            Receiving of data finishes.
```

VCMARK

Function

Vision + Conveyor synchronization function command

Performs manual marking on detected workpiece.

Classification

Option function Command

Format

VCMARK （<Channel number>）

Example

VCMARK(1)

IF VCMARK(1) = = 1 THEN GOTO LP

Explanation & caution

Normally, workpieces that are found are automatically marked, and the marked workpieces are excluded from subsequent searches. (Synchronization cannot be performed twice for the same workpiece.)

This function performs the above marking process manually. To enable this function, the CONVEYOR.PAR function switch must be set as shown below.

[C003] Conveyor function switch

= 0 0 0 1 0 0 ← Manual marking enabled

（If set to 0: Disabled, automatic marking is used as in normal operation）

Sample

```
PROGRAM VCPROG
   IF VCNEXT(1) == OFF THEN GOTO JP        Detection of workpiece in detection area
   MOVE VCPSWLD1 + POINT(, ,50.0)          Detected workpiece point data
   GOTO JP2
JP:
   MOVE HOME + POINT(, ,50.0)
JP2:
VCFUNC = ON                                Data reception start
CONV( N, A, B, C, D )
LP:
SYNC(1)                                    Synchronization is started if a workpiece is
      :                                    found in the workpiece detection area
VCMARK(1)                                  Manual marking
USYNC(1)                                   Synchronization end
GOTO LP
VCFUNC = OFF                               Data reception end
END
```

VCNEXT

Function

Vision + Conveyor synchronization function command

Searches for workpiece within detection area during conveyor asynchronization

Classification

Option function Command

Format

VCNEXT (<Channel number>)

Example

IF BCNEXT(1) = = OFF THEN GOTO HMPS

Explanation & caution

This function searches from the first workpiece in the head track to the last workpiece in the tail track only during conveyor asynchronization. If a workpiece is found, the return value changes to ON, and the world coordinate system position posture of the found workpiece is output to VCPSWLD1 (or VCPSWLD2).

Marking is not performed when manual marking is set to disabled (=automatic marking).

Sample

```
PROGRAM VCPROG
   IF VCNEXT(1) == OFF THEN GOTO JP          Detection of workpiece in detection area
   MOVE VCPSWLD1 + POINT(, ,50.0)            Detected workpiece point data
   GOTO JP2
JP:
   MOVE HOME + POINT(, ,50.0)
JP2:
VCFUNC = ON                                  Data reception start
CONV( N, A, B, C, D )
LP:
SYNC(1)                                      Synchronization is started if a workpiece is
      :                                      found in the workpiece detection area

VCMARK(1)                                    Manual marking
USYNC(1)                                     Synchronization end
GOTO LP
VCFUNC = OFF                                 Data reception end
END
```

VCPOS

Function

Vision + Conveyor synchronization function system variable

Shows the workpiece position indexed by SYNC command.

Classification

System variable (Option function)

Format

MOVES VCPOS

Example

MOVES VCPOS

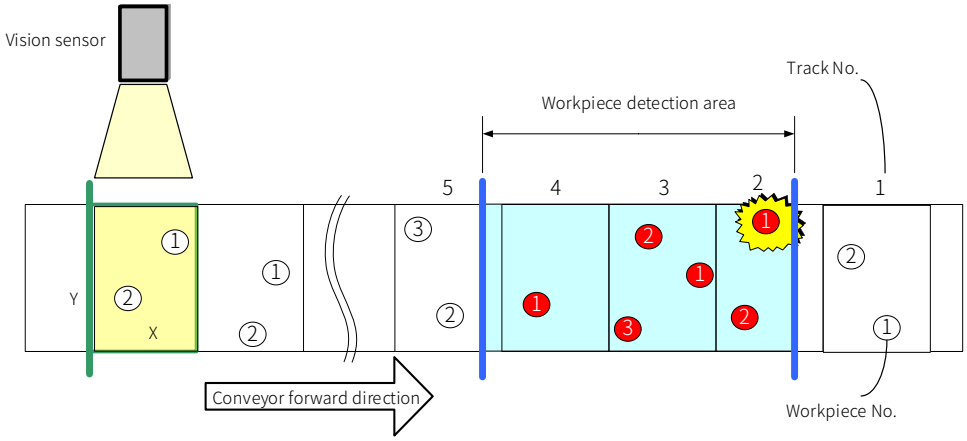
MOVES VCPOS + POINT(, ,100)

P0 = VCPOS

Explanation & caution

Specifies the target motion position during "vision sensor + conveyor" synchronization. The workpiece in the optimal position is automatically selected from the workpieces on the conveyer, and its position data is set to VCPOS.

- VCPOS signifies the target motion position which is exclusively used in the "vision sensor + conveyor" synchronization function. NEVER use it in other than the "vision sensor + conveyor" synchronization (i.e., between the SYNC and UNSYNC commands). Otherwise, the robot may move to an unexpected position, or the alarm saying "002–079: PointOutOfRange" occurs.
- The workpiece position data, which is selected from the current synchronized conveyer channel, is set to VCPOS.
- Only reading of this system variable is possible. Writing is not allowed.
- Data structure:
  VCPOS is the position type variable. The format is the same as that of the position type variable (X, Y, C, Z, T, Posture).
- Initial value:
  The initial value of VCPOS target motion position is "0.0" for all of X, Y, C, Z, T and posture.
- Detection timing and set value of VCPOS:
  If a workpiece has entered or already entered the workpiece detection area after execution of SYNC command or VCSEEK command, a workpiece located most downstream in the area is detected and this coordinate position is set in VCPOS.
- Calculation using VCPOS:
  It is possible to specify the target position which is offset from this target position, using the POINT command.
  Example) MOVES VCPOS＋POINT（ , , 25 ）



Sample

```
PROGRAM VISCONV2
        VCFUNC = ON
        MOVE A                          Moves to waiting position.
        SYNC(1)                         If a workpiece exists in the workpiece detection
                                        area, synchronization starts.
        MOVES VCPOS + POINT(,,100)      Moves to the position above the workpiece
                                        position as detected by SYNC command.
        MOVES VCPOS                     Moves to the workpiece position detected by
        :                               SYNC command.
                                        Picks up the workpiece.
        MOVES VCPOS + POINT(,,100)      Moves to the position above the workpiece
                                        position as detected by SYNC command.
        UNSYNC(1)                       Conveyor synchronization ends.
        MOVE HOME + POINT( , , 50)
        MOVE HOME
        :                               Places the workpiece.
        MOVE HOME + POINT( , , 50)
        WAIT MOTION >= 100
        PRINT 10.0-TIMER,CR
END
```

MEMO

<Motion Commands during the conveyer synchronization>

Following 4 motion commands are available during the conveyer synchronization.

MOVES： Straight line interpolation

MOVEC： Circular interpolation

MOVE ： PTP(point to point) motion command

MOVEJ： Arch motion command

MOVE or MOVEJ motion commands make more high-speed handling possible even during the conveyer synchronization. Please be careful of the following points.

- *If a value other than FREE is specified for CONFIG or if a value different from the CONFIG of the previous operation is specified, "2-069: CONFIG Error" occurs.*
- *The MOVE and MOVEJ operation commands generate a path so that the axis that requires the longest travel time is set to the maximum speed (when SPEED=100). Also, during conveyor synchronization, the conveyor speed is added to this operation speed. As a result, if each joint operation direction matches the conveyor travel direction, "8-041 to 8-046: Axis1 to Axis6 Exceed Speed (Joint Axis Speed Limit Exceeded)" or "8-461 to 8-466: Axis1 to Axis6 Motor speed over (Motor speed exceeded)" occurs. If one of these errors occurs, use the SPEED command to reduce the robot operation speed.*

VCPSWLD1 to 2

Function

Vision + Conveyor synchronization function system variable

Outputs the point data for the detected workpiece in the world coordinate system

Classification

System variable (Option function)

Format

VCPSWLD1

Example

P0=VCPSWLD 1

MOVE VCPSWLD 2

Explanation & caution

The SYNC, VCSEEK, and VCNEXT commands are used to output the position data for the indexed workpiece in the world coordinate system. This variable can be referred to during conveyor synchronization/asynchronization.

Sample

```
PROGRAM VCPROG
   IF VCNEXT(1) == OFF THEN GOTO JP        Detection of workpiece in detection area
   MOVE VCPSWLD1 + POINT(, ,50.0)          Detected workpiece point data
   GOTO JP2
JP:
   MOVE HOME + POINT(, ,50.0)
JP2:
VCFUNC = ON                                Data reception start
CONV( N, A, B, C, D )
LP:
SYNC(1)                                    Synchronization is started if a workpiece is
                                           found in the workpiece detection area
      :
VCMARK(1)                                  Manual marking
USYNC(1)                                   Synchronization end
GOTO LP
VCFUNC = OFF                               Data reception end
END
```

VCSEEK

Function

Command for confirming next workpiece presence or absence and detecting next workpiece.

Classification

Option function Command

Format

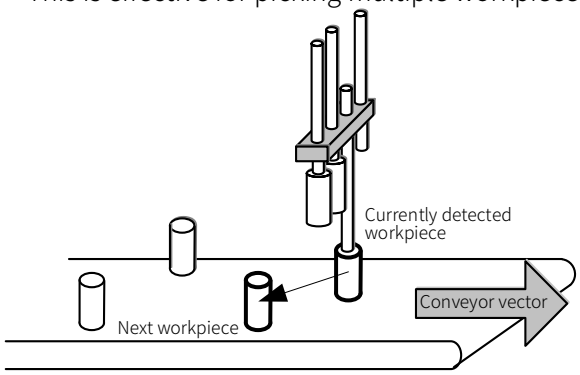
VCSEEK == <O N> or <O F F>

Example

IF VCSEEK == OFF THEN GOTO LP1

Explanation & caution

- The VCSEEK command uses the SYNC command when indexing the next workpiece that is further downstream from the workpiece currently being indexed during the process from workpiece detection to conveyor synchronization. When this command is executed, the operation target position of the next workpiece is automatically set to "VCPOS".
- After this command is executed, a search is conducted for the next workpiece in the workpiece detection area, and if the next workpiece is found, the position data is set to VCPOS.
- This is effective for picking multiple workpieces while tracking the conveyor.



- When VCSEL1 to 3 (conditions for detecting "vision sensor + conveyor" synchronization, such as workpiece type and workpiece status) is used, it is possible to discriminate and group the workpieces of specified type alone.
  For details, see the descriptions on VCSEL1 to 3 given later.

- The role of VCSEEK:

  A) Judgment of next workpiece presence or absence

  Replies the presence or absence of the next workpiece in the workpiece detection area during conveyor synchronization.

  Next workpiece is present. ⇒ VCSEEK == ON

  Next workpiece is absent. ⇒ VCSEEK == OFF

  B) Updating target motion position "VCPOS"

  When the next workpiece is present, its target motion position is set in VCPOS.

- As shown in the table below, this function can be used together with IF statement (condition judgment command). It cannot be used jointly with WAIT command or ON command, however.

Condition judgment command	Use
IF VCSEEK == ON THEN GOTO LP	Permitted
WAIT VCSEEK == ON	Prohibited
ON VCSEEK == ON DO	Prohibited

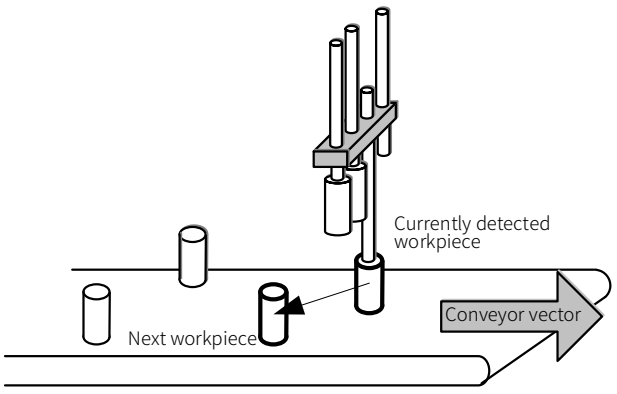
- The VCSEEK command functions normally only during the "vision sensor + conveyor" synchronization mode (SYNC to UNSYNC).
- This command can be used during conveyor synchronization after processing of SYNC command.
- Only reading of this system variable is possible. Writing is not allowed.
- After execution of VCSEEK command, only reading of this system variable is possible. Writing is not allowed.
- VCSEEK is executed against the current channel of conveyor in synchronization.

Sample

```
PROGRAM VISCONV2B                          Multiple-workpiece (four (4) pcs.)
                                           pick-and-place program
ENABLE NOWAIT
    PASS = 60
    VCFUNC = ON
    CONV(1,A,B,0,0)
    MOVE HOME + POINT(,,OFST)
    SYNC(1)
    I = 0
LP:
    I = I + 1
    ENABLE PASS
    MOVES VCPOS + POINT(,,OFST)
    DISABLE PASS
    MOVES VCPOS
    '***** PICK-UP WORK PIECE *****
    DELAY 0.1
    ENABLE PASS
    MOVES VCPOS + POINT(,,OFST)
    WAIT MOTINO >= 100
    IF I >=  4 THEN GOTO USNC
    IF VCSEEK == ON THEN GOTO LP
USNC:                                      Conveyor synchronization end
    UNSYNC(1)
    MOVE P0 + POINT(,,OFST)
    DISABLE PASS
    MOVE P0
    DELAY 0.1
    '***** PLACE WORK PIECE *****
    ENABLE PASS
    MOVE P0 + POINT(,,OFST)
    DISABLE PASS
    MOVE HOME + POINT(,,OFST)
END
```

If the number of workpieces exceeds the maximum value, conveyor synchronization finishes.

If next workpiece exists, the robot moves to the next workpiece position. If not, conveyor synchronization finishes.



VCSEL1 to 3

Function

Designation of conditions for detecting "vision sensor + conveyor" synchronization (workpiece type and workpiece status).

Classification

System variable (Option function)

Format

VCSEL 1 = <Integer>

Example

VCSEL 1 = 1 0

Explanation & caution

The TS5000 controller can receive three (3) different workpiece additional information from the vision sensor. It is possible to identify only desired workpieces by applying the detecting conditions of VCSEL1 to 3 to this additional information. (This function is useful when sorting and detecting workpieces.)

- Only reading of this system variable is possible. Access is not allowed.
- Joint use of this command with motion command by using WITH statement is not allowed.
- VCSEL1 to 3 set the detecting condition to the current conveyor channel in synchronization.

Workpiece data format:

The format of workpiece data that can be received from the vision sensor is predetermined, as shown below. (For details, see Para. 4.2.)

Name	Description
x	X coordinate data of workpiece
y	Y coordinate data of workpiece
c	C coordinate data of workpiece
Opt1	Additional data 1 of workpiece
Opt2	Additional data 2 of workpiece
Opt3	Additional data 3 of workpiece

Among the above six (6) elements, Opt1 to Opt3 are the additional information which can be specified for each workpiece by the user at his or her discretion.

Example) Opt1: Workpiece type data 1 (Workpiece profile)
Opt2: Workpiece type data 2 (Color information)
Opt3: Workpiece quality data 3 (Workpiece accepted, rejected, etc.)

How to set VCSEL1 to 3:
As shown above, Opt1 to 3 are linked with VCSEL1 to 3, respectively. If the number identical with the workpiece additional data number as shown above is programmed beforehand, only workpieces belonging to specific type or satisfying specific conditions can be detected when detecting workpieces, using the SYNC or VCSEEK command.

Name	Description	
x	X coordinate data of workpiece	VCSEL1
y	Y coordinate data of workpiece	VCSEL1
c	C coordinate data of workpiece	VCSEL2
Opt1	Additional data 1 of workpiece	
Opt2	Additional data 2 of workpiece	
Opt3	Additional data 3 of workpiece	

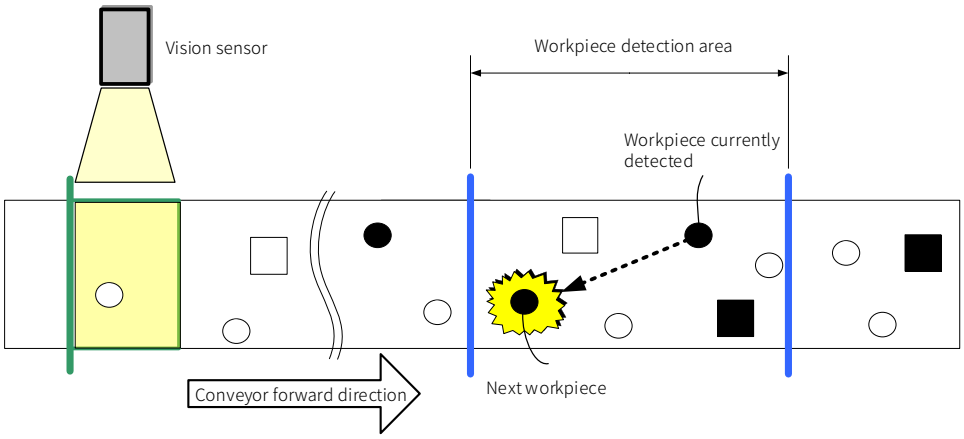
For instance, assume that the following data are set in workpiece additional data 1 to 3.

Opt1: Type of workpiece (1: Round, 2: Square)
Opt2: Color of workpiece (1: White, 2: Black)
Opt3: Workpiece status (1: OK, 2: NG)

Now, when the following three (3) conditions are programmed,

VCSEL1=1 Round workpiece
VCSEL2=2 Black workpiece
VCSEL3=1 NG workpiece

in the detection command (SYNC command or VCSEEK command) to be executed after this, only workpieces satisfying all conditions of VCSEL1 to 3 are collected.



・ How to set "free status" for VCSEL1 to 3:

When zero (0) is specified for VCSEL1 to 3, the free status mode takes effect and conditional detection is not executed.
For instance, assume that the following data are specified for workpiece additional data 1 to 3.

Opt1: Workpiece type data 1 (Workpiece profile ⇒ 1: Round, 2: Square)
Opt2: Workpiece type data 2 (Color information 1: White, 2: Black)
Opt3: Workpiece quality data 3 (Workpiece accepted, rejected, etc. ⇒ 1: OK, 2: NG)

Example 1) When detecting only black NG workpieces, neglecting workpiece type data 1:
VCSEL1=0 Workpiece type data 1 (Workpiece profile) → Free
VCSEL2=2 Workpiece type data 2 (Color information) → Black workpiece
VCSEL3=1 Workpiece quality data 3 (Workpiece accepted, rejected, etc.) → NG workpiece

Example 2) When detecting all workpieces, neglecting workpiece type data 1 to 3:
VCSEL1＝0 Workpiece type data 1 (Workpiece profile) → Free
VCSEL2＝0 Workpiece type data 2 (Color information) → Free
VCSEL3＝0 Workpiece quality data 3 (Workpiece accepted, rejected, etc.) → Free

Sample

```
PROGRAM VISCONV2                    Program for discriminating and detecting
                                    specified workpieces
    CONV( 1, A, B, 0, 0 )
    VCSE L 1 = 1                    "1: Round" is set for detecting condition 1
                                    (workpiece profile).
    VCFUNC = ON
    MOVE A                          Moves to waiting position.
    SYNC(1)                         If a round workpiece exists in the workpiece
                                    detection area, synchronization starts.
    MOVES VCPOS + POINT(,,100)      Moves to the position above the workpiece
                                    position as detected by SYNC command.
    MOVES VCPOS                     Moves to the workpiece position detected by
                                    SYNC command.
        :                           Picks up the workpiece.
    MOVES VCPOS + POINT(,,100)      Moves to the position above the workpiece
                                    position as detected by SYNC command.
    UNSYNC(1)                       Conveyor synchronization end.
    MOVE HOME + POINT( , , 50)
    MOVE HOME
        :                           Places the workpiece.
    MOVE HOME + POINT( , , 50)
END
```

VCVTYPE1 to 2

Function

Virtual vision conveyor synchronization: Selecting of set data

Classification

Option function Command

Format

VCVTYPE1 = < Integer >

Example

VCVTYPE1 = 1

Explanation & caution

Virtual vision conveyor synchronization uses the fixed data defined in “CONVEYOR.PAR [C005] Data set group for virtual vision conveyor tracking” to perform vision conveyor synchronization.
A total of 10 types of fixed data can be registered. Specify a number from 1 to 10 for the tables where the used data is defined.
VCVTYPE1: For CH1 conveyor
VCVTYPE2: For CH2 conveyor

Sample

```
PROGRAM VCPROG1
      :
VCFUNC = ON                    Data reception start
CONV( N, A, B, C, D )
LP:
VCVTYPE1 = 1                   Selection of number of table used
      SYNC(1)                  Synchronization is started if a workpiece is
                               found in the workpiece detection area
        :
      USYNC(1)                 Synchronization end
        :
GOTO LP
VCFUNC = OFF                   Data reception end
END
```

VCWSTAT1 to 3

Function

Refers to workpiece additional information (workpiece type and status) of workpiece currently detected.

Classification

System variable (Option function)

Format

VCWSTAT1

Example

```
IF VCWSTAT1 == 1 THEN GOTO LP
PRINT VCWSTAT3，CR
A=VCWSTAT1
```

Explanation & caution

Each workpiece data has three (3) additional information (workpiece type and status). Additional information of currently detected workpiece can be output to VCWSTAT1 to 3 by means of SYNC command or VCSEEK command.

When VCWSTAT1 to 3 is used, workpiece of many kinds running on the conveyor can be randomly picked up and sorted out by type.

- Only access of this system variable is possible. Writing is not allowed.
- Joint use of this command with motion command by using WITH statement is not allowed.
- VCWSTAT1-3 can refer the additional information from current conveyor channel in synchronization.

Sample

```
PROGRAM VISCONV2
        CONV( 1, A, B, 0, 0 )
        VCFUNC = ON
        VCSEL1 = 0
        MOVE A
        SYNC(1)
        TYPE = VCWSTAT1                    Workpiece additional data 1 of detected
                                           workpiece is saved in variable "TYPE".
        CONFIG=LEFTY
        MOVES VCPOS + POINT(,,100)
        MOVES VCPOS
        DELAY 0.2
'***** PICK-UP WORK PIECE *****
        MOVES VCPOS + POINT(,,100)
        UNSYNC(1)
        MOVE    P(TYPE)    + POINT( , , 50)
        MOVE    P(TYPE)
        DELAY 0.2
'***** PLACE WORK PIECE *****
       MOVE    P(TYPE)    + POINT( , , 50)
       WAIT MOTION >= 100
       PRINT 10.0-TIMER,CR
END
```

Workpiece set position changes with the workpiece additional data number of detected workpiece.

5.3 Vision + conveyer SCOL2 language programming notes

Please make program taking care of the following points.

1. The language that can be used for 1/2CH individual conveyer is as follows.
   CONV(N), SYNC(N), and USYNC(N) (N: Channel number)

2. The common language for 1/2CH conveyer is as follows.
   VCSEL1～3, VCFUNC, VCWSTAT1～3, VCSEEK, CNVSTAT

3. "2. The common language" becomes effective for the current channel in synchronization.

Operation

The utility function used for the "vision sensor + conveyor" synchronization function, and the procedures for stopping, resetting and restarting operations are described.

6. Operation

Operation can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

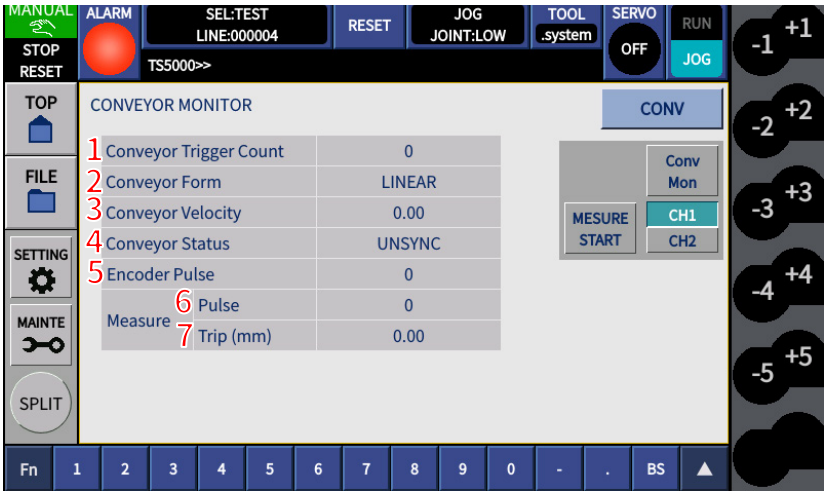
6.1 Conveyor Synchronization Utility

This paragraph describes the conveyor utility function used in the "vision sensor + conveyor" synchronization mode.
(There are a total of five TP5000 screens and seven TP1000 screens.)

6.1.1 Conveyor Monitor Screen (CONV_MON)

This paragraph deals with the utility function for performing various monitoring operations of the conveyor synchronization function.

<For TP5000>

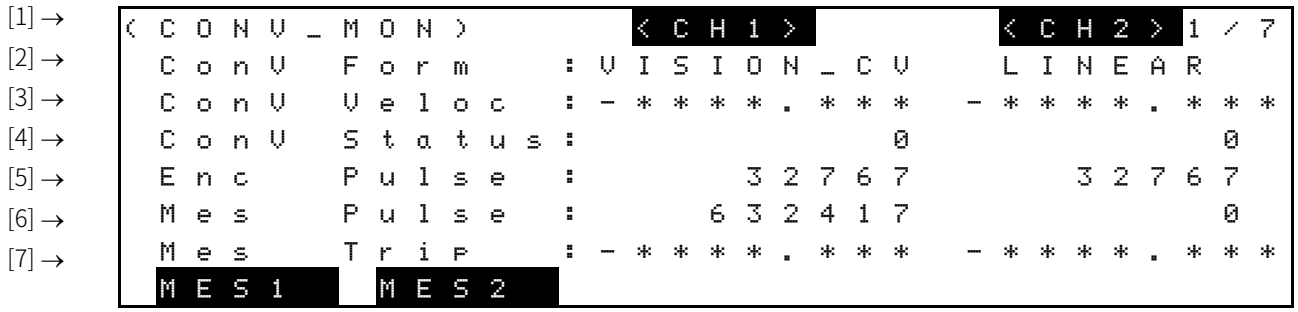


Descriptions of the elements in the screen are shown in the table below.

No. in figure	Description
1	Shows the number of times that the conveyor trigger signal was ON.
2	Shows the conveyor synchronization method. “LINEAR” → Linear “RESERVE” → Reserved “VISION_CV” → Vision + Conveyor synchronization
3	Shows the conveyor speed.
4	Shows the conveyor status. 0: Not synchronized 1: Waiting for synchronization signal 2: Synchronization delayed 3: Accelerated speed synchronization in progress 4: Constant-speed synchronization in progress 5: Decelerated speed synchronization in progress 6: Synchronization end process in progress -1: Conveyor wire disconnected
5	Shows the pulse counter value of the conveyor encoder.
6	Shows the total number of pulses of the measured conveyor encoder.
7	Shows the conveyor travel that was calculated from the travel per encoder pulse and number of pulses.

<For TP1000>

[Utility] → [Conv]



[1] Trigger signal monitor:
When the conveyor trigger signal has been turned on, <CH1> and/or <CH2> are inversely displayed.

<CH1> : Trigger signal OFF

<CH1> : Trigger signal ON

[2] Conv Form: Type of conveyor synchronization

LINEAR → Straight line

ARC (+) → Arc (CW)

ARC (–) → Arc (CCW)

VISION_CV → Vision sensor + conveyor synchronization

[3] Conv Veloc: Conveyor speed

[4] Conv Status: Conveyor status

0: During asynchronization

1: Waiting for synchronization signal

2: During synchronization delay

3: During acceleration for synchronization

4: During constant-speed for synchronization

5: During deceleration for synchronization

6: During processing of synchronization end

–1: Conveyor breakage

[5] Enc Pulse: Pulse counter value of the conveyor encoder

[6] Mes Pulse: Total number of pulses of the conveyor encoder
When you press the [MES1] (F1) or [MES2] (F2) function key, measurement of total encoder pulses starts.

[7] Mes Trip: Travel distance of the conveyor
Real travel distance of the conveyor is indicated in mm. When you press the [MES1] (F1) or [MES2] (F2) function key, measurement of conveyor travel distance starts.

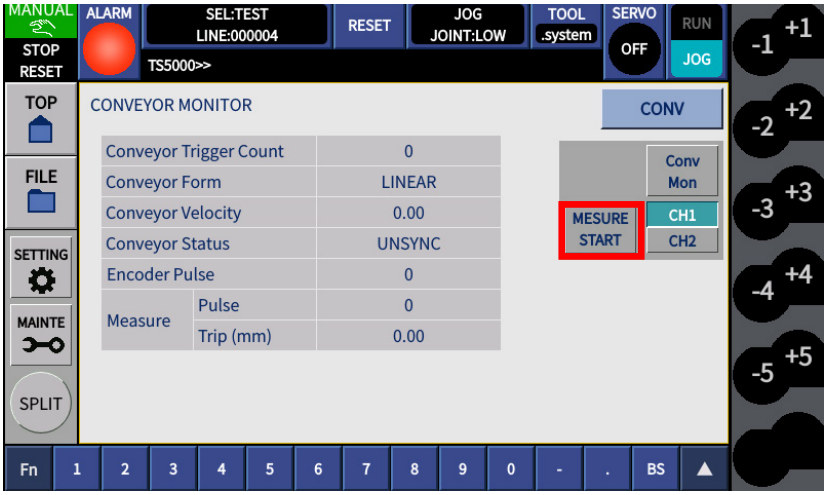
How to measure the total number of pulses of conveyor encoder and conveyor travel distance

<For TP5000>

1 Tap the channel to be measured.

2 Tap "Start Measurement".

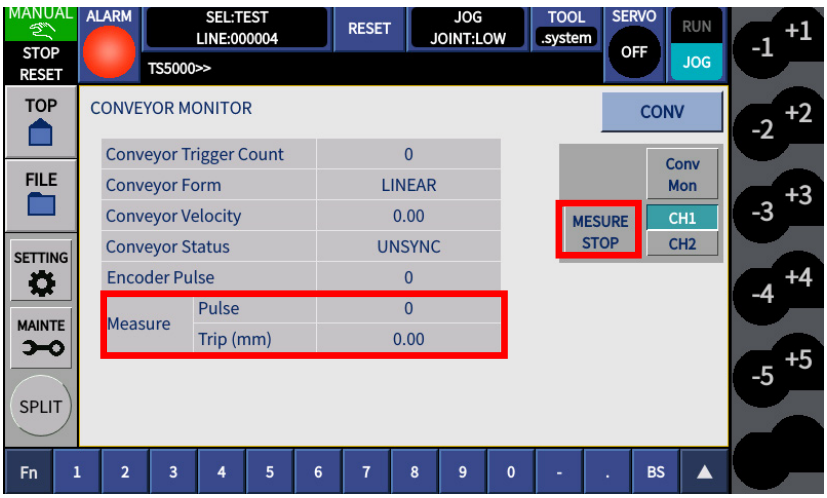
Start measurement of the encoder pulse.
(When tapped, the button name changes to "End Measurement".)



3 Move the conveyor by an amount of ΔL only.

4 Tap "End Measurement".

Exit measurement. Updating of "Pulses" and "Distance" under "Measurement" is stopped.



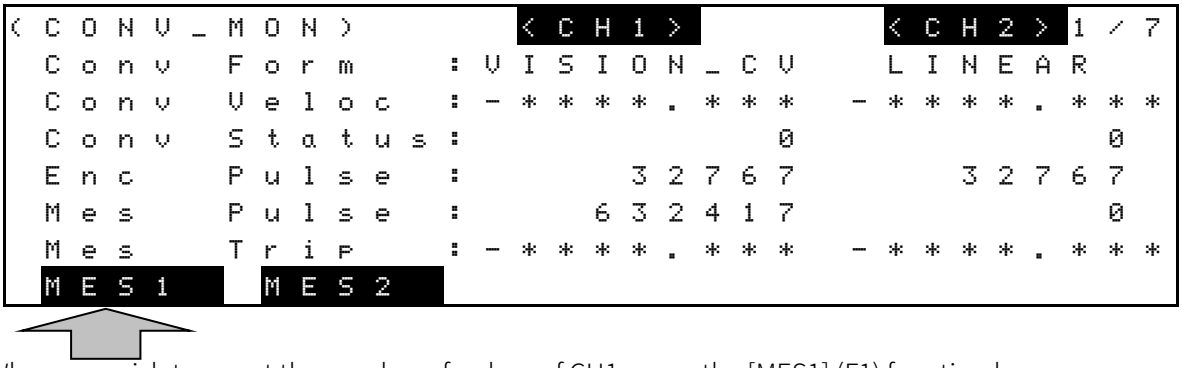
MEMO

・The value for "Distance" under "Measurement" is not displayed correctly unless the conveyor parameter is set. To display this parameter, the value must be set for the travel per pulse [C000] Conveyor Parameter that was calculated from the measured value for "Pulse" under "Measurement" and the conveyor travel distance (actual measured value). For details on how to set the parameter, see "3.3. [C000] CONVEYOR PARAMETER (Travel value per pulse of encoder)".

<For TP1000>

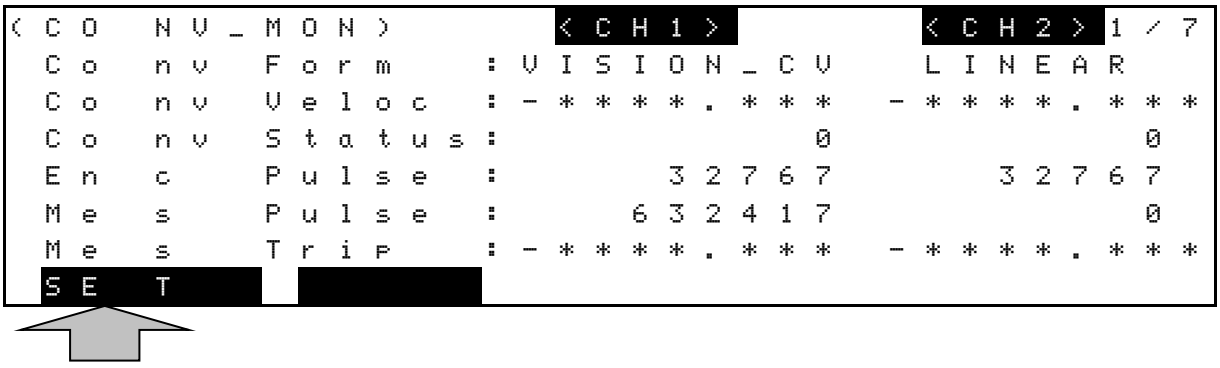
[Utility] → [Conv]

1 Press the [MES1] (F1) or [MES2] (F2) function key corresponding to the channel through which you wish to measure.

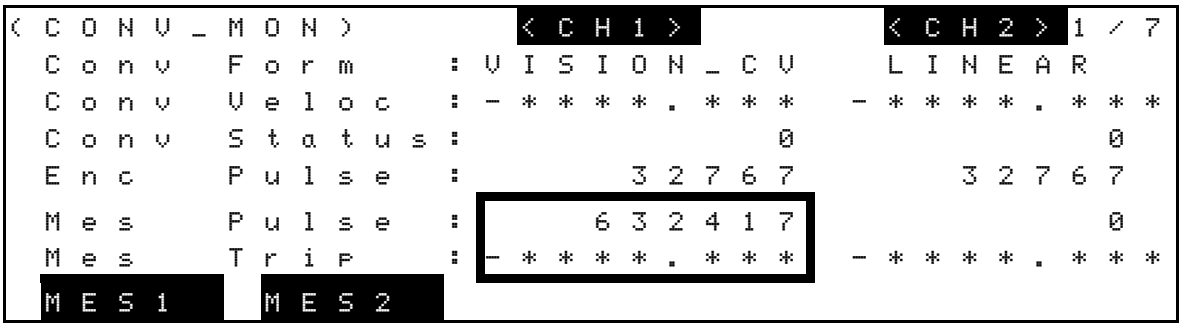


When you wish to count the number of pulses of CH1, press the [MES1] (F1) function key.

2 When you wish to count the number of pulses of CH1, press the [MES1] (F1) function key. Likewise, when you wish to count the number of pulses of CH2, press the [MES2] (F2) function key. After moving the conveyor by "ΔL", press the [SET] function key.



3 Press the [SET] key, and the number of pulses measured "Mes Pulse" and the value converted from the number of pulses measured to the travel distance in mm "Mes Trip" are displayed. To measure again, press the [MES1] (F1) or [MES2] (F2) key and repeat Steps (1) and (2) above.

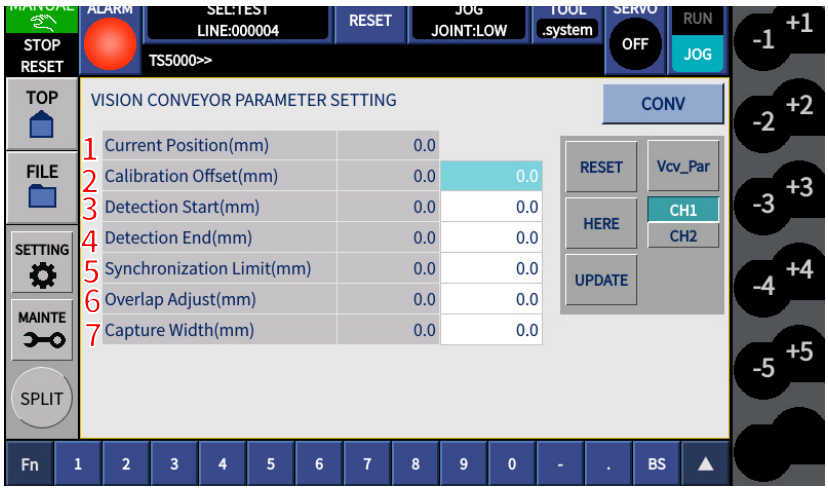


MEMO

・The value of "Mes Trip" is not displayed correctly unless the conveyor parameter is specified. To display this parameter, the value must be set for the travel per pulse [C000] Conveyor Parameter that was calculated from the "Mes Pulse" value that was measured and the conveyor travel distance (actual measured value). For the parameter setting procedures, see "3.3. [C000] CONVEYOR PARAMETER (Travel value per pulse of encoder)".

6.1.2 Vision Conveyor Parameter Setting Screen (VCV_PAR)

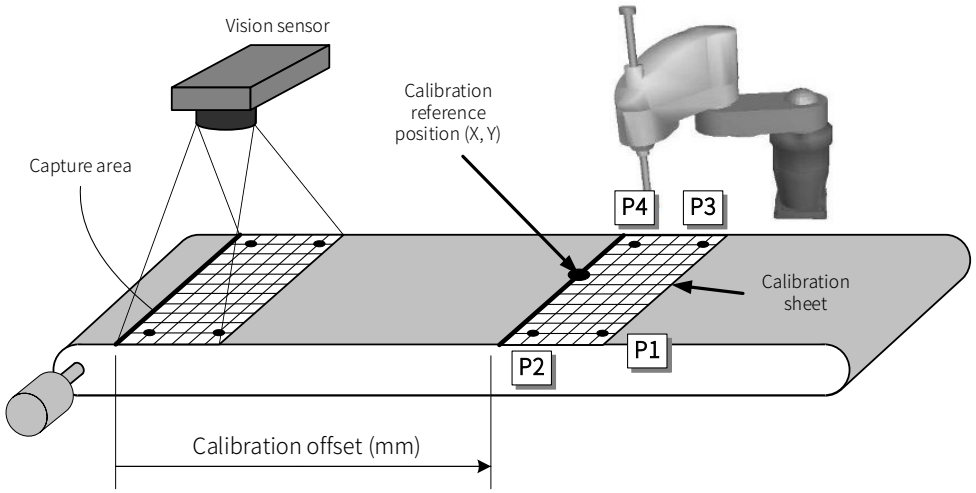
<For TP5000>



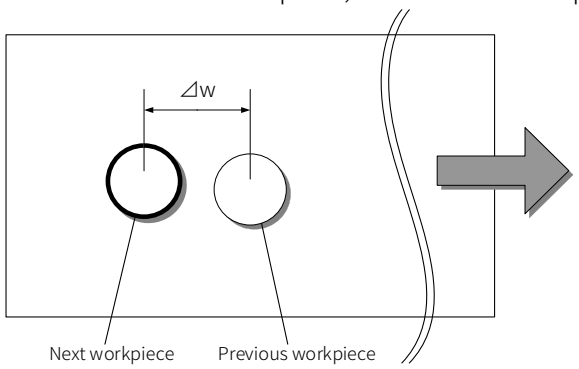
Descriptions of the elements in the screen are shown in the table below.

No. in figure	Description
1	Conveyor current position (mm)
2	[C001] Vision + Conveyor Tracking Parameter [1)]
3	[C001] Detection area start position [mm]
4	[C001] Detection area end position [mm]
5	[C001] Conveyor Sync limit position [mm]
6	[C001] Overlap allowance [mm] [2)]
7	[C001] Capture width [mm] [3)]

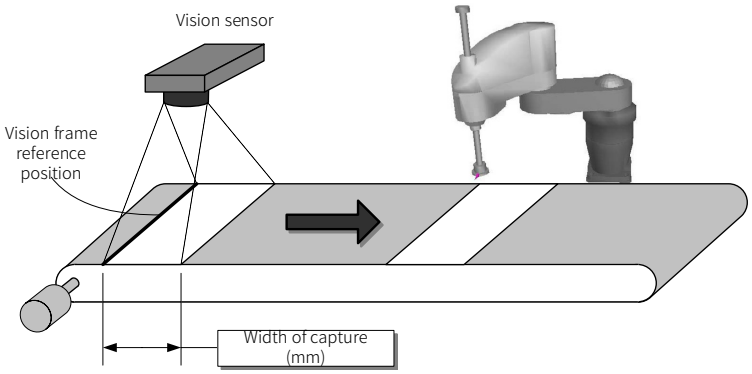
1) The calibration offset (mm) of the conveyor parameter "[C001] Vision + Conveyor Tracking Parameter" can be set. For details on the setting procedure, see "4.3.5 Register of Calibration Offset (mm)".



2) If the distance "⊿w" between the next workpiece being indexed and previous workpiece that was indexed is less than the "Overlap tolerance (mm)", the previous and next workpieces are assumed to be the same workpiece, and the next workpiece is ignored.



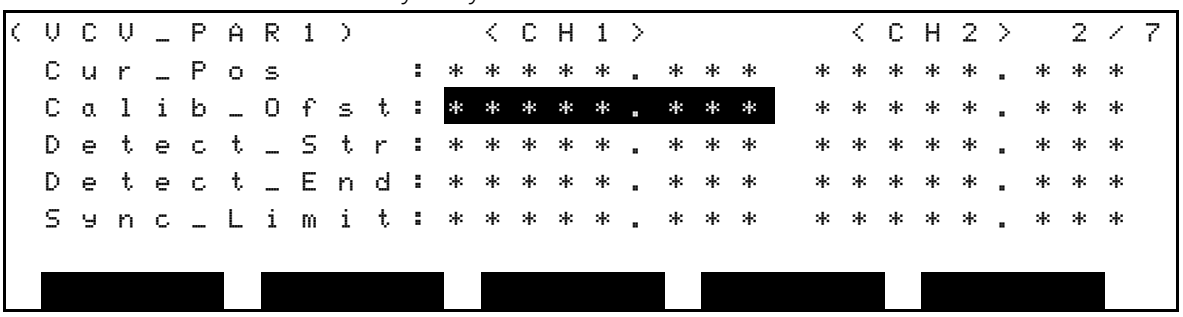
3) Set the width (mm) of the image that was captured by the vision sensor. The "Capture width" is the distance that was measured on the conveyor movement direction side with the vision frame origin as the base point.



<For TP1000>

·VCV_PAR1

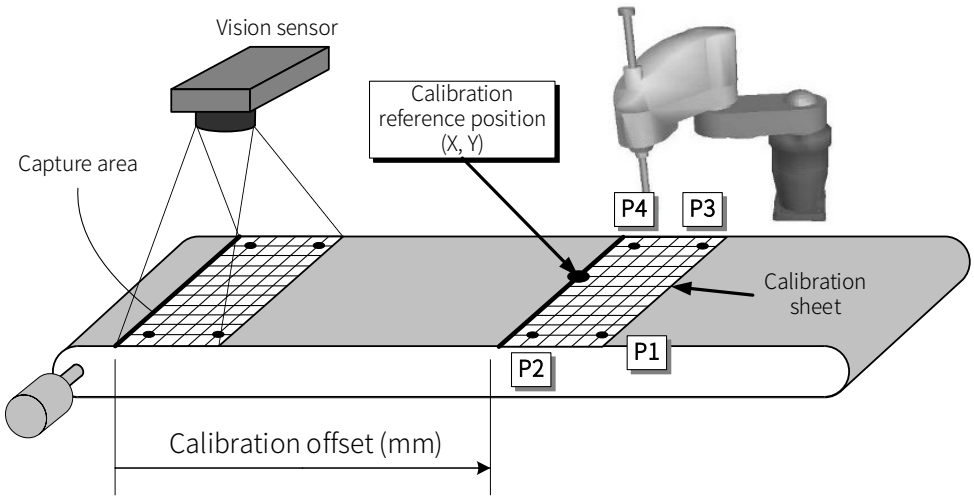
Press the <NEXT> key, and the system calls the utility display where you can edit various parameter values used for the "vision sensor + conveyor" synchronization function.



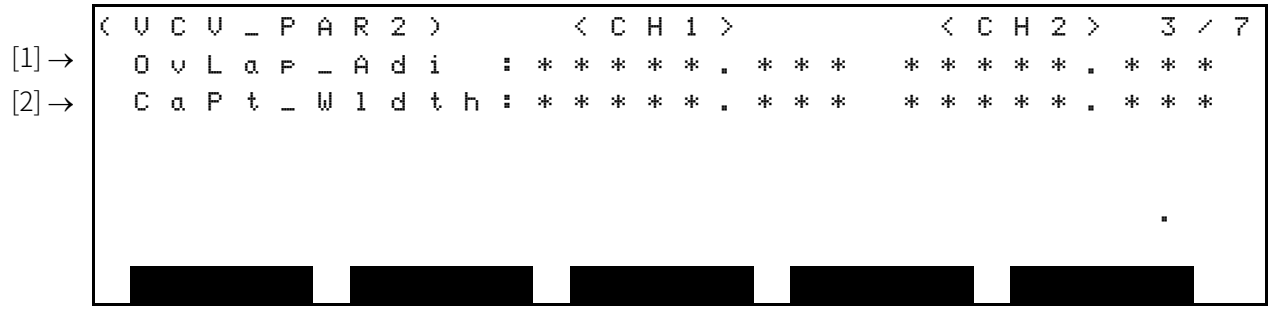
[1] Cur_Pos: Present conveyor position (mm)

[2] Calib_Ofst: Calibration offset (mm)

It is possible to set the calibration offset (mm) of conveyor parameter "[C001] Vision + Conveyor Tracking Parameter" . For details on the setting procedures, see “4.3.4 Register of Vision Sensor Calibration Points”.

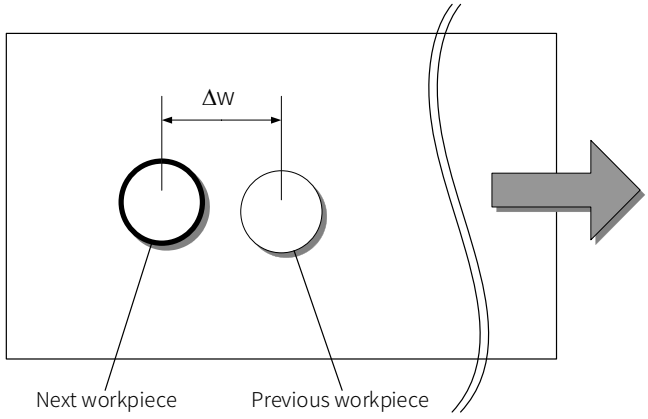


·VCV_PAR2



[1] Overlap_Adj: Overlap tolerance (mm)

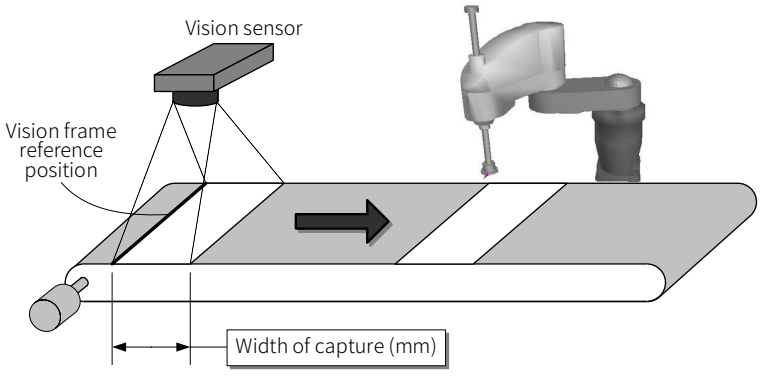
If the distance "Δw" between the workpieces of ongoing and previous detection is not greater than the overlap allowance (mm), the system interprets that the next and previous workpieces are the same workpiece, and ignores the next workpiece.



[2] Capture_Width: Capture width (mm)

Specify the width of an image captured by the vision sensor in mm.

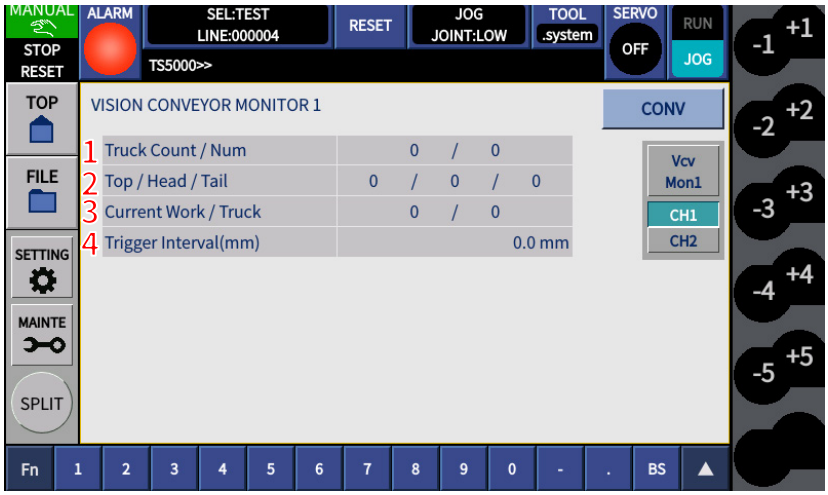
The width of capture is the distance measured in the conveyor travel direction, taking the vision frame reference position as the start position.



6.1.3 Vision Conveyor Monitor 1 Screen (VCV_MON1 -1, VCV_MON1-2)

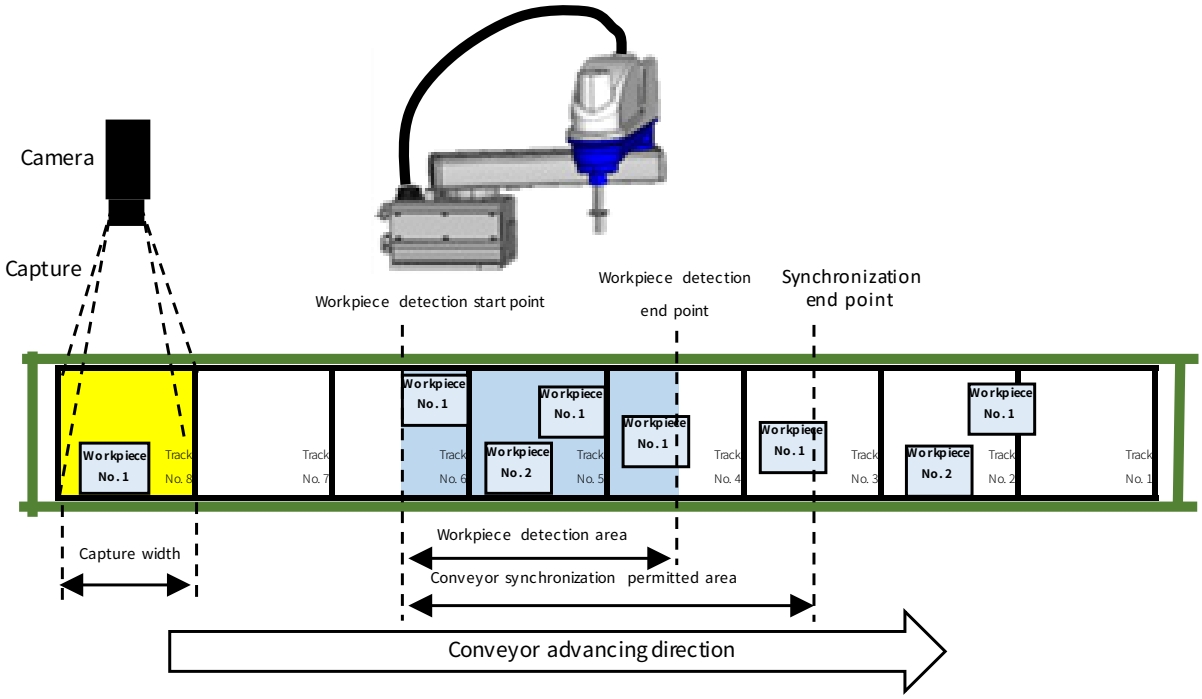
This paragraph refers to the internal counter monitor function during the "vision sensor + conveyor" synchronization mode.

<For TP5000>



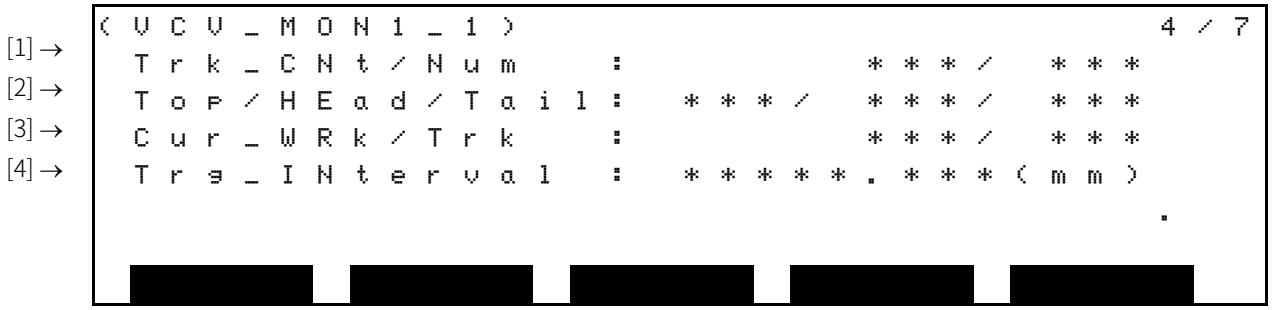
Descriptions of the elements in the screen are shown in the table below.

No. in figure	Description	Example in following figure
1	Track count: Latest track number Total tracks: Total amount of tracking data on the conveyor	Track count: Track No. 8 Total number of tracks: 8
2	Conveyor start: Starting track number within the conveyor synchronization permitted area Workpiece leading edge: Starting track number within the workpiece detection area Workpiece trailing edge: Final ending track number within the workpiece detection area	Conveyor start: Track No. 3 Workpiece leading edge: Track No. 4 Workpiece trailing edge: Track No. 5
3	Current workpiece: "Workpiece No." for workpiece currently being indexed Current track: "Track No." for workpiece currently being indexed	Current workpiece: Workpiece No. 1 Current track: Track No. 4
4	Shows the conveyor travel distance from the previous trigger signal until the current trigger signal is received.	---

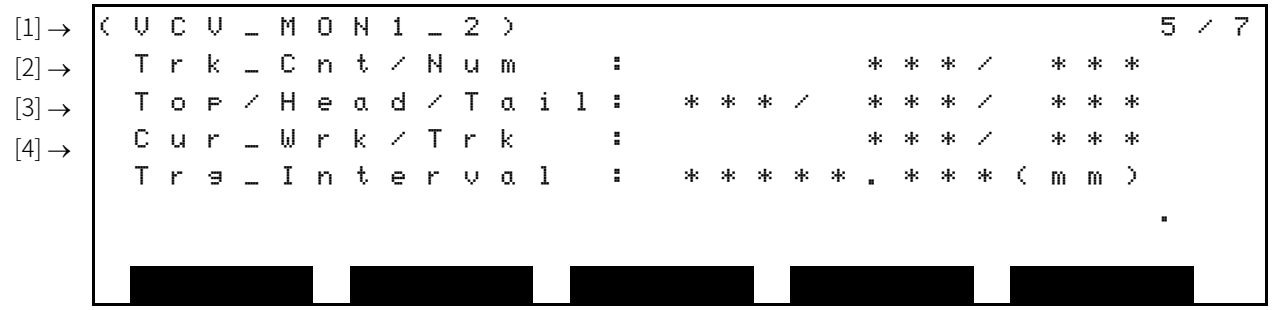

Camera
Capture
Workpiece detection start point
Workpiece detection end point
Synchronization end point
Workpiece No.1
Workpiece No.2
Track No.8
Track No.7
Track No.6
Track No.5
Track No.4
Track No.3
Track No.2
Track No.1
Capture width
Workpiece detection area
Conveyor synchronization permitted area
Conveyor advancing direction

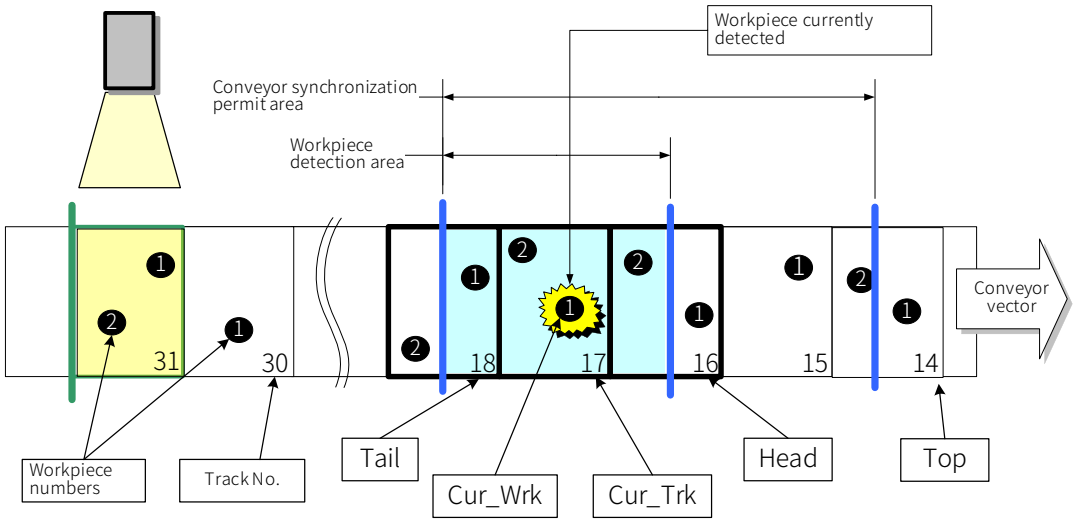
<For TP1000>

For CH1



For CH2





[1] Trk_Cnt/Num

The newest track number [1)] and total number of tracking data existent on the conveyor are displayed.

1) Index number of tracking data.

[2] Top/Head/Tail

Top: The leading track number of the conveyor synchronization permit area is displayed.

Head: The leading track number of the workpiece detection area is displayed.

Tail: The last track number of the workpiece detection area is displayed.

[3] Cur_Wrk/Trk

Cur_Wrk: The workpiece number [2)] of the workpiece currently detected is displayed.

Cur_Trk: The track number of the workpiece currently detected is displayed.

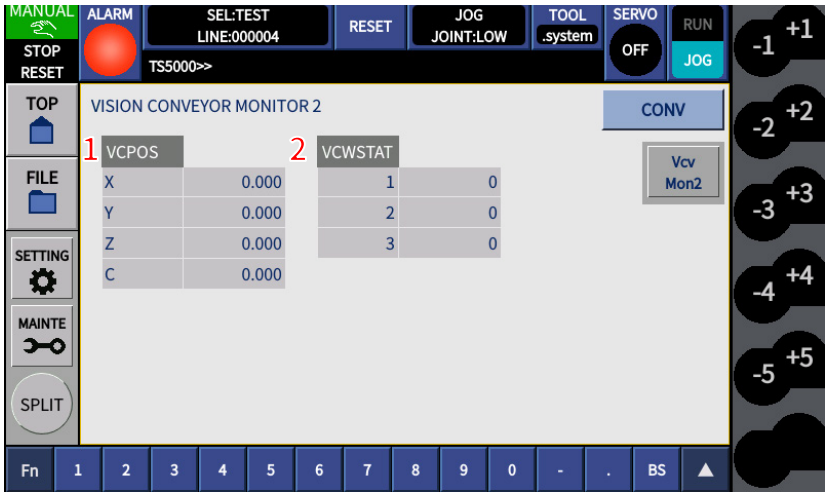
2) Workpiece index number existent in tracking data.

[4] Trg_Interval

The conveyor travel distance until the current trigger signal is accepted after the previous trigger signal is displayed.

6.1.4 Vision Conveyor Monitor 2 Screen (VCV_MON2)

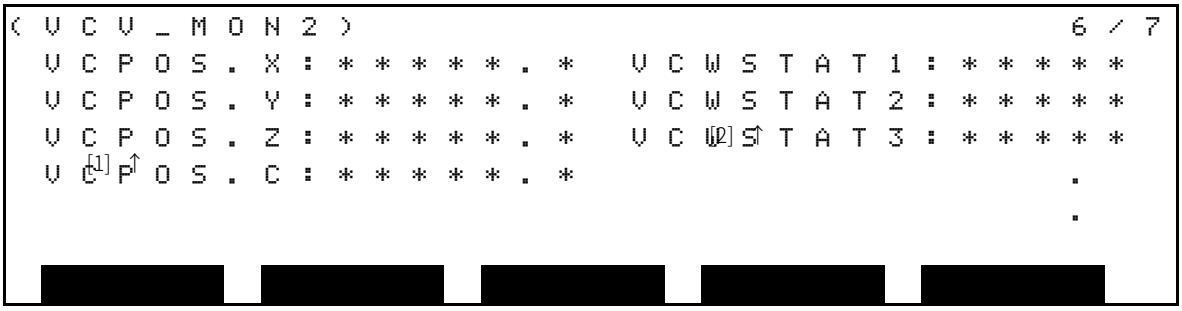
<For TP5000>



Descriptions of the elements in the screen are shown in the table below.

No. in figure	Description
1	Shows the movement target position of the workpiece that is currently being indexed. (The same values as "VCPOS" in SCOL2 language are displayed.)
2	References the "Workpiece additional information (product type data, workpiece status)" for the workpiece that is currently being indexed. (The same values as "VCWSTAT1" to "VCWSTAT3" in SCOL2 language are displayed.)

<For TP1000>



[1] VCPOS. X/VCPOS. Y/VCPOS. Z/VCPOS. C

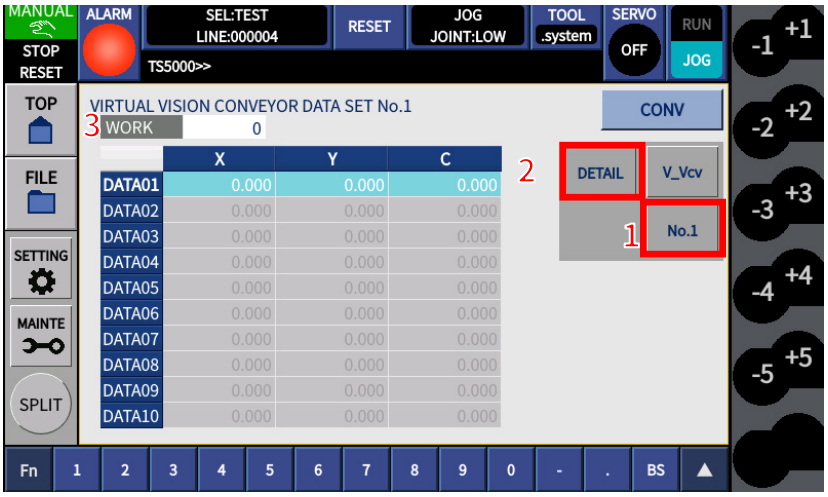
The target motion positions of the workpiece detected currently are displayed. (The same values as in SCOL2 language "VCPOS" are displayed.)

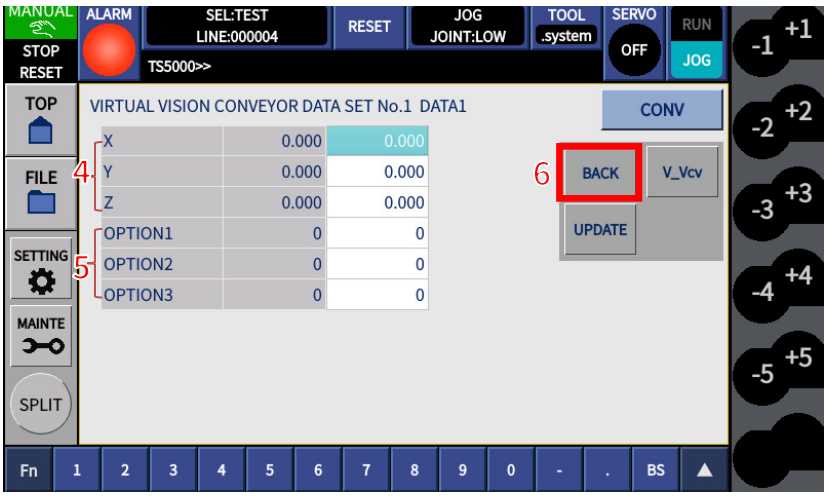
[2] VCWSTAT1 to 3

Workpiece additional information (workpiece type and status) for the workpiece currently detected can be referred to. (The same values as in SCOL2 language "VCWSTAT1 to 3" are displayed.)

6.1.5 Virtual Vision Conveyor Data (Virtual VCV data)

<For TP5000>

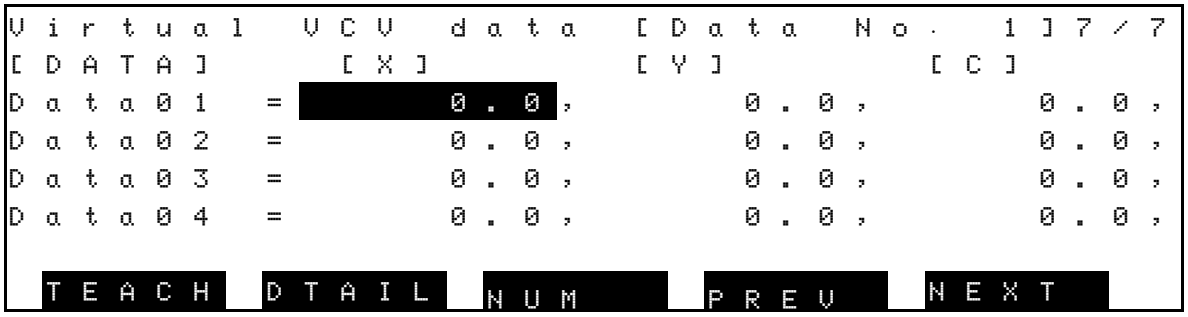




Descriptions of the elements in the screen are shown in the table below.

No. in figure	Description
1	Switches display of the data set used in virtual vision conveyor synchronization. (This is used for "VCVTYPE1" and "VCVTYPE2" in SCOL2 language.)
2	Shows details of the selected data.
3	Sets the quantity of workpieces used for a selected data set number. This is set in the range from 0 to 10.
4	Shows the data used as "Movement target position" when performing virtual vision conveyor synchronization. (The same value as "VCPOS" in SCOL2 language is displayed.)
5	Shows the data used as "Workpiece additional information (product type data, workpiece status)" when performing virtual vision conveyor synchronization. (The same values as "VCWSTAT1" to "VCWSTAT3" in SCOL2 language are displayed.)
6	Returns to the original screen.

<For TP1000>



・X / Y / C

This displays the data used for the "Operation target position" during virtual vision conveyor synchronization. (The same value as the SCOL2 language "VCPOS" is displayed.)

・OPT1 to 3

This displays the data used for "Workpiece additional information (product type data, workpiece state)" during virtual vision conveyor synchronization. (The same values as the SCOL2 language "VCWSTAT1" to "VCWSTAT3" are displayed.)

6.2 Stop Operation

The following stop operations are available for the TS5000 controller.

- STOP command and external stop signal input
- BREAK command
- Master mode switch
- STOP command of SCOL2 language
- MODE selection (STEP/MOTION)
- FIELD HOLD command

} A. Stop with servo ON

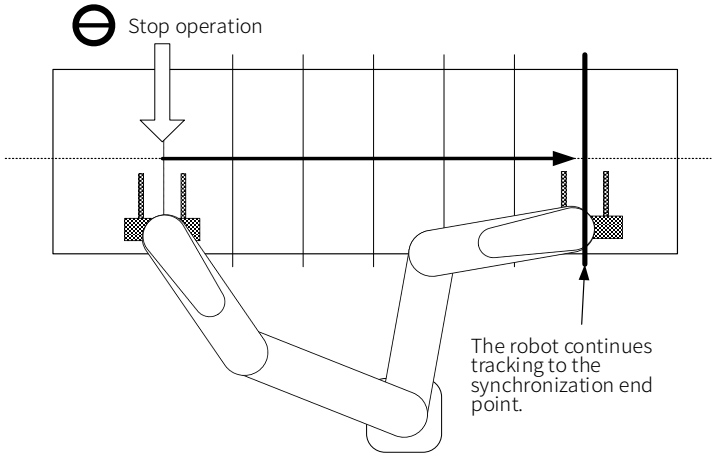
- Emergency stop
- Servo OFF (external servo OFF input signal)

} B. Stop with servo OFF

When the stop operation is executed during conveyor synchronization (i.e., while the robot is moving by tracking the conveyor speed), the results are as follows:

A. Stop with servo ON

If the stop operation is executed during conveyor synchronization, the program stops, but the robot will not stop and continues tracking the conveyor to the conveyor synchronization end point.



NOTICE	
 **REQUIRED**	・ If the robot is kept intact under this condition, the robot stops at the conveyor synchronization end point. As a result, the robot hand may collide with a workpiece on the synchronized conveyor. To prevent this, the system should be constructed, fully considering the following matters. [1] If the conveyor or other external equipment has entered an emergency stop state, the robot should also enter an emergency stop state. [2] If the robot has entered an emergency stop state, the conveyor or other external equipment should also enter an emergency stop state. [3] If the robot program has stopped due to some reason (error stop/stop operation by operator), the conveyor should be stopped also.

B. Stop with servo OFF

The robot stops instantaneously to stop the synchronization processing instantly.

When performing the stop operation during conveyor synchronization, strictly observe the following precautions, fully taking notice of the above.

MEMO

- *Even if the program stopped during conveyor synchronization, continuous operation is possible by using the restart function just after the program stop. In principle, however, after the stop operation, be sure to reset the operation, move the robot to the safe place by manual guidance, reset the program, then resume the operation. (For details, see Para. 6.3.)*

NOTICE



REQUIRED

- Unless an abnormal error is caused, DO NOT perform “B. Stop with servo OFF” during conveyor synchronization. Otherwise, the robot hand will collide with a workpiece on the synchronized conveyor.
  To avoid the above accident, construct the system, taking notice of the following matters.

[1] If the conveyor or other external equipment has entered an emergency stop state, the robot should also enter an emergency stop state.

[2] If the robot has entered an emergency stop state, the conveyor or other external equipment should also enter an emergency stop state.

[3] If the robot program has stopped due to some reason (error stop/stop operation by operator), the conveyor should be stopped also.

6.3 Procedures for Reset to Restart

When the reset operation is executed after the stop operation done during conveyor synchronization, the internal processing as shown below is executed.

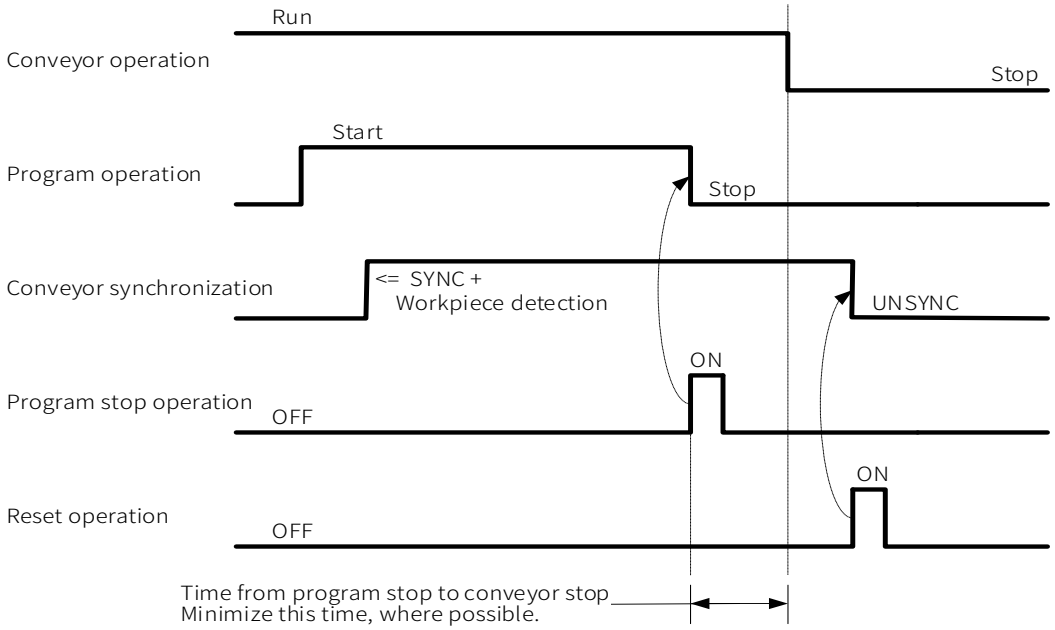
1 The processing of conveyor synchronization finishes.

(The UNSYNC command (i.e., synchronization end command) is executed internally.)

2 All tracking data saved (i.e., track number, workpiece number, workpiece coordinates, etc.) are cleared.

The reset operation comes in the three (3) types; program reset, step reset and cycle reset.

After the stop operation as described in the preceding paragraph is executed during conveyor synchronization, be sure to perform the reset operation before restarting the program. When executing the reset operation, however, be sure to strictly observe the following procedures. Otherwise, the robot hand may collide with a workpiece on the synchronized conveyor.



MEMO

- *After the stop operation and before the reset operation, be sure to stop the conveyor. The time covering from the program stop to the conveyor stop as indicated in the above figure should be as short as possible.*
- *If the stop operation in "B. Stop with servo OFF" has been effected, the time covering from the program stop to the conveyor stop as indicated in the above figure should be almost zero (0).*

6.4 Conveyor Stop

If the number of encoder pulses counted is zero (0) during the time designated by the construct parameter after the start of conveyor synchronization following the synchronization trigger signal ON, the error saying "001–048: Conveyor Stop" generates. This is the error of level 1, and the program is executed continuously without being stopped.

The conveyor (encoder) speed fluctuation should be within ±7.5% to the average constant speed of the conveyor being used. Otherwise, precise synchronization accuracy cannot be assured, and the workpiece or hand, etc., may be damaged accordingly.

Troubleshooting

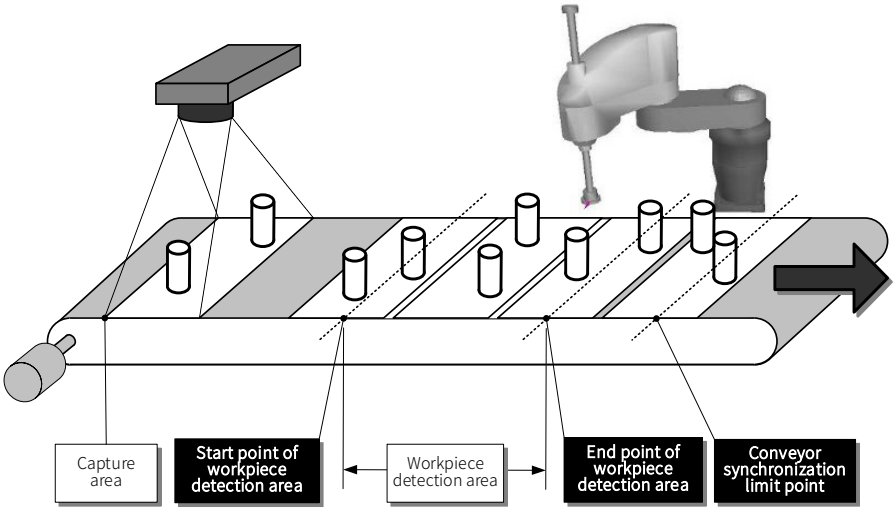
The self-diagnosis function and troubleshooting of the "vision sensor + conveyor" synchronization function are described.

7. Troubleshooting

7.1 Error Message Table

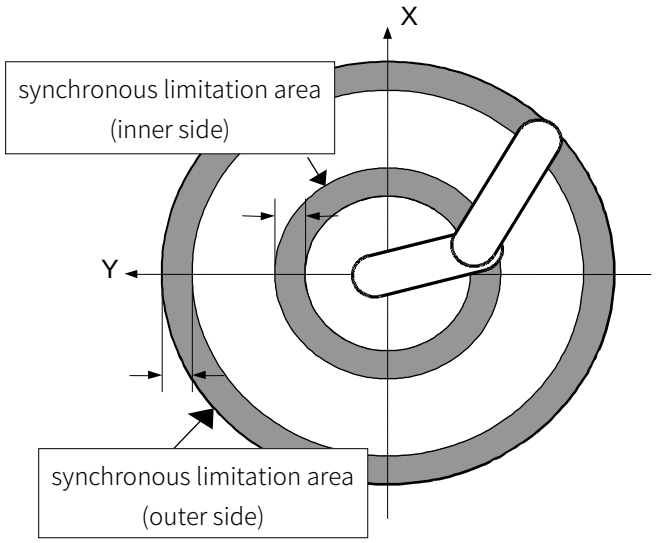
If an error has been detected by the self-diagnosis function, appropriate error message is displayed on the teach pendant, as shown below. Identify the cause and take appropriate measures according to the following table.

Error code	Error message, possible causes and remedies
001-047	**“Conveyor Line Breakage”** The encoder cable of the specified conveyor number is broken or the encoder is defective. Check for the wiring. Even after this error occurred, the robot continuously tracks the conveyor based on the conveyor (encoder) speed just before the error generation, without being stopped. If the encoder cable breakage has been detected during conveyor synchronization, be sure to recess the robot to the safe place. Otherwise, the robot may collide with the workpiece or conveyor. This error will not be reset unless the power is turned off. Locate and remedy the broken part, then turn the power on again.
001-048	**“Conveyor Stop”** The conveyor stop time has reached the judgment time specified by the parameter. • The conveyor was stopped when the UNSYNC command has been executed.
001-147	**“COM Buffer Over”** “[C000] Type of conveyor synchronization” is not set.
001-177	**“VCV Comm Process Abnormal”** Communication data has been sent from the vision sensor multiple times before the trigger signal turned on. → Confirm again the data transmission timing from the vision sensor and the trigger signal transmission timing. (For details, see Para. 4.1.)

Error code	Error message, possible causes and remedies
001-178	**“VCV Comm data format error”** The format of the data reached from the vision sensor is incorrect. (Data other than the numerical value has reached. The format of the numerical data is incorrect. The number of data is incorrect.) → Confirm again the data format sent from the vision sensor. (For details, see Para. 4.2.)
001-179	**“VCV Too many triggers”** The number of tracking data that can be saved exceeds the maximum value. → Increase the capture width of the vision sensor and reduce the number of capture counts. → Or reduce the distance from the capture reference point to the conveyor synchronization limit point. 
001-180	**“VCV Comm Process Timeout”** The "1-180: VCV Comm Process Timeout" error occurs when the data reception cannot be completed within [C001] Data reception time out setting time. When [C001] {Data reception timeout [ms]} is "0.0", this function becomes disabled.
002-033	**“Conveyor Parameter Error”** • The parameter value exceeds the permissible range.

<table>
<tr><th>Error code</th><th>Error message, possible causes and remedies</th></tr>
<tr><td rowspan="2">004-051</td><td><h2>“Conveyor sync limit area”</h2></td></tr>
<tr><td>

The robot goes into the "conveyer synchronous limitation area" whilst the conveyor synchronization. Execute program reset and error reset after moving the robot outside the limitation area. Review the program and the conveyer parameter setting when occurring again.
Specify the "conveyer synchronous limitation area" parameter on "CONVEYOR.PAR: [C000] Conveyor parameter -> { Limitation width of synchronization }" in mm.
This error can be reset.
After the error occurs, execute "PROGRAM RESET" by all means before resuming operations.



</td></tr>
<tr><td rowspan="2">004-052</td><td><h2>“VCV Comm port closed”</h2></td></tr>
<tr><td>The communication port of Ethernet (IP7 or IP8) closed during the vision sensor + conveyer synchronization. Please confirm whether wiring and the vision device of Ethernet work normally.</td></tr>
</table>